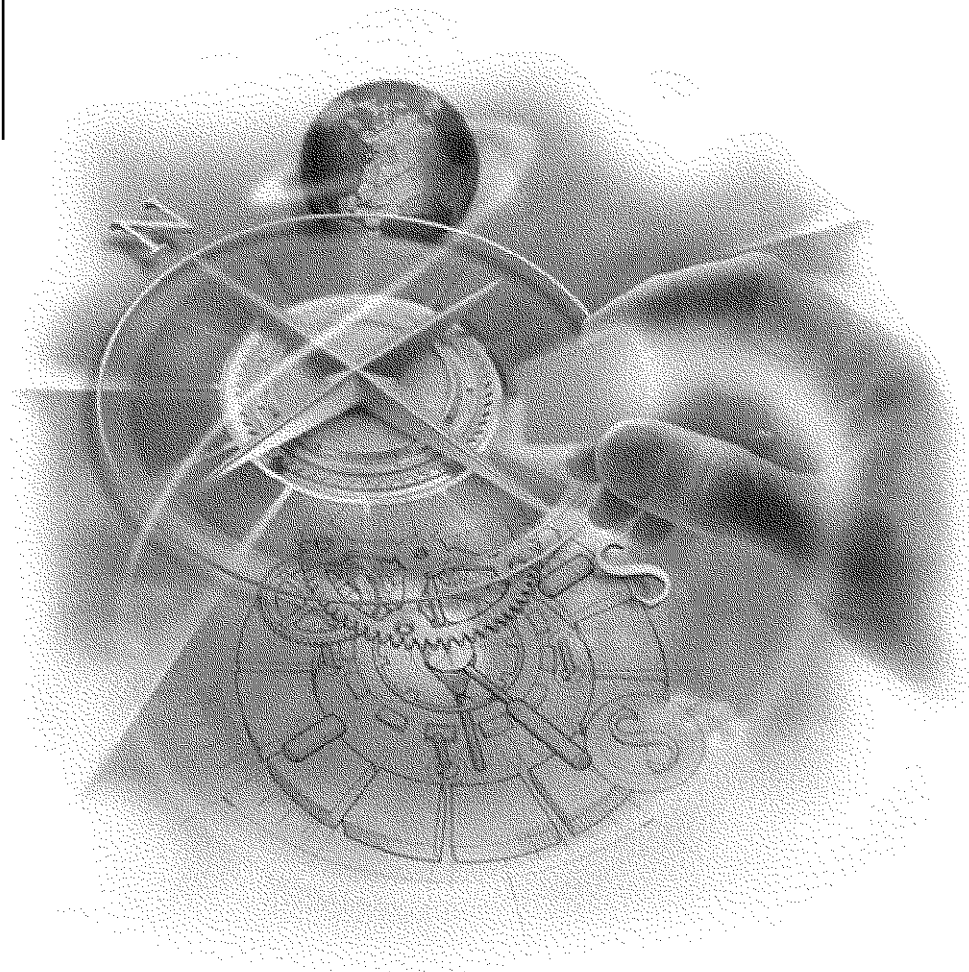


NDS™ Schema Reference



Novell Developer Kit

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Preface

NDS is based on a set of rules which define the following:

- ♦ The types of objects that can exist in an NDS tree
- ♦ The possible locations of these objects in the NDS tree
- ♦ The information (stored as attributes) that can be and must be maintained about the object

This set of rules is called the schema. The NDS Schema Reference describes these rules and the items that have been defined for the NDS 8 operational schema. This includes the object classes, the attribute definitions, and the attribute syntax definitions which NDS defines. This document also describes the object classes and attribute definitions that NetWare 5 defines when it is installed.

NDS 8 adds the following:

- ♦ A container class, domain, that accepts most leaf objects as subordinate objects
- ♦ Effective class status for the Person and Organizational Person classes
- ♦ An ndsLoginProperties class which allows Person, Organizational Person, Organization, and Organizational Unit classes to inherit all the attributes required for logging in to NDS

NDS eDirectory has enhanced the support for LDAP v3 and the LDAP schema. An LDAP name section has been added to the attribute and class definitions that are automatically made available for LDAP access. The other definitions that do not conform to LDAP naming conventions must be mapped to allow LDAP access.

The NDS schema is extensible, which means developers can add to the types of objects and attributes maintained by NDS.

1

Schema Concepts

This chapter describes the rules that govern the schema and its various data types. For general rule structure and rule information, see “Schema Structure” on page 27. For information about its three data types, see

- ♦ “Object Classes” on page 30
- ♦ “Attribute Type Definitions” on page 58
- ♦ “Attribute Syntax Definitions” on page 64

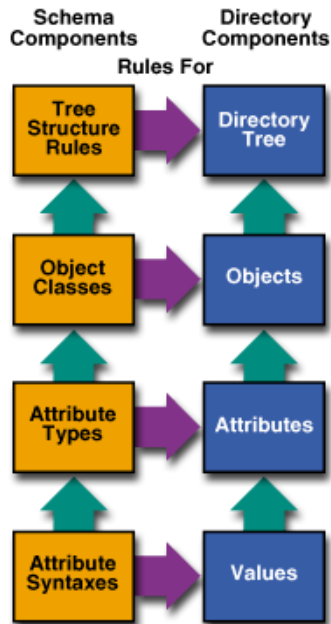
For information on extending these data types to include other object classes and attributes, see “Schema Extensions” on page 67.

Schema Structure

The schema defines the set of rules that govern the types of objects that can exist in an NDS™ tree. Each object belongs to an object class that specifies which attributes can be associated with the object. All attributes are based on a set of attribute types that are, in turn, based on a standard set of attribute syntaxes.

The NDS schema not only controls the structure of individual objects, but it also controls the relationship among objects in the NDS tree. The schema rules allow some objects to contain other subordinate objects. Thus the schema gives structure to the NDS tree.

The figure below shows how the schema components and the NDS components are interrelated. The vertical arrows indicate the structure dependencies from the basic building blocks up to the NDS schema and the NDS tree, respectively. The horizontal arrows denote the schema rules that apply to the respective NDS components.



The attribute syntaxes define the primary data types for values stored in the NDS tree. Attribute types are defined from the attribute syntaxes and define the possible attributes an object can have. Object classes are defined using a subset of the possible attributes and determine the types of objects that can be in the tree. The tree structure rules define how the object classes can be organized and nested in the tree, and therefore determine the tree's structure.

Objects that can contain other objects are called container objects. Container objects are the building blocks of the NDS tree. Objects that cannot contain other objects are known as noncontainer or leaf objects. Leaf objects comprise the actual network resources such as a user, a server, or a printer.

See Also:

- ♦ "Schema Components" on page 29
- ♦ "Object Classes" on page 30
- ♦ "Attribute Type Definitions" on page 58
- ♦ "Attribute Syntax Definitions" on page 64

Structure Rules

The Structure Rules define the relationships between objects in the NDS tree. These relationships are defined by two properties:

- ♦ “Containment Classes” on page 35
- ♦ “Naming Attributes” on page 33

Containment classes determine where the object can appear in the NDS tree relative to other classes of objects. Naming attributes-sometimes called Named By attributes-determine how the object is named.

Distribution of the NDS Schema

The NDS schema is global. Each server stores a replica of the schema in its entirety. The schema replica is stored separately from the partitions that contain directory objects. Changes to any one schema replica are propagated to the other replicas. You can perform modifications to the schema only through a server that stores a writable replica of the root partition. Servers storing read-only replicas of the root partition can read but not modify schema information.

Schema Components

The NDS schema consists of two basic components:

- ♦ Object Classes
- ♦ Attribute Types

The set of rules that controls the creation of a particular object is called an object class. Each object class is defined in terms of attributes. An attribute is a specific piece of information that exists for an object. (Attributes are called properties by NetWare utilities and documentation.)

For example, NDS contains an object class for users, called User object. This User object class defines many attributes (more than 80), including attributes for such items as the user's name, telephone number, address, and group memberships.

Attributes are defined in terms of a base set of data types called attribute syntaxes. The attribute syntaxes define the primary data types for values stored in the NDS database.

For example, some attributes such as Password Minimum Length or Minimum Account Balance take integer values while other attributes such as a user's Full Name or Given Name take string values.

NDS has a set of built-in classes and attribute types that accommodate general categories of network objects such as organizations, users, and devices. This set is called the base schema. To this schema, Novell developers have added objects and attributes to form a standard schema for NetWare 5.x. As a developer, you can build on the standard schema to create new classes and new attributes for objects. However, these new classes and attributes must be defined in terms of the existing syntaxes. Defining new syntaxes is not allowed.

See Also:

- ♦ “Schema Structure” on page 27
- ♦ “Object Classes” on page 30
- ♦ “Attribute Type Definitions” on page 58
- ♦ “Attribute Syntax Definitions” on page 64

Object Classes

Object classes define the types of objects that can exist in the NDS database. Database entries are created by selecting an object class and then supplying the required attribute information for the entry. For example, to create an entry for a user, you must select the User object class and then supply a name for the user.

In the base schema, all object classes are nonremovable; that is, they cannot be deleted or in any other way removed from the schema. Object classes that extend the schema are removable.

An object class is defined by its characteristics and consists of the following kinds of information:

- ♦ “Naming Attributes” on page 33 (for naming and containment)
- ♦ “Containment Classes” on page 35
- ♦ “Super Classes” on page 42
- ♦ “Object Class Flags” on page 48
- ♦ “Mandatory and Optional Attributes” on page 49

- ♦ “Default ACL Templates” on page 50

An object class does not have to specify definitions for all characteristics, because it can inherit characteristics from super classes. See “Class Inheritance Rules” on page 45.

See Also:

- ♦ “Novell Object Class Extensions” on page 289
- ♦ “Class Construction Rules” on page 71

Effective and Noneffective Classes

Object classes can be either effective or noneffective.

- ♦ Effective class—you can actually create an instance of the defined object in the NDS tree.
- ♦ Noneffective class—the class is used only to define other classes. You cannot create an object of a noneffective class

The Computer class, for example, is an effective class. You could create a Computer object on the NDS tree using the Computer class. The Device class is a noneffective class. You could not create a Device object because it would have no real function. However, the Device class is a super class of the Computer class and helps to define the attributes needed by the Computer class.

Most of the object classes in the base schema are effective classes. Since effective classes are the active building blocks from which an NDS tree is created, their structure rules must be complete. This means that the naming attributes and containment classes cannot be ambiguous.

For example, if naming attributes or containment classes are not specified for a new effective class, they are inherited from the new class's super classes. If the new effective class inherits from multiple super classes, the naming attribute and containment classes must be identical. If they aren't identical, the structure rules conflict and are ambiguous. In this case, an effective class must define its naming attributes and containment classes.

If the structure rules are incomplete or ambiguous, NDS automatically flags the class as noneffective. The effective or noneffective flag is assigned to a class when it is originally defined. The value cannot be modified after the class is created.

The noneffective classes are not active and thus cannot be used to create objects in an NDS tree. They are typically used as super classes to define class information that is shared by multiple effective classes. The effective classes can then inherit the class information from the noneffective super class rather than repetitively defining it.

The base schema defines the following noneffective classes:

- ♦ Device
- ♦ ndsLoginProperties
- ♦ Resource
- ♦ Server

Top is the one special case for the Effective flag. Although Top is flagged as an effective class, no object can be created from the Top class.

Reading Class Definitions

The “Novell Object Class Extensions” on page 289 section lists the name of each class in alphabetical order. The name is followed by a brief description of the purpose of the class and whether the class is effective or noneffective.

Each object class has the following information defined:

Class Flags

The class flags determine whether the object is a leaf or container object and whether object class is effective or noneffective. See “Object Class Flags” on page 48 for a description of these flags.

Super Classes

Objects inherit information from classes listed in this section. All object classes must have one or more super classes, except Top (page 263), which is a super class to all classes. The super classes are listed in a hierarchical manner, with the super class at the bottom of the list being the immediate super class from which the current class inherits.

Containment

Objects of the class can be created only as subordinates in the NDS tree to objects of the classes listed here. An object of the class cannot be subordinate to any object of a class that is not listed here.

Named By

The partial name or Relative Distinguished Name (RDN) of objects of the class consists of at least one of the attributes listed here. These attributes can be either mandatory or optional attributes, but at least one must be given a value when creating an object of the class. If the only Named By attribute is optional, it is in effect mandatory.

Attributes listed in this section will also be listed in the Mandatory Attributes and Optional Attributes sections. For example, a User object is named by Common Name, which is a mandatory attribute.

Default ACL Template

Every expanded class definition has an ACL attribute (inherited from Top (page 263)). This attribute holds information about which trustees have access to the object itself (entry rights) and which trustees have access to the attributes for the object. This information is stored in sets of information containing the trustee name, privileges, and the affected attribute (entry, all attributes, or a specific attribute). For example, the default template for AFP Server (page 74) is that the creator of an object has the supervisor right on [Entry Rights].

Some object classes define a default set of values for their ACL. Objects also inherit default ACL values from their super classes. Therefore, every object class inherits a default ACL template from Top (page 263). When an object is created, its ACL contains the values that are in the default ACL template for that object. There are two cases where the ACL values are different:

- ◆ Your code overrides the default values.
- ◆ The creator of the object has effective rights comparable to those in the default template. In this case, the rights are not granted explicitly.

Naming Attributes

Objects are identified by their own names and the names of their parent objects. An object's name is called its partial name or Relative Distinguished Name (RDN). For example, a user's partial name might be the following:

CN=Fred

The full name of an object, which includes the names of its parent objects, is called the complete name or Distinguished Name (DN). For example, a user's complete name may be the following:

CN=Fred.OU=Client.OU=Engineering.O=Novell

An object's DN is determined by all the objects to which it is subordinate, but only the immediate parent's object class must appear in the object's containment class list. Hence, structure rules effectively control the formation of Distinguished Names.

Naming attributes are part of the Structure Rules (page 29) of the schema and are listed under the Named By headings in the object class definitions found in "Novell Object Class Extensions" on page 289.

Naming attributes have the following characteristics:

- ♦ "Naming Attribute Rules" on page 34
- ♦ "Multi-Valued Naming Attributes" on page 34
- ♦ "Shareable Naming Attributes" on page 35
- ♦ "Naming Attribute Inheritance" on page 35

See Also:

- ♦ "Containment Classes" on page 35
- ♦ "Super Classes" on page 42

Naming Attribute Rules

Each class has one or more attributes designated as naming attributes. These attributes can be either mandatory or optional attributes, but at least one must be given a value when creating an object of that class. If the only naming attribute is declared as optional, it is, in effect, mandatory.

Naming attributes specify the rules for the partial name of the object. For example, Organization objects are named by the O (Organization Name) attribute. This attribute is the only attribute value that can appear in an organizational entry's partial name.

Multi-Valued Naming Attributes

Naming attributes can be multi-valued; in other words, more than one name (value) can be added to the naming attribute. For example, an organization can have both "Testing" and "Engineering" as values for the O (Organization Name) attribute. However, only the first value is used in search operations.

Some object class definitions specify multiple naming attributes. For example, the Locality object class is named by the L (Locality Name) and S (State or Province Name) attributes. Thus, an RDN for locality can include just an L (Locality Name) attribute, just an S (State or Province Name) attribute, or both attributes. For example, the name for the Provo, Utah locality could be

- ♦ L=Provo
- ♦ S=Utah
- ♦ L=Provo + S=Utah

The last example uses both attributes with a plus sign (+) to indicate where the second attribute's value begins. When the type specifiers (in this case, L and S) are used as shown, the name is referred to as a typed name. A typeless name has the following format: "Provo+Utah".

Shareable Naming Attributes

A naming attribute does not necessarily reflect the class an object belongs to. Many classes, such as Computer, User, and Server, are named by their CN (Common Name) attribute. In such names, the naming attribute itself does not indicate which class the object belongs to, but the value of the naming attribute might suggest the nature of the object. However, some naming attributes are closely tied to specific classes. For example, the C (Country Name) attribute is used to name only Country objects.

Naming Attribute Inheritance

Naming attributes for effective classes must follow the inheritance rules. Effective classes can inherit naming attributes only if the naming attributes of the super classes are identical and do not conflict. If they are different and therefore ambiguous, the effective class must define its own naming attributes. Noneffective classes may have ambiguous naming attributes, but they often define the naming attributes so subordinate objects can inherit them. For example, the Server class defines naming attributes that are inherited by the AFP Server, NCP Server, CommExec, Messaging Server, and Print Server classes.

Containment Classes

Objects that can contain other objects are called container objects or parent objects. Container objects are the branches of the NDS tree, are part of the

Structure Rules (page 29) of the schema, and provide a structure that is similar to a directory in a file system. Objects that cannot contain other objects are called noncontainer or leaf objects. Leaf objects represent the actual network resources that perform some function in the NDS tree, such as users, printers, modems, servers, or volumes.

The following topics contain additional information about containment classes:

- ♦ “Containment Class Rules” on page 36
- ♦ “Containment Classes in the Base Schema” on page 37
- ♦ “Containment of Leaf Objects” on page 40
- ♦ “Containment Classes and Inheritance” on page 41

See Also

- ♦ “Naming Attributes” on page 33
- ♦ “Structure Rules” on page 29
- ♦ “Object Classes” on page 30

Containment Class Rules

For each object class, a list of containment classes specifies where an object of that class may appear in the hierarchical structure of the NDS tree. An object can be immediately subordinate to only those objects whose classes appear in the containment list of the object's expanded class definition. An expanded class definition includes all the characteristics defined for the class plus all the characteristics that the class can inherit from super classes.

Effective classes can inherit containment classes from super classes only if the inheritance does not make containment ambiguous. If the inherited containment is ambiguous, the class must define containment. Class-defined containment overrides containment defined for super classes.

Effective classes are those object classes that can be used to create entries in the NDS database. Noneffective classes cannot be used to create entries and are used by the schema so that multiple object classes can inherit a common set of schema characteristics. Noneffective classes can have ambiguous containment.

Containment classes limit the possible locations of an object in the NDS tree, thus restricting the order and types of partial names that appear in the object's

complete name. Containment helps to ensure that the NDS tree expands in a consistent and logical fashion. For example, an Organization object can be either the topmost object of the NDS tree or subordinate to the Tree Root object. A User object can be subordinate to an Organization object but not to a Tree Root object. Before users can be added to an NDS tree, the tree must contain either an Organization object or an Organizational Unit object which are the containment classes for the User object.

While helping to control the structure of the NDS tree, containment classes must also be flexible enough to accommodate a variety of organizational situations. An example is the relationship between the Organization and Locality classes. Each class specifies the other as a containment class. This allows an administrator to decide which hierarchical order best represents the company's organization.

Containment Classes in the Base Schema

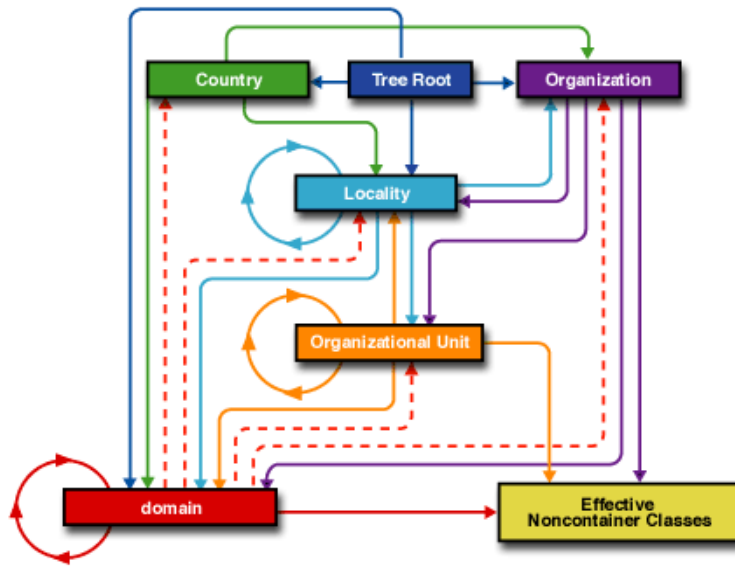
The following table shows the classes that can contain other objects and the object types that they can contain.

Object Class	Contained Classes
Tree Root (page 266)	Country
	domain
	Organization
Country (page 105)	domain
	Locality
	Organization
Locality (page 141)	domain
	Locality
	Organization
	Organizational Unit

Object Class	Contained Classes
Organization (page 202)	domain Locality Organizational Unit Leaf objects
Organizational Unit (page 215)	domain Locality Organizational Unit Leaf objects
domain (page 119)	domain Country Locality Organization Organizational Unit Leaf objects

The ability for Country, Locality, Organization, and Organizational Unit objects to contain domain objects comes with the installation of NDS 8. The ability of domain objects to contain Country, Locality, Organization, and Organizational Unit objects does not come through installation. The schema must be expanded to this functionality with a schema option in DSRepair.

The following figure presents a graphical view of the NDS containment structure. This view shows the containment classes and the object classes that they can contain and that can contain them. Object classes that cannot contain other objects (leaf objects) are collectively shown as noncontainer classes. The object class Top is not shown in the graphical view because Top is used for schema hierarchy and inheritance but not for NDS tree hierarchy.



Tree Root, Organization, and Country are shown on the same level because they all can be the topmost object in the tree. Tree Root has arrows pointing to both Country and Organization because Country and Organization can be, but are not required to be, subordinate to Tree Root.

Lines that have arrows pointing up indicate that these objects can contain each other. For example, an Organizational Unit object can contain a Locality object, and a Locality object can contain an Organizational Unit object.

Objects with circular lines indicate that they can contain objects of the same type as themselves. For example, one Organizational Unit object can contain another Organizational Unit object.

The base schema in NDS 8 adds one new container object, domain. A domain can be contained by all the other containers in the base schema: Tree Root, Country, Organization, Locality, Organizational Unit, and domain. It can contain all the leaf objects that Organization Unit and Organization can.

The domain object can also contain all the other base container objects, except Tree Root. This containment is shown with a dotted red line with an arrow pointing up to these objects. The line is not solid because this functionality is not automatically added to the schema with an NDS 8 upgrade. This functionality must be added to the schema by running a schema option in

DSRepair, and the other servers in the NDS tree must be running NDS 5.17, 6.01, or higher.

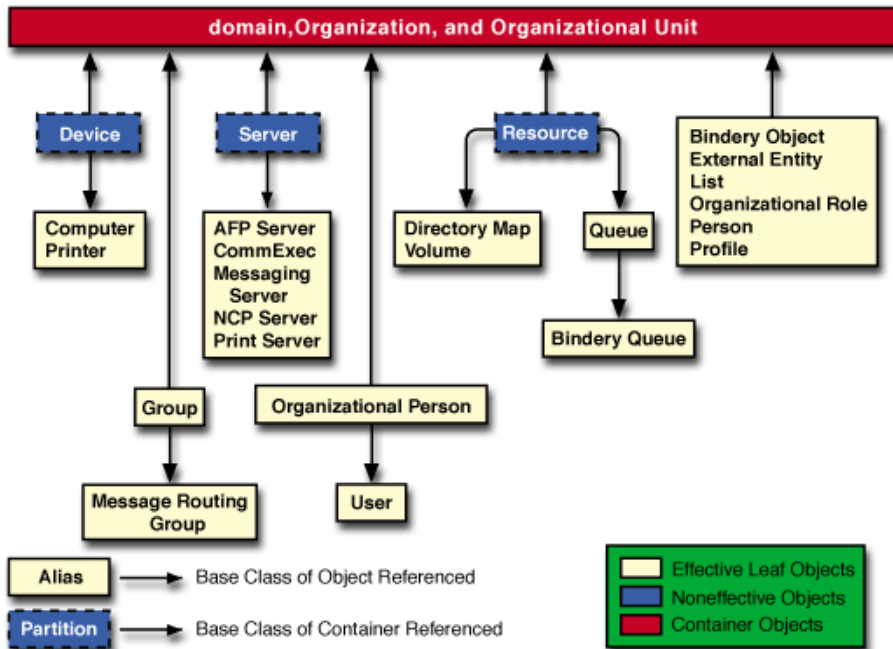
Containment of Leaf Objects

Most leaf classes and noneffective classes in the base schema are contained by the domain, Organization, and Organizational Unit classes. The following table lists the exceptions.

Object Class	Classes Contained By	Class Defined For
Alias (page 77)	Special case	Inherits containment from referenced object. Since an Alias can reference a container object, it can inherit the containment rules of the container object.
Partition (page 219)	Special case	Inherits containment from the object that is the root object of the partition. In NDS 8, Partition becomes an Auxiliary class and no longer requires any containment rules.
Unknown (page 269)	Special case	Any.

Noneffective classes cannot be used to create objects in NDS, but they are often used to define containment classes for other object classes to inherit. Effective classes can define containment for themselves and for subordinate classes.

The following figure is a graphical view of how the leaf objects obtain their containment classes. The arrows pointing up to the container objects indicate which object class declared the containment class. Arrows pointing down to a leaf object indicate the objects that inherit the containment classes. Effective classes are represented by shapes with solid lines and noneffective classes by shapes with dotted lines.



One effective object class is unique: Alias. It is shown at the bottom of the figure because Alias inherits its containment classes from the object that it references. Since all leaf objects have domain, Organization, and Organizational Unit as their containment classes, an Alias will usually inherit these containment classes. However, an Alias can reference a container class, and when it does, the Alias inherits the container's containment classes.

The Partition class is like the Alias class. Since any container class can be the root object in a partition, the Partition class inherits its containment classes from that root object.

The ndsLoginProperties class is not shown because it is a noneffective class, defines no containment classes, and inherits no containment class from its super class, Top. Thus, ndsLoginProperties is like Top in that they both do not affect containment classes of any objects in the NDS tree.

Containment Classes and Inheritance

Containment classes create the hierarchy of the NDS tree and determine where an instance of an object can be created in the NDS tree. Once an instance of an object (or an entry) is created in the NDS tree, the entry inherits rights from its container objects and the container objects are part of the entry's

Distinguished Name. However, the object classes in the schema do not inherit anything from their containment classes.

Object classes can inherit containment definitions, but such inheritances come from the schema's super class structure.

Super Classes

Super classes create the hierarchy of the schema and determine the characteristics that an object class can inherit from another object class. Inheritance simplifies the rules of the schema because it allows some characteristics to be defined once, while multiple object classes can use and enforce these common characteristics.

The following topics describe the other aspects of super classes:

- ♦ “Root Schema Object” on page 42
- ♦ “Super Class Rules” on page 43
- ♦ “Class Hierarchy” on page 43
- ♦ “Class Inheritance Rules” on page 45

See Also:

- ♦ “Mandatory and Optional Attributes” on page 49
- ♦ “Object Class Flags” on page 48
- ♦ “Default ACL Templates” on page 50
- ♦ “Structure Rules” on page 29

Root Schema Object

The Top object class is the root of the schema. Since all other object classes inherit characteristics from the Top class, the Top class specifies information that pertains to all other classes. For example, the Top class defines the following optional attributes:

- ♦ ACL—contains access control information
- ♦ Back Links—contains reference information, used by NetWare 4.x to connect objects with their external references
- ♦ Last Referenced Time—contains a time stamp for the last time the object was referenced

- ♦ Obituary—contains information about objects that are being deleted, renamed, or moved
- ♦ Used By—contains Distributed Reference Link information, used by NetWare 5.x to connect objects with their external references

NDS™ uses these attributes to maintain information. Since these attributes are defined for Top, all object classes inherit these attributes. Entries in the NDS tree have them available whenever NDS needs to assign a value to one.

Super Class Rules

Each object class must define an object class as its super class. Super classes cannot be recursive; therefore, an object class cannot list itself as a super class. The complete definition of each object class is derived from the characteristics of the object class itself plus the characteristics of all classes in its super class lineage. Hierarchies of classes develop through class inheritance in this manner. The classes at the top of the hierarchy provide general characteristics, while those at the bottom become more and more specialized. The complete set of rules for an object class is called the expanded class definition.

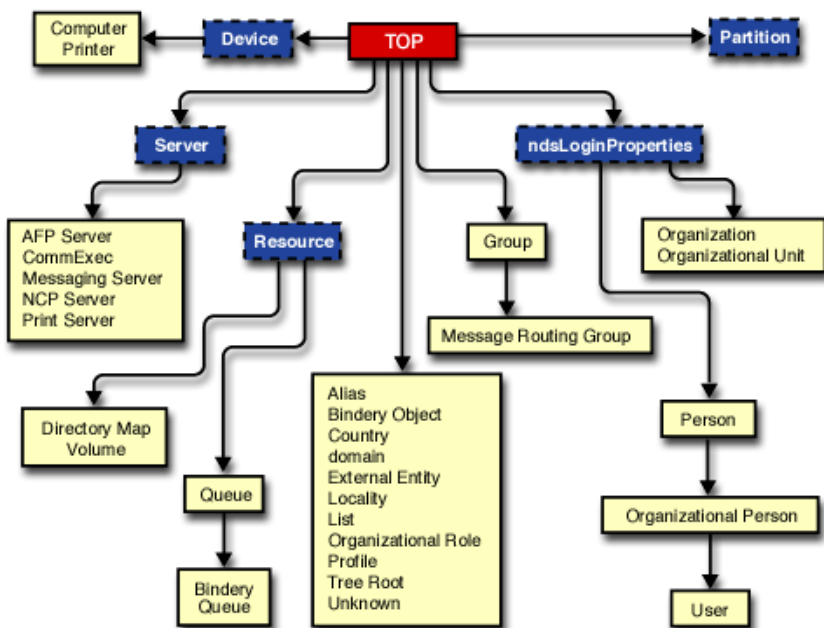
The object class from which an entry is created is called the entry's base class. The expanded class definition for an object class includes the base class and the sum of the information specified by all its super classes. For the purpose of searching the NDS tree, an entry is considered a member of all of its super classes. For example, the base class for creating a user is the User class. The User class inherits from the following super classes: Organizational Person, Person, and Top.

Although the schema is stored with the rest of the NDS database, schema data is logically separated from the NDS tree and must be accessed through different functions. Also, the schema's class hierarchy does not necessarily form a simple tree graph because a class can list more than one class as a super class. Listing multiple classes as a super class is called multiple inheritance. (None of the objects in the NDS base schema uses multiple inheritance.)

Class Hierarchy

The following figure is a single graphical view of the NDS base schema, showing the object classes in the structure of the class hierarchy. In this view of the object classes, the arrows show the direction of flow for inheritance from super classes. An object class inherits the rules and attributes defined by all its super classes, but does not inherit from its subordinates.

Effective object classes are represented by shapes with solid lines and noneffective object classes are represented by shapes with dotted lines.



The Top class is an effective class, but it is a special super class because it cannot be used to define an instance of an object.

The figure illustrates the purpose of noneffective classes. For example, the Server class (noneffective) defines those characteristics shared by all servers; the effective classes (AFP server, NCP Server, Print Server, etc.) define only those characteristics that are particular to that type of server.

Effective classes can inherit from effective classes. In NDS 8, the Person and Organizational Person classes are effective. The User class inherits from both of them.

Class Inheritance Rules

While a class automatically inherits some characteristics in the NDS schema, a class can select to inherit or block the inheritance of other characteristics. The schema enforces the following inheritance rules:

- ♦ A class must declare another class as its super class. The class then automatically inherits any super classes of its defined super class. (Top is the only class that has no super class.)
- ♦ A class can, but is not required to, define mandatory or optional attributes. The class, however, always inherits all the attributes, both mandatory and optional, of its super classes.
- ♦ A class can, but is not required to, define a default ACL template. The class always inherits all the default ACL templates of its super classes. Classes that extend the schema cannot define new default ACL templates.
- ♦ A class can inherit containment classes and naming attributes, but if the class defines them, any definitions made in super classes are not applied to the class.

These class inheritance rules are illustrated in the following figure by the relationship between the classes Top (page 263), Device (page 113), and Computer (page 101). Computer has Device as a super class and inherits the features defined by Device. Device and Computer inherit from Top, because Top is a super class of Device.

Class	Top	Device	Computer
Super Classes	Top	Top	Device
Containment		Organization Organizational Unit	
Named By		CN (Common Name)	
Mandatory Attributes	Object Class	CN (Common Name)	
Optional Attributes	ACL Authority Revocation Back Link Bindery Property CA Private Key CA Public Key Certificate Revocation Certificate Validity Interval Cross Certificate Pair Equivalent To Me Last Referenced Time Obituary Reference Revision Used By	Description L (Locality Name) Network Address O (Organization Name) OU (Organizational Unit Name) Owner See Also Serial Number	Operator Server Status
Default ACL Template	[Creator] Supervisor [Entry Rights]		

The Computer class inherits all of the attributes from the Device class but adds several attributes to accommodate the needs of computers. A different type of device, such as a modem, would add different attributes from a computer.

In the example, the Computer class does not define containment or naming attributes, so it inherits from the Device class.

- ♦ Containment classes: Organization and Organizational Unit
- ♦ Naming attribute: CN (Common Name)

It also inherits the default ACL template that the Top class defines. This ACL enables the creator of an object to be the supervisor of the object.

According to the object class definition, a Computer object could appear only as a subordinate of objects belonging to either the Organization or the Organizational Unit class. The object would be recognized by its common name and might be assigned various optional attributes. In this case, Computer refines the definition of the Device to serve a particular class of devices.

Graphical View Explanation

The “Graphical View of Object Class Inheritance” on page 349 represents the NDS base schema. It shows the object classes in the structure of the class hierarchy, rather than in alphabetical order as presented in “Novell Object Class Extensions” on page 289. This graphical view does not contain the object classes and attributes added during a default NetWare[®] 5 installation or and NDS 8 installation.

Each object class is represented by a box that contains the name of the object class, its immediate super class, and a listing of the containment rules and attributes that are defined for that object class. This view does not show the default ACL templates that are listed in “Novell Object Class Extensions” on page 289, nor does it provide comments about the attributes.

In this graphical view, the object classes that are placed above an object class are its super classes. The object classes that are below an object class are its subordinates. An object class inherits all attributes defined by all of its super classes and any naming or containment rules that the class does not define. The arrows show the direction of flow for inheritance.

Objects inherit from their super classes but they do not inherit from their subordinates. For example, Device (page 113) inherits from Top (page 263), but it does not inherit from Computer (page 101). However, Computer inherits from Top and from Device.

Each object class has the following information listed:

Super Classes. Objects of the class inherit information types and attributes from classes listed here. All object classes must have one or more super classes, except Top, which is a super class to all classes. (In this view, Top is shown multiple times, since it is a super class to all classes.) These listings just list the object's immediate super class.

Containment. Objects of the class can be created as subordinates in the NDS tree to objects of the classes listed here. An object of the class cannot be subordinate to any object that is of a class not listed here. If no classes are listed, the object inherits its containment classes from its super classes.

Named By. The partial name or Relative Distinguished Name (RDN) of objects of the class consists of at least one of the attributes listed here. These naming attributes can be either mandatory or optional attributes, but at least one must be given a value when creating an object of the class. If the only naming attribute is optional, it is in effect mandatory. If no naming attributes are listed, the class inherits its naming attributes from its super classes.

Mandatory and Optional Attributes. All attributes are either mandatory or optional. If an attribute is mandatory, a value must be assigned to that attribute. If an attribute is optional, an assigned value is not required, unless it is the only naming attribute.

Object Class Flags

Besides basic information about containment classes, naming attributes, mandatory and optional attributes and super classes, NDS uses a set of flags to define allowable class operations. Developers, when extending the schema, can set the following flags on object class definitions (For LDAP names, see Object Class Flags in *LDAP and NDS*).

Flag	Description
Container	Indicates whether the object can contain other objects. The flag is turned On for those object classes that are designated as container classes. The flag is turned Off for all leaf object classes.

Flag	Description
Effective	Indicates whether an object class is effective or not. The effective flag is turned On for those classes which can be used both to provide definition and to create objects. The effective flag is turned Off for those classes which provide definition but cannot be used to create objects
Auxiliary	Indicates that the object class is an auxiliary class. When this flag is turned on, the Effective and Container flags must be turned off. This flag is new in NDS 8. For more information on auxiliary classes, see “Auxiliary Classes” on page 51
Non-removable	Indicates whether the object class can be removed from the schema. The flag is turned On for objects that cannot be removed. The flag is turned Off for object classes that can be removed. All base schema object classes are flagged nonremovable. Object classes added to extend the schema are the only classes that can have the nonremovable flag turned Off In NDS 8, developers can turn this flag On. In previous NDS versions, the flag was reserved for NDS

NDS controls and sets the following flags.

Flag	Description
Ambiguous Containment	Indicates whether the object class has clearly defined containment classes. As a general rule, noneffective classes can be created with ambiguous containment, but effective classes must have nonambiguous containment. Ambiguous containment occurs when an object inherits non-identical containment classes from different super classes. Only in special cases can effective classes be created with ambiguous containment. The Alias class object is one of these special cases since it needs to inherit the containment classes of its reference object class. For most object classes in the base schema, the Ambiguous Container flag is turned Off. It is turned On for object classes Top, Alias, and Partition.

Flag	Description
Ambiguous Naming	Indicates whether the object class has clearly defined naming attributes. As a general rule, noneffective classes can be created with ambiguous naming, but effective classes must have nonambiguous naming attributes. Ambiguous naming occurs when an object inherits non-identical naming attributes from different super classes. Only in special cases can effective classes be created with ambiguous naming. The Alias class object is one of these special cases since it needs to inherit the naming attributes of its reference object class. For most object classes in the base schema, the Ambiguous Naming flag is turned Off. The only object classes where this flag is turned On are Top, Alias, and Partition.
Operational	Indicates that NDS requires this object class to exist. This flag is turned ON for object classes that NDS must have in order to operate correctly.

Mandatory and Optional Attributes

An attribute is a single piece of information that is stored in the NDS database about an object. The attributes assigned to an object class can be mandatory or optional:

- ♦ If an attribute is mandatory, a value must be assigned to the attribute before an instance of the object can be created.
- ♦ If an attribute is optional, a value does not need to be assigned to create an instance of the object. The only exception is an optional naming attribute. If the optional naming attribute is the only attribute used for naming the object, this optional attribute becomes a mandatory attribute.

A class inherits all the mandatory and optional attributes defined for its super classes (see “Novell Object Class Extensions” on page 289). There is no way to block the inheritance.

A client cannot associate an attribute with an object unless the attribute is listed among the mandatory or optional attributes of the object's expanded class definition. If a client must associate an attribute with a particular object

and the attribute is not specified by the object class, the client must extend the schema by either

- ♦ Adding the new attribute to the class as an optional attribute
- ♦ Defining a new class that inherits from the original class and adds the new attribute as an optional attribute

See Also:

- ♦ “Attribute Type Definitions” on page 58
- ♦ “Structure Rules” on page 29
- ♦ “Super Classes” on page 42
- ♦ “Default ACL Templates” on page 50

Default ACL Templates

Every object in the NDS tree has an ACL attribute. This attribute holds information about which trustees have access to the object itself (entry rights) and which trustees have access to the attributes for the object. This information is stored in sets of information containing

- ♦ The trustee name
- ♦ The affected attribute-[Entry Rights], [All Attributes Rights], or a specific attribute
- ♦ The privileges

Default ACL templates are defined for specific classes in the base schema and provide a minimum amount of access security for newly created objects. Only base schema objects can have default ACL templates. Developers extending the schema cannot create default ACL templates for new objects.

Since the Top object class defines a default ACL template, all object classes inherit a default ACL template. The ACL defined for Top allows the object that creates another object the right to supervise the created object. This ACL ensures that every object added to an NDS tree has a supervisor.

An object inherits the default ACL templates that are defined for any of the object's super classes. For example, the NCP Server object inherits default ACL templates from Top and Server, and then defines one for itself.

Developers extending the schema cannot create templates that overwrite or add to the templates in the base schema. However, when an object is created

in an NDS tree, the creation process can set the object's ACLs to any value, including one that changes a value that comes from a default ACL template.

See Also:

- ♦ “Class Inheritance Rules” on page 45
- ♦ “Super Classes” on page 42
- ♦ “Novell Object Class Extensions” on page 289

Auxiliary Classes

Auxiliary classes are a special type of object class definition that have been added to the schema in NDS 8.

Auxiliary classes are dynamic classes that can be added to the Object Class attribute of individual objects. When the auxiliary class is added, the object inherits all the attributes of the auxiliary class while retaining all of its own attributes. When the auxiliary class is removed from the object, the auxiliary class attributes are removed from the object and the object is no longer permitted to use those attributes.

In other words, the attributes allowed on an object are the union of the attributes defined for the following:

- ♦ Object's base class
- ♦ Current auxiliary classes
- ♦ Super classes of its base class and auxiliary classes

For example, NDS 8 adds a `dcObject` auxiliary class to the schema. This auxiliary class allows all objects in the NDS database to support the `dc` attribute for LDAP naming conventions—if the class is added to the object. When added or removed, this attribute has the following effects:

- ♦ Add this auxiliary class to the containers that are in a user's distinguished name, and LDAP applications can find the user with a domain name search.
- ♦ Remove the auxiliary class from the container objects, and the `dcObject` attributes (mandatory or optional) and their values are removed from the objects.

The following sections describe the major differences between auxiliary classes and standard object classes:

- ♦ “Attribute Additions with Auxiliary Classes” on page 52
- ♦ “Auxiliary Classes and Object Class Rules” on page 54
- ♦ “Required Rights” on page 55
- ♦ “Backwards Compatibility” on page 55

Attribute Additions with Auxiliary Classes

NDS allows mandatory attributes for a class definition such as User to be added only when the class definition is first created, and once attributes are added to a class definition, NDS does not allow them to be removed. Auxiliary classes add flexibility by allowing attributes to be added and removed, but they are added to and removed from an existing NDS object rather than to or from the class definition.

Auxiliary classes are added to individual instances of the object in the database. For example, suppose you have the following conditions:

- ♦ An auxiliary class called Pager Users with attributes for the Pager Number, Pager Codes, and Pager Is Alphanumeric
- ♦ Four users: Kim, Chris, Lynn, and Terry
- ♦ Two users with pagers: Kim and Chris

As system administrator, you assign the Pager User class to Kim and Chris. The objects would have the following classes and attributes.

Name	Base Class and Attributes	Auxiliary Class and Attributes
Kim	User	Pager User
	Account Balance, Last Login Time, Password Required, etc.	Pager Number, Pager Codes, Pager Is Alphanumeric
Chris	User	Pager User
	Account Balance, Last Login Time, Password Required, etc.	Pager Number, Pager Codes, Pager Is Alphanumeric
Lynn	User	
	Account Balance, Last Login Time, Password Required, etc.	

Name	Base Class and Attributes	Auxiliary Class and Attributes
Terry	User Account Balance, Last Login Time, Password Required, etc.	

Two months later, Kim switches job assignments with Lynn and gives Lynn the pager. You remove the Pager User class from Kim and all the Pager User attributes are deleted from Kim. You then add the Pager User class to Lynn who gains the attributes required for pagers, enabling you to add the appropriate values. The objects would then have the following classes and attributes.

Name	Base Class and Attributes	Auxiliary Class and Attributes
Kim	User Account Balance, Last Login Time, Password Required, etc.	
Chris	User Account Balance, Last Login Time, Password Required, etc.	Pager User Pager Number, Pager Codes, Pager Is Alphanumeric
Lynn	User Account Balance, Last Login Time, Password Required, etc.	Pager User Pager Number, Pager Codes, Pager Is Alphanumeric
Terry	User Account Balance, Last Login Time, Password Required, etc.	

Auxiliary Classes and Object Class Rules

The object class rules have been modified to allow auxiliary classes to have features that no other class type can have in the NDS schema. The following paragraphs explain how auxiliary classes use object class flags, super classes, containment classes, mandatory and optional attributes, and naming attributes.

Object Class Flags. Auxiliary classes do not support all the possible object class flags. When creating an auxiliary class, the only flag that should be set

is the auxiliary class flag. If developers attempted to turn on any of the following flags when the auxiliary class is defined, creation of the auxiliary class fails:

- ♦ **Container**—an auxiliary class cannot be a container class. Since auxiliary classes can be added and removed from an object, an auxiliary class cannot contain other objects. Object containment rules need to be stable.
- ♦ **Effective**—an auxiliary class cannot be an effective class.

The operational, ambiguous containment, and ambiguous naming flags can be set, but NDS ignores their settings.

Super Classes. Auxiliary classes are not required to have a super class. They may declare other classes as their super class, but auxiliary classes should not declare Top as their super class.

If an auxiliary class does have a super class, NDS adds the super class to the object's Object Class attribute and flags them and the base auxiliary class so that they can be deleted only if the base auxiliary class is removed from the object. The object inherits all the attributes defined for the super classes of the auxiliary class.

Containment Classes. Auxiliary classes cannot define containment.

Mandatory and Optional Attributes. Auxiliary classes can have mandatory attributes, optional attributes, or both. If you add mandatory attributes to an auxiliary class, the application that allows the user to add the auxiliary class to an object must also prompt the user for values or supply the values for the mandatory attributes. NDS will not add an auxiliary class to an object without values for all mandatory attributes.

Naming Attributes. Auxiliary classes can define naming attributes, which can be either optional or mandatory. If an auxiliary class attribute is used to name an object, the object must be renamed to use a non-auxiliary class attribute before the auxiliary class can be removed from the object.

Required Rights

To add the auxiliary class to the schema, the user needs the standard rights required to extend the schema: Write rights to a Read/Write partition of the root partition of the NDS tree.

To add an auxiliary class to, or delete an auxiliary class from, an object in the NDS database, the user needs Write rights to that object's Object Class attribute.

Backwards Compatibility

NDS versions prior to NDS 8 do not know about auxiliary classes. NDS 8 servers will send auxiliary class information and auxiliary attribute information only to NDS 8 servers. To servers running previous versions, NDS modifies the information to make it compatible. Special modifications have been made for the following operations.

Replica Synchronization in a Mixed Replica Ring. Changes to objects are synchronized to all servers in a replica ring. If the replica ring contains both NDS 8 servers and servers with previous versions of NDS, NDS must send the auxiliary class information in a manner that is compatible with the previous releases. Since an auxiliary class adds attributes to an object that previous versions on NDS consider illegal, NDS 8 servers make the following modifications before sending objects with auxiliary classes to servers with previous versions of NDS:

- ♦ **Object Class.** The object's class is changed to Unknown object.
- ♦ **AuxClass Object Class Backup.** This attribute is added to the object and all the information from the object's Object Class attribute is stored in the attribute.

When an NDS 8 server receives an Unknown object with an AuxClass Object Class Backup attribute, the server has the information needed to restore the object to its base class and to restore the object's auxiliary class information.

If many objects are using auxiliary classes, replicas on servers with previous versions of NDS will not be particularly useful because they will contain so many Unknown objects. If system administrators are going to add auxiliary classes to objects, they should be encouraged to include only NDS 8 servers in the replica ring.

Schema Synchronization in a Mixed NDS Tree. Schema changes are synchronized from the root of the NDS tree down to its branches. Since an NDS tree can have NDS 8 servers near the root, with NetWare 5 or 4.11 servers in the middle, and an NDS 8 server below them, NDS must be able to send information about auxiliary classes in a manner that is compatible with previous versions of NDS and with sufficient clues that an NDS 8 server can recreate an auxiliary class from the information. To accomplish this, NDS

must make three characteristics of auxiliary classes compatible with previous versions:

- ♦ **Auxiliary Class Flag.** This is a new object class flag for NDS 8, and NDS 8 uses it to recognize which classes are auxiliary classes. Since previous versions of NDS do not recognize this flag, NDS 8 servers send auxiliary class definitions as standard class definitions with one additional attribute, the Auxiliary Class Flag attribute, that contains the auxiliary class flag information. When an NDS 8 server receives a class definition with this attribute, the NDS 8 server knows it should remove the attribute from the class definition and recreate an auxiliary class from the class definition.
- ♦ **Super Classes.** Versions of NDS previous to NDS 8 require all classes to have a super class. To make auxiliary classes compatible with these rules, NDS 8 servers send Top as the super class of any auxiliary class which has declared no super class. When an NDS 8 server receives a class definition with the Auxiliary Class Flag attribute and with Top as its super class, the NDS 8 server removes Top as its super class.
- ♦ **Object Class Attribute.** In versions of NDS previous to NDS 8, the Object Class attribute is a Read-Only attribute. When NDS 8 servers send the definition of this attribute to servers with previous versions of NDS, the NDS 8 servers include the Read-Only constraint. When NDS 8 servers receive the definition for this attribute from a server with previous versions of NDS, they remove the Read-Only constraint from the definition.

Backup. The backup routines in NDS 8 are compatible with existing backup applications. They perform the same data conversions that NDS uses for replica and schema synchronization: the replica synchronization conversions for backing up objects and attributes, and the schema synchronization conversions for backing up schema definitions. Information backed up in this manner can be restored, without loss of data, to either an NDS 8 server or a server running an earlier version.

Class Definitions. Since existing applications that read class definitions do not understand auxiliary classes, the read class definition routines have been modified. These routines perform the same data conversions as the schema synchronization routines and display auxiliary classes as regular classes with an Auxiliary Class Flag attribute. Only applications that have been updated to be compatible with NDS 8 can display auxiliary class definitions with an auxiliary object class flag.

When to Use Auxiliary Classes

Auxiliary classes have the following advantages over modifying existing class definitions:

- ♦ **Efficient.** Auxiliary classes can be applied to some instances of a class definition and not applied to other instances of that class definition. For example, if you are developing a laptop management utility and want to add a few attributes to the users who have laptops, adding these attributes to the User class would not be as efficient as creating an auxiliary class with the attributes. Since all users in a company might not be laptop owners, modifying the User class definition adds attributes that will never be used by some objects. The auxiliary class option is efficient because the attributes are added only to the objects that have laptops.
- ♦ **Flexible.** Auxiliary classes can be used on objects from multiple class definitions. In NetWare 4.x, all logged in users were User objects. With the changes made in NDS 8 and higher, users can now authenticate as other object types such as Person, inetOrgPerson, and residentialPerson objects. An auxiliary class allows you to add your attributes to objects from any of these classes or any other class in the schema.
- ♦ **Separate from Class Definitions.** Auxiliary classes and their attributes are added to the objects rather than to the class definitions. Therefore, when the auxiliary class is removed from the objects, the class definition doesn't retain any unnecessary, unused attributes. The auxiliary class attributes were never added to the class definition, so they don't need to be removed.
- ♦ **Removable.** The attributes in an auxiliary class are removable from the schema. When you modify an existing class definition, the attributes you add cannot be removed without deleting the class definition. All the classes in the base schema, such as User and Person, are nonremovable. Thus, any attributes added to these classes are permanent additions.

As your product evolves and you change your attributes, you cannot remove your obsolete attributes from base schema classes. With an auxiliary class, you can find all the objects in the NDS tree with your auxiliary class, add your new auxiliary class to these objects, transfer any old attribute values to the new attributes, and then delete your old auxiliary class and its obsolete attributes.

- ♦ **LDAP Conformance.** Modifying objects with auxiliary classes conforms to LDAP standards. LDAP expects class definitions to remain unmodified once the definition of the class has been published. NDS has

allowed a more dynamic schema, with the notion that the attributes defined for the User object in one tree will probably never match all the attributes defined for the User object in another tree. Applications in this environment must read the schema definitions rather than assuming schema definitions.

The NDS method for conforming to LDAP standards is to use auxiliary classes where possible, and when not, to modify a non-effective super class which allows the child class to inherit the attributes without the attributes becoming part of its definition.

Auxiliary classes cannot be used in products that require the following:

- ♦ Compatibility with NDS trees in which all the servers are using versions of NDS earlier than NDS 8
- ♦ The new attributes will be used in every instance of a class definition

Attribute Type Definitions

All attributes found in an NDS tree consist of an attribute type and an attribute value, which can be multi-valued. The attribute type identifies the nature of information the attribute stores, and the value is the stored information.

The attribute type definition

- ♦ Identifies the attribute syntax used for the value
- ♦ Specifies the constraints that are imposed on the syntax

These constraints are also known as attribute flags. Attributes are assigned to objects according to the object's class definition.

An example of an attribute type is CN (Common Name) (page 400), which uses the Case Ignore String syntax. The CN (Common Name) attribute constrains this syntax to a range of from 1 to 64 elements.

Attribute types can be added to the NDS schema. However, once an attribute type has been created, it can't be modified.

Attribute types can be removed from the NDS schema, but only if the attribute is not part of the base schema and only if the attribute type isn't assigned to a class. All attribute types in the base schema are always flagged nonremovable.

See Also:

- ♦ "Base Attribute Definitions" on page 361

- ♦ “Novell Attribute Extensions” on page 729

Attribute Syntaxes

The attribute syntax controls the type of information that can be stored in the value of an attribute. For example, the syntax determines whether the attribute stores integer, string, or stream data. The attribute's syntax must be selected from the set of predefined attribute syntaxes. The syntax also controls the type of compare operations that can be performed on the value. See “Attribute Syntax Definitions” on page 64 for more information.

Attribute Constraints

The attribute constraints restrict the information that can be stored in the data type and constrain the operations of NDS and NDS clients. The constraints specify whether the attribute

- ♦ Allows only a single value or multiple values
- ♦ Has a range or size limit to the value
- ♦ Is synchronized immediately, at the next scheduled interval, or never
- ♦ Is hidden or viewable
- ♦ Is writable or read-only

The attribute constraints are flags, which are either TRUE or FALSE, and they can only be set when the attribute definition is created. Since there are more than dozen, they have been functionally grouped. For LDAP names for these flags, see *Attribute Flags in LDAP and NDS*.

Reading the Attribute. These flags determine who can read the attribute's information.

Name	Description
Hidden Attribute	In NDS version 6.xx and below, marks the attribute as usable only by the NDS server.
	In NDS version 7.xx and above, marks the attribute as usable by NDS and the applications running on the NDS server.
	If FALSE, clients can see the attribute.

Name	Description
Public Read	Indicates that anyone can read the attribute without read privileges being assigned. You cannot use inheritance masks to prevent an object from reading attributes with this constraint. If FALSE, NDS rights determine who can read the value of the attribute. If TRUE, NDS skips all rights checking, making access to the data extremely efficient.
Server Read	Indicates that Server class objects can read the attribute even though the privilege to read has not been inherited or explicitly granted. You cannot use inheritance masks to restrict servers from reading attributes with this constraint. The client cannot set or modify this constraint flag and thus cannot modify the attribute.

Modifying the Attribute.The following flags regulate who can modify the attribute's value.

Name	Description
Read Only Attribute	Prevents clients from remotely modifying the attribute. The NDS server and applications running on it create and maintain these attributes. Clients can read the attribute's value. If FALSE, clients can remotely modify the attribute.
Write Managed	Requires users to have supervisor rights to the object before they can add or delete the object as a value for this attribute. This flag only works on attributes which have a DN in the syntax. It is used on attributes such as Security Equals, Group Membership, and Profile Membership.

Synchronizing the Attribute.The following flags regulate how changes to the attribute's value affect NDS synchronization.

Name	Description
Per Replica	Marks the attribute so that the information in the attribute is not synchronized with other replicas. Modifications to the attribute never schedule synchronization and are never synchronized to other replicas.
Schedule Sync Never	Allows the attribute's value to change without scheduling synchronization. Synchronization occurs at the next regularly scheduled synchronization cycle or when another event triggers synchronization Set this flag to TRUE if the change in the attribute's value can wait ten to thirty minutes to be propagated.
Sync Immediate	Schedules synchronization within 10 seconds when the value of the attribute changes. Set this flag to TRUE if the change in the attribute's value needs to be immediately propagated or changed throughout the NDS tree. If FALSE, the attribute is synchronized at the next synchronization interval.

If all of these synchronizing flags are false, NDS synchronizes the data at the slow synchronization level set on the server that contains the replica holding the change.

Constraining the Attribute Values. The following flags regulate the type of data that the attribute can store.

Name	Description
Single Valued Attribute	Indicates that the attribute has a single value, with no order implied. If FALSE, the attribute is multi-valued.
Sized Attribute	Indicates that the attribute has an upper and lower boundary. The first number indicates the lower boundary and the second, the upper boundary. This flag should be set only on attributes with integer and string syntaxes. If FALSE, the attribute has no length or range limits.

Name	Description
String Attribute	Labels the attribute as a string type. NDS sets this constraint on all attributes that use a string for their syntax. Naming attributes must have this constraint. If FALSE, the attribute isn't a string and cannot be used as a naming attribute.

Removing an Attribute. These flags control whether the attributes are removable from the schema.

Name	Description
Nonremovable Attribute	Prevents the attribute from being removed from the schema: ♦ In NDS version 6.xx and below, clients cannot set this constraint flag. ♦ In NDS version 7.xxx and above, clients can set this flag when the attribute is created. All base attribute definitions have the nonremovable flag set to TRUE If FALSE, the attribute can be removed if it hasn't been assigned to a class.
Operational	Indicates that NDS uses the attribute internally and requires the attribute to function correctly. Also used for LDAP compatibility.

Attributes and Classes

When an attribute is first defined, it is not associated with any object class. You must create an association with an object class before the attribute can be useful.

In other words, you could create a set of attributes such as Given Name, Initials, Surname, Telephone Number, and EMail Address. By themselves, they aren't particularly useful. Then you define an object class such as Person, and have it include these attributes. The attributes now take on a meaning and give dimension to the object class.

Attribute Type Abbreviations

For convenience, NDS uses abbreviations for the name types that are used most often. The following table shows the accepted abbreviations for these attributes.

Attribute Type	Abbreviation
Country Name	C
Organization Name	O
Organizational Unit Name	OU
State or Province Name	S
Locality Name	L
Common Name	CN
Street Address	SA

Reading NDS Attribute Type Definitions

The Base Attribute Definitions (page 361) section lists the names of each attribute type in alphabetical order. The name of the attribute is followed by a brief description of the attribute's purpose. Valid abbreviations for the attribute appear in parentheses next to the attribute name. Additionally, you will find the following information:

Syntax

The name of the syntax for this attribute type. See “Attribute Syntax Definitions” on page 919 for the syntax specification.

Constraints

Any constraints that apply to this attribute type. For a description of the constraints, see *Attribute Constraint Flags (NDS Core Services)*.

Used In

The object class definitions which require or allow an attribute of this type when creating that class of object.

Remarks

These remarks can include further restrictions, how to use the attribute, references to related documents, etc.

Attributes are assigned to objects according to the object's class definition. For more information about the specific attributes an object class uses, see “Novell Object Class Extensions” on page 289.

Attribute Syntax Definitions

An attribute syntax defines a standard data type which an attribute uses to store its values in the NDS tree. The syntax definitions are static definitions represented in basic C-code format. For example, the schema includes the following attribute syntaxes:

`SYN_CI_STRING`

The Case Ignore String syntax is used in attributes whose values are strings and where the case (upper or lower) is not significant.

`SYN_INTEGER`

The Integer syntax is used for attributes whose values are signed integers.

For a complete list, see “Attribute Syntax Definitions” on page 919.

Attribute type definitions are built on attribute syntaxes. Developers extending the schema can create new attribute types using these syntaxes, but they cannot create any new syntax definitions.

Matching Rules

An attribute syntax consists of a single data type. Matching rules indicate the characteristics that are significant when comparing two values of the same syntax. There are three primary matching rules:

Equality

To match for equality, two values must be identical, use the same attribute syntax, and conform to the data type of the attribute syntax. Most syntaxes specify a match for equality. NDS™ checks that the values being matched conform to the data type of the syntax. NDS will not attempt to match two values if the syntax does not specify a match for equality.

Ordering

To match for ordering, a syntax must be open to comparisons of less than, equal to, and greater than. For example, 50 is less than 100, and N is greater than B.

Substrings

To match substrings, a syntax must be open to search and comparison patterns that include the asterisk (*) wildcard. For example, in a syntax using substring matching, N*V*L would match NAVAL, NAVEL, or NOVEL.

An approximate comparison rule can be used in searches and comparisons on syntaxes with lists of strings and syntaxes with multiple fields and an ID field:

Strings

The approximate rule determines whether a string is present in a syntax with a string list.

IDs

The approximate rule determines whether an ID matches the ID in a corresponding field while ignoring the other fields in the syntax. Although most of the API structures for syntaxes require an object name, NDS replaces these names with IDs in the comparison and search operations.

This NDS approximate matching rule is quite different from the LDAP approximate matching rule. In LDAP, approximate matching means that the names “sound” similar. NDS does not support this type of matching.

A syntax can specify one or more of these matching rules. For example, the Case Ignore String syntax specifies matching rules of equality and substrings.

A syntax can also specify qualifiers for comparison which ignore characters such as dashes, leading spaces, trailing spaces, and multiple consecutive internal spaces. All string syntaxes use comparison operations that ignore extra spaces. Other qualifiers allow only digits or only printable characters.

Reading Syntax Definitions

The “Attribute Syntax Definitions” on page 919 section lists the names of each syntax in alphabetical order. The name of the syntax is followed by a brief description of how the syntax is used.

Each syntax defines the following information:

Syntax ID

A 32-bit integer used as an identifier in Application Programming Interface (API) functions that transfer attribute values in and out of NCP™ message buffers. The syntax IDs are defined in `nwdsdefs.h`.

API Data Structure

A C structure supported by the NDS API.

Transfer Format

The format the syntax takes when it is transferred on the wire.

Matching Rules

The rules for matching two values that comply with the syntax.

Used In

A list of attribute type definitions that use the syntax.

Remarks

Additional information that may include information concerning comparisons, explanations of structure members, and qualifiers.

To determine which syntax is used for a particular attribute type, refer to the attribute's syntax specification in the "Base Attribute Definitions" on page 361.

Schema Extensions

The NDS schema is extensible. Developers can define new object classes and attributes. In fact, the NetWare operating system and Novell utilities and applications extend the schema. You can perform the following operations on the NDS schema:

- ♦ Create, read, and delete attribute type definitions
- ♦ Create, read, and delete object class definitions
- ♦ List containable classes
- ♦ Modify class definitions
- ♦ Read attribute syntax definitions

- ♦ Read attribute syntax IDs

However, you cannot create or modify an attribute syntax. The attribute syntaxes are the most fundamental building blocks of NDS and cannot be altered.

To extend the NDS schema, you must have Admin equivalent rights to a server containing a Read/Write replica of the root partition. In addition, if you are extending the schema from a workstation client, you must initialize the Unicode tables.

See Also:

- ♦ “Distribution of the NDS Schema” on page 29

Class Definition Creation

Before creating a new object class, you must make the following decisions:

- ♦ What should you call the new object class? To register the name with Novell, see “Registering Attribute Types and Class Definitions” on page 68. For naming rules, see “Valid Class and Attribute Names” on page 70.
- ♦ Which class flags does the object class require? For a list, see “Object Class Flags” on page 48.
- ♦ Which class or classes will you assign as the super classes? For information, see “Super Classes” on page 42.
- ♦ Do you need to define containment for the class? For information, see “Containment Classes” on page 35.
- ♦ Do you need to define naming attributes for the class? For information, see “Naming Attributes” on page 33.
- ♦ Does the class require any mandatory attributes that cannot be inherited from its super classes?
- ♦ Does the class require any optional attributes that cannot be inherited from its super classes?

Your code should create new attributes before the new class so that the attributes can be assigned when the class is created. Otherwise, you will have to modify the class after the attributes are created. Mandatory attributes cannot be added later; they must be added to the class when the class is created.

The schema can be extended with many of the interfaces in the NDK such as NDS Libraries for C, LDAP Service Provider for JNDI, NDS Service Provider for JNDI, Novell Controls for ActiveX, Beans for Novell Services, and Novell ConsoleOne.

Registering Attribute Types and Class Definitions

When you define new attribute types or new object class definitions, you must register them with Developer Support to ensure uniqueness. To register, you should go to the Novell Developer Support Web site (<http://developer.novell.com/support/schreg2c.htm>), fill out the form, and submit the information.

If you cannot use the form, call Developer Support at 1-800-REDWORD (1-800-733-9673). If you cannot use an 800 number, the international number is 1-44-801-861-5588.

When you register, you receive a unique prefix that you prepend to the names of your new attribute and class definitions. You also receive two ASN.1 (Abstract Syntax Notation One) identifiers: one for object classes and one for attribute definitions. These IDs can be expanded to include as many unique IDs as you need for object classes and attributes.

In NetWare 4.11 and below, object class and attributes can have, but are not required to have, an ASN.1 ID. In NetWare 5.x, ASN.1 IDs are required for applications to pass Novell certification. NDS 8 verifies that the ASN.1 is BER encoded. ASN.1 IDs serve as a common syntax for transferring information between two end systems.

Prefix

The prefix, which can be from 3 to 8 characters long, serves two purposes. It guarantees that your attribute and class names are unique. We keep a database of registered prefixes to ensure that no two companies select the same prefix. Since no two class definitions can share the same name, the unique prefix ensures that the name is unique.

The prefix also identifies which attributes and classes belong to your application. In the past, we used a convention that separated the prefix from the descriptive name with a colon. Since LDAP does not support colons in schema definition names, we are now using a convention which lowercases the prefix and capitalizes the first letter of each word in the descriptive part of

the name. Don't use spaces between words because LDAP naming rules do not support spaces.

For example, if you registered the ABC prefix with Novell and used it to create a Reading Group class definition, we would recommend the following name which conforms to both NDS and LDAP naming conventions:

abcReadingGroup

ASN.1 IDs or OIDs

When you register for a prefix and an OID, we assign you a number similar to the following:

2.16.840.1.113719.2.888.4.12.1

These numbers have the following meanings:

2 — joint-iso-ccitt

16 — Country

840 — US

1 — Organization

113719 — Novell

2 — external application (Novell internal applications receive a 1)

888 — unique subarc for your application

4 — type identifier (4 = attribute; 5 = syntax; 6 = class; 100 = LDAP extension; 101 = LDAP control)

12 — unique number for attribute

1 — version

After the subarc number, you are free to assign and subdivide the OID as best fits your application. However, we recommend that you use the format described above for the next few subdivisions. If you want to conform to standards and supply meaningful OIDs, the type identifier is important and needs to be used according to established conventions.

We also recommend starting the attribute and class count with 1 (not 0) and leaving the version number for first release blank (don't use zero). For the next release, use 1 for the first revision and increment the version number with each subsequent release. Add the version number to the classes that are modified and to the classes and attributes that are new for that version. The version number of previous classes and attributes which have not been modified should not be incremented to the new version.

Valid Class and Attribute Names

For a new class name or attribute name to be valid NDS name, it must fit two criteria:

- ♦ The name cannot exceed 32 characters. Spaces are allowed but are counted in the 32-character limit. Spaces are not recommended since they are not allowed in LDAP schema names.
- ♦ The name must be unique in its level of hierarchy in the NDS tree. Names are case insensitive, although case can be used for easier visual discrimination.

For example, “Accounting” and “accounting” would be considered as duplicates and not allowed as two class names or two attribute names, although differences in capitalization would appear on the screen. However, NDS allows the same name to be used for a class name and an attribute name. Thus Accounting could be the name of a class, and accounting the name of an attribute.

NDS has defined a number of classes and attributes with the same name, for example User (Class) and User (Attribute), Queue (Class) and Queue (Attribute), and Resource (Class) and Resource (Attribute).

If you want your application to work with LDAP (Lightweight Directory Access Protocol) applications, your schema extensions should conform to the LDAP naming conventions, which are more restrictive than NDS schema naming conventions. LDAP schema names must conform to the following rules:

- ♦ Use alpha-numeric characters. LDAP allows one hyphen in a name, the only non-alpha-numeric character allowed, and it cannot start the name. The hyphen can be used only once in the name. We recommend that names contain only alpha-numeric characters without a hyphen.
- ♦ Start with an alpha character. Numeric characters cannot start a name.
- ♦ Do not include spaces.
- ♦ Create a unique name for the type (class or attribute). Names are not case sensitive.
- ♦ Do not create a name with more than 32 characters. This is an NDS restriction. LDAP allows longer names, but NDS does not currently support names longer than 32 characters.

We recommend that you lowercase the prefix, and then capitalize the initial word of the attribute’s or class’s descriptive name. For example, if we assigned

you the prefix of *adb* for an address book application, the following object class names are possible:

adbGroup
adbConferenceRoom

Class Construction Rules

Below is a list of rules that regulate the construction of new object classes. If you need to define new classes, pay close attention to these rules.

1. Object class definitions cannot be recursive. An object cannot have itself as a super class.
2. Only classes with complete structure rules can be flagged effective and used to create objects. That is, the super classes, containment, and naming attributes must be complete.
3. An effective class can be constructed in three ways:
 - ♦ The class defines its own structure rules.
 - ♦ The class inherits structure rules from its super classes.
 - ♦ The class defines part of the structure rules (such as naming) and inherits the other part of the structure rules (such as containment) from a super class.
4. For a class that defines its own structure rules, any structure rules that might be inherited from its super classes are ignored.
5. If structure rules of an effective class are inherited, they must be nonambiguous (for more information, see “Effective and Noneffective Classes” on page 31).

2

Base Object Class Definitions

This chapter lists alphabetically all the object classes that NDS installs with NDS eDirectory 8.5 on all platforms : NetWare, NT, Solaris, and Linux.

- ♦ For object class definitions that are added in a default NetWare installation or with the downloadable schema files, see “Novell Object Class Extensions” on page 289.
- ♦ For an explanation of the types of information included about each class, see “Reading Class Definitions” on page 32.

AFP Server

Identifies objects that provide AFP services.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.0

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	AFP Server
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

AFP Server	Inherited from Top	Inherited from Server
(None)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

AFP Server
Serial Number (page 677)
Supported Connections (page 687)

Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Inherited from Server

Account Balance (page 363)	Private Key (page 643)
Allow Unlimited Credit (page 365)	Public Key (page 648)
Description (page 410)	Resource (page 658)
Full Name (page 429)	Security Equals (page 673)
Host Device (page 447)	Security Flags (page 674)
L (Locality Name) (page 458)	See Also (page 675)
masvAuthorizedRange (page 520)	Status (page 683)
masvDefaultRange (page 521)	SvcInfo (page 903)
masvProposedLabel (page 524)	SvcType (page 904)
Minimum Account Balance (page 534)	SvcTypeID (page 905)
Network Address (page 556)	User (page 714)
O (Organization Name) (page 595)	Version (page 719)
OU (Organizational Unit Name) (page 606)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

Alias

Defines alias objects which reference other object classes. The aliased object class determines how the alias is named and where it can be contained.

Type: Effective

NDS Operational: Yes

LDAP Name

alias

ASN.1 ID

2.5.6.1

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	On
Ambiguous Container	On
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Alias
Containment	(special)	Alias

Rule	Class/Attribute	Defined For
Named By	(special)	Alias

Mandatory Attributes

Alias	Inherited from Top
Aliased Object Name (page 362)	Object Class (page 601)

Optional Attributes

Alias
(None)

Inherited from Top
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424) GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

An alias is an object that references the Distinguished Name of another object in the NDS tree. An object in the NDS tree can have zero or more aliases. Several Alias entries can point to the same object entry. Only object entries can have aliases; thus, aliases of aliases are not permitted. An object entry does not have to be a leaf entry to have an Alias. However, Alias entries cannot have subordinates and are always leaf objects.

NDS uses the Aliased Object Name attribute in an Alias entry to identify and to find the corresponding object entry. The object class Alias does not specify naming attributes for alias entries, nor does the class define where alias entries may be contained in the NDS tree. NDS enforces the naming and containment rules mandated by the base class of the object to which the alias points.

applicationEntity

Represents an application entity.

Type: Effective

LDAP Name

applicationEntity

ASN.1 ID

2.5.6.12

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	applicationEntity

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	applicationEntity
	Locality (page 141)	applicationEntity
	Organization (page 202)	applicationEntity
	Organizational Unit (page 215)	applicationEntity
Named By	CN (Common Name) (page 400)	applicationEntity

Mandatory Attributes

applicationEntity	Inherited from Top
CN (Common Name) (page 400) presentationAddress (page 637)	Object Class (page 601)

Optional Attributes

applicationEntity	
Description (page 410) L (Locality Name) (page 458) O (Organization Name) (page 595)	OU (Organizational Unit Name) (page 606) See Also (page 675) supportedApplicationContext (page 686)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For more information, see RFC 2256.

applicationProcess

Represents an LDAP application process.

Type: Effective

LDAP Name

applicationProcess

ASN.1 ID

2.5.6.11

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	applicationProcess

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	applicationProcess
	Locality (page 141)	applicationProcess
	Organization (page 202)	applicationProcess
	Organizational Unit (page 215)	applicationProcess
Named By	CN (Common Name) (page 400)	applicationProcess

Mandatory Attributes

applicationProcess	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

applicationProcess	
Description (page 410)	OU (Organizational Unit Name) (page 606)
L (Locality Name) (page 458)	See Also (page 675)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For more information, see RFC 2256.

Audit:File Object

Contains the audit records for an audited object.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.12.6.1.0

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Audit:File Object
Containment	Country (page 105)	Audit:File Object
	Locality (page 141)	Audit:File Object
	Organization (page 202)	Audit:File Object
	Organizational Unit (page 215)	Audit:File Object
	Top (page 263)	Audit:File Object
	Tree Root (page 266)	Audit:File Object

Rule	Class/Attribute	Defined For
Named By	CN (Common Name) (page 400)	Audit:File Object

Mandatory Attributes

Audit:File Object	Inherited from Top
Audit:Contents (page 370)	Object Class (page 601)
Audit:Policy (page 375)	
CN (Common Name) (page 400)	

Optional Attributes

Audit:File Object	
Audit:A Encryption Key (page 368)	Audit:Link List (page 373)
Audit:B Encryption Key (page 369)	Audit:Path (page 374)
Audit:Current Encryption Key (page 371)	Description (page 410)
Audit:Type (page 376)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

Bindery Object

Used to represent an object that has been created by the Bindery Emulator to emulate a Bindery object.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.2

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Bindery Object
Containment	domain (page 119)	Bindery Object
	Organization (page 202)	Bindery Object
	Organizational Unit (page 215)	

Rule	Class/Attribute	Defined For
Named By	Bindery Type (page 388)	Bindery Object
	CN (Common Name) (page 400)	Bindery Object

Mandatory Attributes

Bindery Object	Inherited from Top
Bindery Object Restriction (page 383)	Object Class (page 601)
Bindery Type (page 388)	
CN (Common Name) (page 400)	

Optional Attributes

Bindery Object
(Special)
Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The special case notation for optional attributes indicates that any attribute type which has been defined for the schema and is not listed in the “Mandatory Attributes” section for a Bindery Object is optional for a Bindery Object. See “Base Attribute Definitions” on page 361 for a list of attribute types defined for the base schema.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

Bindery Queue

Represents an object that has been created by the Bindery Emulator to emulate a user-defined queue object.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.3

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	Queue (page 241)
	Queue (page 241)	Bindery Queue

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Resource (page 245)
	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)
Named By	Bindery Type (page 388)	Bindery Queue
	CN (Common Name) (page 400)	Bindery Queue

Mandatory Object Class

Bindery Queue	Inherited from Top	Inherited from Resource	Inherited from Queue
Bindery Type (page 388)	Object Class (page 601)	CN (Common Name) (page 400)	Queue Directory (page 652)

Optional Attributes

Bindery Queue
(None)
Inherited from Top
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424) GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Inherited from Resource

Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Inherited from Queue

Device (page 414)	Server (page 678)
Host Server (page 449)	User (page 714)
Network Address (page 556)	Volume (page 720)
Operator (page 603)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Root]	Read	[All Attributes]	Bindery Queue

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The queue management service (QMS) requires the Host Server attribute to identify the NCP server that holds this queue.

The Queue directory names the subdirectory where this queue's files are stored.

The Operator, Server, and User attributes are used by the QMS as access control lists.

The Operator attribute identifies users or groups that have operator privileges. The Server attribute identifies print servers allowed to service this queue.

The User attribute contains a list of objects that are authorized to use this queue. The server that controls the queue must determine if the user list is maintained by an administrator or if the list is automatically generated by the server. If the user list is used by the server as an access control list, the administrator will usually maintain the list. If the user list is purely informational, reflecting access control information stored elsewhere, the server usually maintains the list.

The See Also attribute might be used to list related queues. For example, two queues, “Fast” and “Slow,” might provide the same set of services, except that “Fast” runs at a higher priority. These two queues might reference each other in their respective See Also attributes.

The Host Resource Name attribute is used when the host's local identification for a resource differs from the global resource identification. For example, a server might recognize “SYS:” as the local name for a volume with the following NDS name:

```
"Project X.Engineering.Acme.US"
```

The L (Locality Name), O (Organization Name), and OU (Organizational Unit Name) attributes are useful when multiple localities, organizations, or organizational units use a single resource. If these attributes contain appropriate values, a search can be initiated for resources associated with a particular locality or organization.

The Network Address attribute (inherited from Resource) acts as a cache for the server's network address. The user can contact the server without having to dereference the Host Server attribute.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

certificationAuthority

Defines an LDAP Certificate Authority.

Type: Auxiliary Class

LDAP Name

certificationAuthority

ASN.1 ID

2.5.6.16

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	(None)	certificationAuthority
Containment	(None)	certificationAuthority
Named By	(None)	certificationAuthority

Mandatory Attributes

certificationAuthority

authorityRevocationList (page 377)

certificateRevocationList (page 396)

cACertificate (page 392)

Optional Attributes

certificationAuthority

crossCertificatePair (page 399)

Default ACL Template

None

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For more information, see RFC 2256.

CommExec

Represents a resource used by NetWare MHS services.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.31

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	CommExec
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

CommExec	Inherited from Top	Inherited from Server
	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

CommExec
Network Address Restriction (page 558)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Server

Account Balance (page 363)	Public Key (page 648)
Allow Unlimited Credit (page 365)	Resource (page 658)
Description (page 410)	Security Equals (page 673)
Full Name (page 429)	Security Flags (page 674)
Host Device (page 447)	See Also (page 675)
L (Locality Name) (page 458)	Status (page 683)
Minimum Account Balance (page 534)	SvcInfo (page 903)
Network Address (page 556)	SvcType (page 904)
O (Organization Name) (page 595)	SvcTypeID (page 905)
OU (Organizational Unit Name) (page 606)	Used By (page 712)
Private Key (page 643)	Version (page 719)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Computer

Represents both computers that host NetWare[®] servers and computers used as client workstations.

Type: Effective

NDS Operational: Yes

LDAP Name

computer

ASN.1 ID

2.16.840.1.113719.1.1.6.1.4

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Device (page 113)
	Device (page 113)	Computer

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Device (page 113)
	Organization (page 202)	Device (page 113)
	Organizational Unit (page 215)	Device (page 113)
Named By	CN (Common Name) (page 400)	Device (page 113)

Mandatory Attributes

Computer	Inherited from Top	Inherited from Device
	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

Computer	
Operator (page 603)	Status (page 683)
Server (page 678)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Device	
Description (page 410)	See Also (page 675)
L (Locality Name) (page 458)	Serial Number (page 677)
Network Address (page 556)	SvcInfo (page 903)
O (Organization Name) (page 595)	SvcType (page 904)
OU (Organizational Unit Name) (page 606)	SvcTypeID (page 905)
Owner (page 608)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The Operator attribute lists individuals or groups that are responsible for day-to-day maintenance of the computer. This can differ from the value of Owner, which can indicate more of an administrative responsibility with respect to the computer.

The Server attribute provides a list of servers that are hosted on this computer. The See Also attribute might be used, in this instance, to identify other related computers assigned to a network.

The L (Locality Name) attribute identifies the physical location of a device. For example, if the device were a printer, the locality might be “Building D, Section 24, by Ed Bender's desk.”

The O (Organization Name) and OU (Organizational Unit Name) might already be present in the device's distinguished name. They are repeated here to aid searching when an organization spans multiple subtrees in the NDS tree. However, NDS does not add these attributes automatically even though they may be present in the device's Distinguished Name. Additional values for the organization or organizational unit can be useful when a device is “co-owned” by multiple organizations.

The LDAP server maps requests for the computer class to this class.

Country

Defines country entries in the NDS tree.

Type: Effective

NDS Operational: Yes

LDAP Name

country

ASN.1 ID

2.5.6.2

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Country

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Country
	Top (page 263)	Country
	Tree Root (page 266)	Country
	(Nothing)	Country
Named By	C (Country Name) (page 390)	Country

Mandatory Attributes

Country	Inherited from Top
C (Country Name) (page 390)	Object Class (page 601)

Optional Attributes

Country
Description (page 410)
searchGuide (page 670)

Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

In NetWare 5.x, Country can appear as the topmost object in an NDS tree or subordinate to Tree Root. Top is a special object and never appears as an object in an NDS tree.

The Description attribute might contain the full name of the country, since the C attribute is restricted to the two letter code defined by ISO 3166.

cRLDistributionPoint

Represents a distribution point for a certificate revocation list.

Type: Effective Class

LDAP Name

cRLDistributionPoint

ASN.1 ID

2.5.6.19

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	cRLDistributionPoint

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	cRLDistributionPoint
	Locality (page 141)	cRLDistributionPoint
	Organizational Unit (page 215)	cRLDistributionPoint
	Organization (page 202)	cRLDistributionPoint
	SAS:Security (page 248)	cRLDistributionPoint
Named By	CN (Common Name) (page 400)	cRLDistributionPoint

Mandatory Attributes

cRLDistributionPoint	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

cRLDistributionPoint	
authorityRevocationList (page 377)	crossCertificatePair (page 399)
cACertificate (page 392)	deltaRevocationList (page 408)
certificateRevocationList (page 396)	

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

dcObject

Identifies an auxiliary class that supplies the domain naming component for existing structural object classes such as Organization, Organizational Unit, and Locality.

Type: Auxiliary Class

NDS Operational: Yes

LDAP Name

dcObject

ASN.1 ID

1.3.6.1.4.1.1466.344

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	(None)	dcObject
Containment	(None)	dcObject

Rule	Class/Attribute	Defined For
Named By	dc (page 406)	dcObject

Mandatory Attributes

dcObject
dc (page 406)

Optional Attributes

dcObject
(None)

Default ACL Template

None

Remarks

For more information about this object, see RFC 2247.

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Device

Represents physical units that can communicate (such as a modem or a printer). At least one of the L (Locality Name), Serial Number, or Owner attributes should be included with the object entry. The choice depends on the type of device.

Type: Effective

NDS Operational: Yes

LDAP Name

device

ASN.1 ID

2.5.6.14

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Device

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Device
	Organization (page 202)	Device
	Organizational Unit (page 215)	Device
Named By	CN (Common Name) (page 400)	Device

Mandatory Attributes

Device	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

Device	
Description (page 410)	See Also (page 675)
L (Locality Name) (page 458)	Serial Number (page 677)
Network Address (page 556)	SvcInfo (page 903)
O (Organization Name) (page 595)	SvcType (page 904)
OU (Organizational Unit Name) (page 606)	SvcTypeID (page 905)
Owner (page 608)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The L (Locality Name) attribute can be used to identify the physical location of a device. For example, if the device were a printer, the locality might be “Building D, Section 24, by Ed Bender's desk.”

The O (Organization Name) and OU (Organizational Unit Name) may already be present in the device's Distinguished Name. They are repeated here to aid searching when an organization spans multiple subtrees in the NDS tree. However, NDS does not automatically add these attributes even though they may be present in the device's Distinguished Name. Additional values for the organization or organizational unit can be useful when a device is “co-owned” by multiple organizations.

In versions of NDS previous to NDS eDirectory 8.5, device is a non-effective class.

Directory Map

Represents the physical name of a file system directory path.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.7

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	Directory Map
Containment	domain (page 119)	Resource (page 245)
	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)
Named By	CN (Common Name) (page 400)	Resource (page 245)

Mandatory Attributes

Directory Map	Inherited from Top	Inherited from Resource
Host Server (page 449)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

Directory Map
Path (page 625)

Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Inherited from Resource
Description (page 410)
Host Resource Name (page 448)
L (Locality Name) (page 458)
O (Organization Name) (page 595)
OU (Organizational Unit Name) (page 606)
See Also (page 675)
Uses (page 718)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The Host Server attribute identifies the server that owns and services the resource. Requests to manipulate a particular resource must usually be directed to the host server.

The Host Resource Name attribute is used when the host's local identification for a resource differs from the global resource identification. For example, a server might recognize SYS as the local name for a volume with the following NDS name:

```
"Project X.Engineering.Acme.US"
```

The L (Locality Name), O (Organization Name), and OU (Organizational Unit Name) attributes are useful when a resource is used by multiple localities, organizations, or organizational units. If these attributes contain appropriate values, a search can be initiated for resources associated with a particular locality or organization.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

domain

Provides a container class for Internet or domain objects that do not correspond to a defined schema class.

Type: Effective

NDS Operational: Yes

LDAP Name

domain

ASN.1 ID

0.9.2342.19200300.100.4.13

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	domain

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	domain
	domain	domain
	Locality (page 141)	domain
	Organization (page 202)	domain
	Organizational Unit (page 215)	domain
	Top (page 263)	domain
	Tree Root (page 266)	domain
Named By	dc (page 406)	domain

Mandatory Attributes

domain	Inherited from Top
dc (page 406)	Object Class (page 601)

Optional Attributes

domain	
associatedName (page 366)	Postal Office Box (page 633)
businessCategory (page 389)	preferredDeliveryMethod (page 636)
Description (page 410)	registeredAddress (page 655)
destinationIndicator (page 412)	SA (Street Address) (page 665)
Facsimile Telephone Number (page 427)	searchGuide (page 670)
internationaliSDNNumber (page 452)	See Also (page 675)
L (Locality Name) (page 458)	Telephone Number (page 696)
Physical Delivery Office Name (page 627)	telexNumber (page 698)
Postal Address (page 629)	telexTerminalIdentifier (page 699)
Postal Code (page 631)	x121Address (page 727)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For more information about this class, see RFC 2247.

When NDS 8 is installed, domain objects can contain all the leaf objects in the operational schema. However, the ability for domain objects to contain the operational container objects such as Country, Locality, Organization, or Organizational Unit does not come automatically. This functionality must be added to the schema by running a schema option in DSRepair.

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

dSA

Represents an LDAP directory service agent.

Type: Effective

LDAP Name

dSA

ASN.1 ID

2.5.6.13

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	applicationEntity (page 80)
	applicationEntity (page 80)	dSA

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	dSA
	Locality (page 141)	dSA
	Organization (page 202)	dSA
	Organizational Unit (page 215)	dSA
Named By	CN (Common Name) (page 400)	applicationEntity (page 80)

Mandatory Attributes

dSA	Inherited from Top	Inherited from applicationEntity
(None)	Object Class (page 601)	CN (Common Name) (page 400) presentationAddress (page 637)

Optional Attributes

dSA
knowledgeInformation (page 457)
Inherited from Top
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424) GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Inherited from applicationEntity

Description (page 410)	OU (Organizational Unit Name) (page 606)
L (Locality Name) (page 458)	See Also (page 675)
O (Organization Name) (page 595)	supportedApplicationContext (page 686)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For more information, see RFC 2256.

External Entity

Represents a non-native NDS object.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.29

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	External Entity
Containment	domain (page 119)	External Entity
	Organization (page 202)	External Entity
	Organizational Unit (page 215)	External Entity

Rule	Class/Attribute	Defined For
Named By	CN (Common Name) (page 400)	External Entity
	OU (Organizational Unit Name) (page 606)	External Entity

Mandatory Attributes

External Entity	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

External Entity	
Description (page 410)	Physical Delivery Office Name (page 627)
EMail Address (page 419)	Postal Address (page 629)
External Name (page 425)	Postal Code (page 631)
Facsimile Telephone Number (page 427)	Postal Office Box (page 633)
L (Locality Name) (page 458)	S (State or Province Name) (page 663)
Mailbox ID (page 516)	SA (Street Address) (page 665)
Mailbox Location (page 518)	See Also (page 675)
OU (Organizational Unit Name) (page 606)	Title (page 701)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	External Name	External Entity

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

This object can be used by services that need to store information about entities outside of the NDS tree. For example, a messaging service that can send messages to E-mail users outside of the NDS tree needs to store address information about those E-mail users.

Such a messaging service can use External Entity objects to store information about E-mail users who exist on other systems. It can then use the objects' names in distribution lists that are List objects.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

Group

Defines values representing an unordered set of names. The names themselves can represent individual objects or other groups of names.

Type: Effective
NDS Operational: Yes

LDAP Name

group
groupOfNames
groupOfUniqueNames

ASN.1 ID

2.5.6.9 (groupOfNames)
2.5.6.17 (groupOfUniqueNames)
2.16.840.1.113719.1.1.6.1.8 (group)

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Group
Containment	domain (page 119)	Group
	Organization (page 202)	Group
	Organizational Unit (page 215)	Group
Named By	CN (Common Name) (page 400)	Group

Mandatory Attributes

Group	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

Group	
businessCategory (page 389)	Mailbox Location (page 518)
Description (page 410)	Member (page 525)
EMail Address (page 419)	O (Organization Name) (page 595)
Full Name (page 429)	OU (Organizational Unit Name) (page 606)
GID (Group ID) (page 433)	Owner (page 608)
L (Locality Name) (page 458)	Profile (page 645)
Login Script (page 511)	Profile Membership (page 646)
Mailbox ID (page 516)	See Also (page 675)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The membership of a group is static; that is, it is explicitly modified by administrative action, rather than dynamically determined each time the group is referred to. The membership of a group can be reduced to a set of individual object's names by replacing each group with its membership. This process can be carried out recursively until all constituent group names have been eliminated, and only the names of individual objects remain.

In general, NDS operations do not perform recursive membership expansion. However, access control resolution effectively expands one level of groups listed in an Access Control List (ACL). Thus, if A is a member of group B, which is in turn listed in an ACL, A gains the access granted to group B. However, if A is a member of group B, which is a member of group C, and C is listed in an access control list, A does not gain the access granted to group C.

Other applications are free to perform recursive group expansion, if they so choose.

The L (Locality Name), O (Organization Name), and OU (Organizational Unit Name) attributes might already be present in the group's Distinguished Name. They are repeated here to aid searching when an organization spans multiple subtrees in the NDS tree. Additional values for the locality, organization or organizational unit may be useful when a group contains members from multiple organizations, organizational units, or localities.

The Owner attribute could be used to contain the name of the group leader or group moderator. This value might not be the same as the set of individuals authorized to modify the group object.

The See Also attribute might be used to list related groups. For example, the groups "Project A Programmers," "Project A Writers," and "Project A Testers" might mention one another in their See Also attributes.

LDAP Group

Defines configuration data that can be applied to a group of LDAP servers.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.1.6.1.34

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	LDAP Group (Class)
Containment	Country (page 105)	LDAP Group (Class)
	Locality (page 141)	LDAP Group (Class)
	Organization (page 202)	LDAP Group (Class)
	Organizational Unit (page 215)	LDAP Group (Class)
Named By	CN (Common Name) (page 400)	LDAP Group (Class)

Mandatory Attributes

LDAP Group (Class)	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

LDAP Group (Class)	
LDAP Allow Clear Text Password (page 464)	LDAP Server List (page 486)
LDAP ACL v11 (page 463)	LDAP Suffix (page 488)
LDAP Anonymous Identity (page 465)	LDAP:otherReferralUsage (page 494)
LDAP Attribute Map v11 (page 466)	LDAP:searchReferralUsage (page 498)
LDAP Class Map v11 (page 468)	Version (page 719)
LDAP Referral (page 478)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

LDAP Server

Defines configuration data for the LDAP server.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.1.6.1.33

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	LDAP Server (Class)
Containment	Country (page 105)	LDAP Server (Class)
	Locality (page 141)	LDAP Server (Class)
	Organization (page 202)	LDAP Server (Class)
	Organizational Unit (page 215)	LDAP Server (Class)
Named By	CN (Common Name) (page 400)	LDAP Server (Class)

Mandatory Attributes

LDAP Server (Class)	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

LDAP Server (Class)	
LDAP Backup Log Filename (page 467)	LDAP Server Idle Timeout (page 485)
LDAP Enable SSL (page 469)	LDAP SSL Port (page 487)
LDAP Enable TCP (page 470)	LDAP TCP Port (page 489)
LDAP Enable UDP (page 471)	LDAP UDP Port (page 490)
LDAP Group (page 472)	LDAP:bindCatalog (page 491)
LDAP Host Server (page 473)	LDAP:bindCatalogUsage (page 492)
LDAP Log Filename (page 474)	LDAP:keyMaterialName (page 493)
LDAP Log Level (page 475)	LDAP:searchCatalogUsage (page 496)
LDAP Log Size Limit (page 477)	LDAP:searchCatalog (page 495)
LDAP Screen Level (page 479)	searchSizeLimit (page 671)
LDAP Search Size Limit (page 481)	searchTimeLimit (page 672)
LDAP Search Time Limit (page 482)	Version (page 719)
LDAP Server Bind Limit (page 484)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

List

Represents an unordered set of names in the NDS tree.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.30

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	List
Containment	domain (page 119)	List
	Organization (page 202)	List
	Organizational Unit (page 215)	List
Named By	CN (Common Name) (page 400)	List

Mandatory Attributes

List	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

List	
Description (page 410)	Member (page 525)
EMail Address (page 419)	O (Organization Name) (page 595)
Full Name (page 429)	OU (Organizational Unit Name) (page 606)
L (Locality Name) (page 458)	Owner (page 608)
Mailbox ID (page 516)	See Also (page 675)
Mailbox Location (page 518)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Object Name	Default Rights	Affected Attributes	Class Defined For
[Root]	Read	Member	List

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Member contains the names of the objects that are members of the list. The members can be individual objects (including Group objects) or the names of other List objects.

Unlike Group membership, List membership does not imply security equivalence.

Locality

Defines geographic locations in the NDS tree.

Type: Effective

NDS Operational: Yes

LDAP Name

locality

ASN.1 ID

2.5.6.3

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Locality

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	Locality
	domain (page 119)	Locality
	Locality (page 141)	Locality
	Organization (page 202)	Locality
	Organizational Unit (page 215)	Locality
Named By	L (Locality Name) (page 458)	Locality
	S (State or Province Name) (page 663)	Locality

Mandatory Attributes

Locality	Inherited from Top
(None)	Object Class (page 601)

Optional Attributes

Locality	
Description (page 410)	SA (Street Address) (page 665)
L (Locality Name) (page 458)	searchGuide (page 670)
S (State or Province Name) (page 663)	See Also (page 675)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

At least one of the naming attributes, L (Locality Name) or S (State or Province Name), must be represented since these are included in the naming rules.

It is recommended that only an object contained by a country use a state or a province name as a naming attribute.

MASV:Security Policy

Defines the class that holds the MASV security policy.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.31.6.2.1

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	MASV:Security Policy
Containment	SAS:Security (page 248)	MASV:Security Policy
Named By	CN (Common Name) (page 400)	MASV:Security Policy

Mandatory Attributes

MASV:Security Policy	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

MASV:Security Policy
Description (page 410)
masvDomainPolicy (page 522)

Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Mandatory Access Control Service (MASV) uses access class labels. These labels represent the sensitivity of the information and the formal authorization of logged-in objects. Access class labels assigned to logged-in objects are called clearances or ranges. When these labels are assigned to resources (such as volumes or partitions), they are called classifications or security labels.

MASV makes use of both classifications and clearances to compute the access rights of a logged-in object to information stored in NDS objects and volumes. MASV works in conjunction with NDS ACLs and file system rights. All the security systems must allow access for an object to access the resource.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

Message Routing Group

Represents a group (or cluster) of messaging servers that have direct connectivity for transferring messages between any two of them.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.28

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Group (page 128)
	Group (page 128)	Message Routing Group
Containment	domain (page 119)	Group (page 128)
	Organization (page 202)	Group (page 128)
	Organizational Unit (page 215)	Group (page 128)

Rule	Class/Attribute	Defined For
Named By	CN (Common Name) (page 400)	Group (page 128)

Mandatory Attributes

Message Routing Group	Inherited from Top	Inherited from Group
(None)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

Message Routing Group
(None)

Inherited from Top
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424) GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Inherited from Group	
Description (page 410)	Member (page 525)
EMail Address (page 419)	O (Organization Name) (page 595)
Full Name (page 429)	OU (Organizational Unit Name) (page 606)
GID (Group ID) (page 433)	Owner (page 608)
L (Locality Name) (page 458)	Profile (page 645)
Login Script (page 511)	Profile Membership (page 646)
Mailbox ID (page 516)	See Also (page 675)
Mailbox Location (page 518)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Self]	Read	[All Attributes]	Message Routing Group
[Self]	Browse	[Entry Rights]	Message Routing Group

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The Member attribute (inherited from Group) lists the messaging servers that belong to the Message Routing Group.

The Owner attribute (inherited from Group) contains the name of the postmaster general of the messaging server's message routing group. The owner has the authority to add a messaging server's name to, or remove a messaging server's name, from the Member list.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

Messaging Server

Represents messaging servers (such as NetWare MHS servers).

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.27

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	Messaging Server
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

Messaging Server	Inherited from Top	Inherited from Server
(None)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

Messaging Server	
Message Routing Group (page 528)	Postmaster (page 635)
Messaging Database Location (page 530)	Supported Gateway (page 688)
Messaging Server Type (page 533)	Supported Services (page 689)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Server

Account Balance (page 363)	Private Key (page 643)
Allow Unlimited Credit (page 365)	Public Key (page 648)
Description (page 410)	Resource (page 658)
Full Name (page 429)	Security Equals (page 673)
Host Device (page 447)	Security Flags (page 674)
L (Locality Name) (page 458)	See Also (page 675)
masvAuthorizedRange (page 520)	Status (page 683)
masvDefaultRange (page 521)	SvcInfo (page 903)
masvProposedLabel (page 524)	SvcType (page 904)
Minimum Account Balance (page 534)	SvcTypeID (page 905)
Network Address (page 556)	User (page 714)
O (Organization Name) (page 595)	Version (page 719)
OU (Organizational Unit Name) (page 606)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)
[Self]	Read/Write	Status	Messaging Server
[Self]	Browse	[Entry Rights]	Messaging Server
[Self]	Read	[All Attributes]	Messaging Server
[Public]	Read	Messaging Server Type	Messaging Server
[Public]	Read	Messaging Database Location	Messaging Server

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

A MHS messaging server picks up messages, which are either submitted by messaging applications (for example, E-mail) or transferred from another messaging server, and delivers them to the recipients. For recipients whose mailboxes are local on the messaging server, the messages are delivered to their mailboxes. Otherwise, the messaging server transfers the messages to another messaging server for eventual delivery to the recipient's mailbox.

A MHS messaging server runs as `mhs.nlm` on a NetWare server. There is no limit to the number of mailboxes it serves, except that mailboxes take up disk space.

A MHS messaging server is represented by a NDS leaf object whose object class is `Messaging Server`.

The `Host Device` attribute (inherited from `Server`) identifies the NCP Server on which the messaging server's software runs.

The `Message Routing Group` attribute names the message routing groups to which the messaging server belongs.

The `Messaging Database Location` attribute names the volume and path (such as `sys:mhs`) on which the message directory resides. MHS messaging servers use a file system subtree to (1) receive messages from applications, other messaging servers and gateways, (2) store messages while they are being routed, (3) store internal control files, and (4) to extract files.

The `Messaging Server Type` attribute identifies the type of the `Messaging Server` object (for example, `MHS`, `GMHS`, `X400`).

The `Postmaster` attribute specifies one or more users who have the privileges to manage the messaging server, such as privileges to remove a mailbox. Postmasters also receive messages about special events in the messaging server, such as messages not being processed.

The `Supported Gateway` attribute specifies the messaging gateways that are connected through the messaging server. It provides messaging connectivity between the MHS messaging system and foreign messaging systems.

The `Supported Services` attribute indicates the messaging capabilities of the server.

The `User` attribute (inherited from `Server`) contains a list users whose mailboxes are serviced by the messaging server. Any effective object that has the `Mailbox ID` and `Mailbox Location` attributes is a valid value for this list.

Adding an object to the User list has the same effect as assigning values to the object's Mailbox Location and Mailbox ID attributes. An administrator can give an object a mailbox by either means.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NCP Server

Represents servers that provide NCP transport and session services.

Type: Effective

NDS Operational: Yes

LDAP Name

ncpServer

ASN.1 ID

2.16.840.1.113719.1.1.6.1.10

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	NCP Server

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

NCP Server	Inherited from Top	Inherited from Server
	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

NCP Server	
cACertificate (page 392)	NDSPKI:Public Key (page 546)
DS Revision (page 418)	NDSPKI:SD Key ID (page 549)
LDAP Server (page 483)	NDSPKI:SD Key Struct (page 551)
Messaging Server (page 532)	NLS:License Service Provider (page 827)
NDSCat:Catalog List (page 802)	Operator (page 603)
NDSCat:Max Threads (page 812)	Permanent Config Params (page 626)
NDSCat:Synch Interval (page 818)	SLP Directory Agent DN (page 889)
NDSPKI:Certificate Chain (page 536)	SAS:Service DN (page 669)
NDSPKI:Parent CA DN (page 544)	Supported Services (page 689)
NDSPKI:Private Key (page 545)	

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Server

Account Balance (page 363)	Private Key (page 643)
Allow Unlimited Credit (page 365)	Public Key (page 648)
Description (page 410)	Resource (page 658)
Full Name (page 429)	Security Equals (page 673)
Host Device (page 447)	Security Flags (page 674)
L (Locality Name) (page 458)	See Also (page 675)
masvAuthorizedRange (page 520)	Status (page 683)
masvDefaultRange (page 521)	SvcInfo (page 903)
masvProposedLabel (page 524)	SvcType (page 904)
Minimum Account Balance (page 534)	SvcTypeID (page 905)
Network Address (page 556)	User (page 714)
O (Organization Name) (page 595)	Version (page 719)
OU (Organizational Unit Name) (page 606)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)

Object Name	Default Rights	Affected Attributes	Class Defined For
[Self]	Supervisor	[Entry Rights]	Server (page 254)
[Public]	Read	Messaging Server	NCP Server

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Individual services on an NCP server do not need distinct NDS names, since they can all share a common NCP session. However, individual resources on a server may require distinct NDS entries. For example, if a server supports file services and queue management services, that server will have only one object in the NDS tree for the server itself. Other NDS entries would denote the individual queues and the volumes for files.

The Supported Services attribute lists NCP-based services or features available at this network address. It should not be used to list other non-NCP services residing on the same host.

The Operator attribute is used by the NCP server as an ACL. If an object is listed in this attribute, that object is allowed to perform remote-console operations.

The NCP Server class is intended to represent both bindery-based and NDS-based NCP servers. The Version attribute (inherited from Server) should distinguish one type of server from the other

The Private Key and Public Key attributes are present if the server is a client of NDS authentication services. The Resource attribute contains a list of resources managed by this service.

The User attribute contains a list of objects that are authorized to use this server. The server must determine if the user list is to be maintained by an administrator, or if the list is automatically generated by the server. If the user list is used by the server as an ACL, the administrator usually maintains the list. If the user list is purely informational, reflecting access control information stored elsewhere, the server usually maintains the list.

The LDAP server maps requests for the ncpServer class to this class.

ndsLoginProperties

Contains all the attributes that are required by objects who log in to the NDS tree.

Type: Noneffective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.33

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	ndsLoginProperties
Containment	(None)	ndsLoginProperties
Named By	(None)	ndsLoginProperties

Mandatory Attributes

ndsLoginProperties	Inherited from Top
(None)	Object Class (page 601)

Optional Attributes

ndsLoginProperties	
Account Balance (page 363)	Minimum Account Balance (page 534)
Allow Unlimited Credit (page 365)	Network Address (page 556)
Group Membership (page 436)	Network Address Restriction (page 558)
Higher Privileges (page 440)	Password Allow Change (page 616)
Language (page 460)	Password Expiration Interval (page 617)
Last Login Time (page 461)	Password Expiration Time (page 618)
Locked By Intruder (page 499)	Password Minimum Length (page 621)
Login Allowed Time Map (page 501)	Password Required (page 622)
Login Disabled (page 502)	Password Unique Required (page 623)
Login Expiration Time (page 503)	Passwords Used (page 624)
Login Grace Limit (page 504)	Private Key (page 643)
Login Grace Remaining (page 505)	Profile (page 645)
Login Intruder Address (page 506)	Profile Membership (page 646)
Login Intruder Attempts (page 507)	Public Key (page 648)
Login Intruder Reset Time (page 509)	Security Equals (page 673)
Login Maximum Simultaneous (page 510)	Security Flags (page 674)
Login Script (page 511)	Server Holds (page 679)
Login Time (page 513)	Timezone (page 700)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

NDSPKI:Certificate Authority

Defines the certificate authority (CA) for the NDS tree.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.48.6.1.1

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NDSPKI:Certificate Authority
Containment	SAS:Security (page 248)	NDSPKI:Certificate Authority
Named By	CN (Common Name) (page 400)	NDSPKI:Certificate Authority

Mandatory Attributes

NDSPKI:Certificate Authority	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

NDSPKI:Certificate Authority	
Host Server (page 449)	NDSPKI:Private Key (page 545)
NDSPKI:Certificate Chain (page 536)	NDSPKI:Public Key (page 546)
NDSPKI:Parent CA (page 543)	NDSPKI:Public Key Certificate (page 547)
NDSPKI:Parent CA DN (page 544)	NDSPKI:Subject Name (page 552)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Certificate authorities issue public/private key pairs and public key certificates.

The Host Server attribute contains the distinguished name of the NCP server that hosts the certificate authority service.

This attribute was added for NetWare 5. NDS 8 replaces this class with the pkiCA class.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDSPKI:Key Material

Defines the object that holds public/private key pairs and a public key certificate for an NCP Server object.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.48.6.1.2

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NDSPKI:Key Material
Containment	Organization (page 202)	NDSPKI:Key Material
	Organizational Unit (page 215)	NDSPKI:Key Material
	SAS:Security (page 248)	NDSPKI:Key Material
Named By	CN (Common Name) (page 400)	NDSPKI:Key Material

Mandatory Attributes

NDSPKI:Key Material	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

NDSPKI:Key Material	
Host Server (page 449)	NDSPKI:Private Key (page 545)
NDSPKI:Certificate Chain (page 536)	NDSPKI:Public Key (page 546)
NDSPKI:Given Name (page 537)	NDSPKI:Public Key Certificate (page 547)
NDSPKI:Key File (page 538)	NDSPKI:Subject Name (page 552)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

An NCP Server object can have multiple NDSPKI:Key Material objects. One should be created for each service that needs one. Each service should have its own so that if the key pair is broken for one service, the other service is not compromised.

The service that is using a NDSPKI:Key Material object must run on the same NCP server that created the NDSPKI:Key Material object. NDSPKI:Key Material objects cannot be shared across servers.

The Host Server attribute identifies the NCP Server object to which the NDSPKI:Key Material object belongs.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDSPKI:SD Key Access Partition

Defines the object that contains security domain key objects.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.48.6.1.101

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NDSPKI:SD Key Access Partition
Containment	SAS:Security (page 248)	NDSPKI:SD Key Access Partition
Named By	CN (Common Name) (page 400)	NDSPKI:SD Key Access Partition

Mandatory Attributes

NDSPKI:SD Key Access Partition	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

NDSPKI:SD Key Access Partition
(None)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDSPKI:SD Key List

Defines the object that holds information about a security domain key.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.48.6.1.102

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NDSPKI:SD Key List
Containment	NDSPKI:SD Key Access Partition (page 168)	NDSPKI:SD Key List
Named By	CN (Common Name) (page 400)	NDSPKI:SD Key List

Mandatory Attributes

NDSPKI:SD Key List	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

NDSPKI:SD Key List	
NDSPKI:SD Key Cert (page 548) NDSPKI:SD Key Server DN (page 550)	NDSPKI:SD Key Struct (page 551)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDSPKI:Trusted Root

Defines the object that contains trusted root certificate objects.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.48.6.1.3

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NDSPKI:Trusted Root
Containment	Country (page 105)	NDSPKI:Trusted Root
	Locality (page 141)	NDSPKI:Trusted Root
	Organization (page 202)	NDSPKI:Trusted Root
	Organizational Unit (page 215)	NDSPKI:Trusted Root
	SAS:Security (page 248)	NDSPKI:Trusted Root
Named By	CN (Common Name) (page 400)	NDSPKI:Trusted Root

Mandatory Attributes

NDSPKI:Trusted Root	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

NDSPKI:Trusted Root
(None)

Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDSPKI:Trusted Root Object

Defines an object that holds a trusted root certificate from a Certificate Authority that is trusted.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.48.6.1.4

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NDSPKI:Trusted Root Object
Containment	NDSPKI:Trusted Root (page 174)	NDSPKI:Trusted Root Object
Named By	CN (Common Name) (page 400)	NDSPKI:Trusted Root Object

Mandatory Attributes

NDSPKI:Trusted Root Object	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)
NDSPKI:Trusted Root Certificate (page 554)	

Optional Attributes

NDSPKI:Trusted Root Object	
Cross Certificate Pair (page 405)	NDSPKI:Not Before (page 542)
Given Name (page 434)	NDSPKI:Subject Name (page 552)
External Name (page 425)	Surname (page 691)
NDSPKI:Not After (page 541)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NSCP:groupOfCertificates

Represents a group of certificates used by Netscape products.

Type: Effective

LDAP Name

groupOfCertificates

ASN.1 ID

2.16.840.1.113730.3.2.31

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NSCP:groupOfCertificates

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	NSCP:groupOfCertificates
	Locality (page 141)	NSCP:groupOfCertificates
	Organization (page 202)	NSCP:groupOfCertificates
	Organizational Unit (page 215)	NSCP:groupOfCertificates
Named By	CN (Common Name) (page 400)	NSCP:groupOfCertificates

Mandatory Attributes

NSCP:groupOfCertificates	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

NSCP:groupOfCertificates	
businessCategory (page 389)	OU (Organizational Unit Name) (page 606)
Description (page 410)	Owner (page 608)
NSCP:memberCertificateDesc (page 582)	See Also (page 675)
O (Organization Name) (page 595)	

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NSCP:mailGroup1

Represents a Netscape mail group.

Type: Effective

LDAP Name

rfc822mailGroup

ASN.1 ID

2.5.6.17

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Group (page 128)
	Group (page 128)	NSCP:mailGroup1

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	NSCP:mailGroup1
	Locality (page 141)	NSCP:mailGroup1
	Organization (page 202)	NSCP:mailGroup1
	Organizational Unit (page 215)	NSCP:mailGroup1
Named By	CN (Common Name) (page 400)	Group (page 128)

Mandatory Attributes

NSCP:mailGroup1	Inherited from Top	Inherited from Group
	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

NSCP:mailGroup1	
Internet EMail Address (page 453)	NSCP:mgrpRFC822mailmember (page 583)
NSCP:mailAlternateAddress (page 850)	NSCP:mailAlternateAddress (page 850)
NSCP:mailForwardingAddress (page 854)	NSCP:mailForwardingAddress (page 854)
NSCP:mailHost (page 855)	NSCP:mailHost (page 855)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Group

Description (page 410)	Member (page 525)
EMail Address (page 419)	O (Organization Name) (page 595)
Full Name (page 429)	OU (Organizational Unit Name) (page 606)
GID (Group ID) (page 433)	Owner (page 608)
L (Locality Name) (page 458)	Profile (page 645)
Login Script (page 511)	Profile Membership (page 646)
Mailbox ID (page 516)	See Also (page 675)
Mailbox Location (page 518)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

In NDS 8, LDAP adds this class for Netscape support. For more information, see the documentation for the Netscape SuiteSpot schema.

This class was redefined for NDS 8 because the NSCP:mailGroup5 class has restricted containment.

NSCP:mailRecipient

Defines a Netscape auxiliary class for users of Netscape mail.

Type: Auxiliary Class

LDAP Name

mailRecipient

ASN.1 ID

2.16.840.1.113730.3.2.3

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	(None)	NSCP:mailRecipient
Containment	(None)	NSCP:mailRecipient
Named By	(None)	NSCP:mailRecipient

Mandatory Attributes

NSCP:mailRecipient
(None)

Optional Attributes

NSCP:mailRecipient	
NSCP:mailAccessDomain (page 849)	NSCP:mailForwardingAddress (page 854)
NSCP:mailAlternateAddress (page 850)	NSCP:mailHost (page 855)
NSCP:mailAutoReplyMode (page 851)	NSCP:mailMessageStore (page 856)
NSCP:mailAutoReplyText (page 852)	NSCP:mailProgramDeliveryInfo (page 857)
NSCP:mailDeliveryOption (page 853)	NSCP:mailQuota (page 858)

Default ACL Template

None

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NSCP:NetscapeMailServer5

Represents a Netscape mail server.

Type: Effective

LDAP Name

netscapeMailServer

ASN.1 ID

2.16.840.1.113730.3.2.24

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	NSCP:NetscapeServer5 (page 193)
	NSCP:NetscapeServer5 (page 193)	NSCP:NetscapeMailServer5

Rule	Class/Attribute	Defined For
Containment	NSCP:NetscapeServer5 (page 193)	NSCP:NetscapeMailServer5
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

NSCP:NetscapeMailServer5	Inherited from Top	Inherited from Server	Inherited from NCSP:NetscapeServer5
(None)	Object Class (page 601)	CN (Common Name) (page 400)	(None)

Optional Attributes

NSCP:NetscapeMailServer5
(None)

Inherited from Top
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424) GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Inherited from Server	
Account Balance (page 363)	Public Key (page 648)
Allow Unlimited Credit (page 365)	Resource (page 658)
Description (page 410)	Security Equals (page 673)
Full Name (page 429)	Security Flags (page 674)
Host Device (page 447)	See Also (page 675)
L (Locality Name) (page 458)	Status (page 683)
Minimum Account Balance (page 534)	SvcInfo (page 903)
Network Address (page 556)	SvcType (page 904)
O (Organization Name) (page 595)	SvcTypeID (page 905)
OU (Organizational Unit Name) (page 606)	User (page 714)
Private Key (page 643)	Version (page 719)

Inherited from NCSP:NetscapeServer5	
NCSP:adminURL (page 565)	NCSP:serverProductName (page 591)
NCSP:administratorContactInfo (page 564)	NCSP:serverRoot (page 592)
NCSP:installationTimeStamp (page 580)	NCSP:serverVersionNumber (page 593)
NCSP:serverHostName (page 590)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

LDAP added this class for Netscape support. For more information, see the documentation for the Netscape schema.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NSCP:NetscapeServer5

Represents a Netscape server.

Type: Effective

LDAP Name

netscapeServer

ASN.1 ID

2.16.840.1.113730.3.2.10

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	NSCP:NetscapeServer5

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	NSCP:NetscapeServer5
	Locality (page 141)	NSCP:NetscapeServer5
	Organization (page 202)	NSCP:NetscapeServer5
	Organizational Unit (page 215)	NSCP:NetscapeServer5
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

NSCP:NetscapeServer5	Inherited from Top	Inherited from Server (Class)
	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

NSCP:NetscapeServer5	
NSCP:adminURL (page 565)	NSCP:serverProductName (page 591)
NSCP:administratorContactInfo (page 564)	NSCP:serverRoot (page 592)
NSCP:installationTimeStamp (page 580)	NSCP:serverVersionNumber (page 593)
NSCP:serverHostName (page 590)	

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Server

Account Balance (page 363)	Public Key (page 648)
Allow Unlimited Credit (page 365)	Resource (page 658)
Description (page 410)	Security Equals (page 673)
Full Name (page 429)	Security Flags (page 674)
Host Device (page 447)	See Also (page 675)
L (Locality Name) (page 458)	Status (page 683)
Minimum Account Balance (page 534)	SvcInfo (page 903)
Network Address (page 556)	SvcType (page 904)
O (Organization Name) (page 595)	SvcTypeID (page 905)
OU (Organizational Unit Name) (page 606)	User (page 714)
Private Key (page 643)	Version (page 719)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

LDAP adds this class for Netscape support. For more information, see the documentation for the Netscape schema.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NSCP:nginfo3

Represents a Netscape class.

Type: Effective

LDAP Name

nginfo

ASN.1 ID

2.16.840.1.113730.3.2.26

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NSCP:nginfo3

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	NSCP:nginfo3
	Country (page 105)	NSCP:nginfo3
	Organization (page 202)	NSCP:nginfo3
	Organizational Unit (page 215)	NSCP:nginfo3
Named By	NSCP:ngComponent (page 859)	NSCP:nginfo3

Mandatory Attributes

NSCP:nginfo3	Inherited from Top
NSCP:ngComponent (page 859)	Object Class (page 601)

Optional Attributes

NSCP:nginfo3	
NSCP:nsaclrole (page 585)	NSCP:nsnewsACL (page 588)
NSCP:nscreator (page 586)	NSCP:nsprettyname (page 589)
NSCP:nsflags (page 587)	NSCP:subtreeACI (page 594)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

In NDS 8, LDAP adds this class for Netscape support. For more information, see the documentation for the Netscape schema.

This class was redefined for NDS 8 because the NSCP:Nginfo2 class uses the wrong naming attribute and has restricted containment.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NSCP:nsLicenseUser

Defines an auxiliary class for users who license Netscape products.

Type: Auxiliary Class

LDAP Name

nsLicenseUser

ASN.1 ID

2.16.840.1.113730.3.2.7

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	(None)	NSCP:nsLicenseUser
Containment	(None)	NSCP:nsLicenseUser
Named By	(None)	NSCP:nsLicenseUser

Mandatory Attributes

NSCP:nsLicenseUser

(None)

Optional Attributes

NSCP:nsLicenseUser

NSCP:nsLicenseEndTime (page 860)

NSCP:nsLicensedFor (page 861)

NSCP:nsLicenseStartTime (page 862)

Default ACL Template

None

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Organization

Defines organization objects in the NDS tree.

Type: Effective

NDS Operational: Yes

LDAP Name

organization

ASN.1 ID

2.5.6.4

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	ndsLoginProperties (page 159)
	ndsLoginProperties (page 159)	Organization

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	Organization
	domain (page 119)	Organization
	Locality (page 141)	Organization
	Top (page 263)	Orgainization
	Tree Root (page 266)	Orgainization
	(Nothing)	Organization
Named By	O (Organization Name) (page 595)	Organization

Mandatory Attributes

Organization	Inherited from Top	Inherited from ndsLoginProperties
O (Organization Name) (page 595)	Object Class (page 601)	(None)

Optional Attributes

Organization

Description (page 410)	NDPS Control Flags (page 766)
Detect Intruder (page 413)	NDPS Default Printer (page 770)
EEmail Address (page 419)	NDPS Default Public Printer (page 771)
Facsimile Telephone Number (page 427)	NDPS Printer Install List (page 775)
Intruder Attempt Reset Interval (page 454)	NDPS Printer Install Timestamp (page 776)
Intruder Lockout Reset Interval (page 456)	NDPS Public Printer Install List (page 779)
L (Locality Name) (page 458)	NDPS Replace All Client Printers (page 780)
LDAPUserCertificate (page 751)	NNS Domain (page 562)
LDAP:ARL (page 752)	Physical Delivery Office Name (page 627)
LDAP:caCertificate (page 753)	Postal Address (page 629)
LDAP:CRL (page 754)	Postal Code (page 631)
LDAP:crossCertificatePair (page 755)	Postal Office Box (page 633)
Lockout After Detection (page 500)	Print Job Configuration (page 638)
Login Intruder Limit (page 508)	Printer Control (page 642)
Login Script (page 511)	S (State or Province Name) (page 663)
Mailbox ID (page 516)	SA (Street Address) (page 665)
Mailbox Location (page 518)	See Also (page 675)
	Telephone Number (page 696)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from ndsLoginProperties

Account Balance (page 363)	Minimum Account Balance (page 534)
Allow Unlimited Credit (page 365)	Network Address (page 556)
Group Membership (page 436)	Network Address Restriction (page 558)
Higher Privileges (page 440)	Password Allow Change (page 616)
Language (page 460)	Password Expiration Interval (page 617)
Last Login Time (page 461)	Password Expiration Time (page 618)
Locked By Intruder (page 499)	Password Minimum Length (page 621)
Login Allowed Time Map (page 501)	Password Required (page 622)
Login Disabled (page 502)	Password Unique Required (page 623)
Login Expiration Time (page 503)	Passwords Used (page 624)
Login Grace Limit (page 504)	Private Key (page 643)
Login Grace Remaining (page 505)	Profile (page 645)
Login Intruder Address (page 506)	Profile Membership (page 646)
Login Intruder Attempts (page 507)	Public Key (page 648)
Login Intruder Reset Time (page 509)	Security Equals (page 673)
Login Maximum Simultaneous (page 510)	Security Flags (page 674)
Login Script (page 511)	Server Holds (page 679)
Login Time (page 513)	Timezone (page 700)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Self]	Read	Login Script	Organization
[Self]	Read	Print Job Configuration	Organization

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Organization might appear as the topmost object in an NDS tree, or it might be subordinate to Tree Root, Country, or Locality. Top is a special object and never appears as an object in an NDS tree.

An organization located directly under the root denotes an international organization. For international organizations, the values of the O (Organization Name) attribute must all be distinct.

Organizational Person

Defines objects representing people employed by, or in some other important way associated with, an organization.

Type: Effective

NDS Operational: Yes

LDAP Name

organizationalPerson

ASN.1 ID

2.5.6.7

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	ndsLoginProperties (page 159)
	ndsLoginProperties (page 159)	Person (page 222)
	Person (page 222)	Organizational Person
Containment	domain (page 119)	Organizational Person
	Organization (page 202)	Organizational Person
	Organizational Unit (page 215)	Organizational Person
Named By	CN (Common Name) (page 400)	Organizational Person
	uniqueID (page 707)	Organizational Person
	OU (Organizational Unit Name) (page 606)	Organizational Person

Mandatory Attributes

Organizational Person	Inherited from Top	Inherited from ndsLoginProperties	Inherited from Person
(None)	Object Class (page 601)	(None)	CN (Common Name) (page 400)
			Surname (page 691)

Optional Attributes

Organizational Person

businessCategory (page 389)	Postal Office Box (page 633)
destinationIndicator (page 412)	preferredDeliveryMethod (page 636)
EMail Address (page 419)	registeredAddress (page 655)
Facsimile Telephone Number (page 427)	roomNumber (page 661)
internationaliSDNNumber (page 452)	S (State or Province Name) (page 663)
L (Locality Name) (page 458)	SA (Street Address) (page 665)
Mailbox ID (page 516)	telexNumber (page 698)
Mailbox Location (page 518)	telexTerminalIdentifier (page 699)
OU (Organizational Unit Name) (page 606)	Title (page 701)
Physical Delivery Office Name (page 627)	uniqueID (page 707)
Postal Address (page 629)	x121Address (page 727)
Postal Code (page 631)	x500UniqueIdentifier (page 728)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from ndsLoginProperties

Account Balance (page 363)	Minimum Account Balance (page 534)
Allow Unlimited Credit (page 365)	Network Address (page 556)
Group Membership (page 436)	Network Address Restriction (page 558)
Higher Privileges (page 440)	Password Allow Change (page 616)
Language (page 460)	Password Expiration Interval (page 617)
Last Login Time (page 461)	Password Expiration Time (page 618)
Locked By Intruder (page 499)	Password Minimum Length (page 621)
Login Allowed Time Map (page 501)	Password Required (page 622)
Login Disabled (page 502)	Password Unique Required (page 623)
Login Expiration Time (page 503)	Passwords Used (page 624)
Login Grace Limit (page 504)	Private Key (page 643)
Login Grace Remaining (page 505)	Profile (page 645)
Login Intruder Address (page 506)	Profile Membership (page 646)
Login Intruder Attempts (page 507)	Public Key (page 648)
Login Intruder Reset Time (page 509)	Security Equals (page 673)
Login Maximum Simultaneous (page 510)	Security Flags (page 674)
Login Script (page 511)	Server Holds (page 679)
Login Time (page 513)	Timezone (page 700)

Inherited from Person

Description (page 410)	Initials (page 451)
Full Name (page 429)	See Also (page 675)
Generational Qualifier (page 431)	Telephone Number (page 696)
Given Name (page 434)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The X.500 standard defines two subclasses of person: Organizational Person and Residential Person. The schema defined by this document does not include Residential Person, but the division of Person from Organizational Person has been maintained for future compatibility with X.500.

Organizational Role

Defines a position or role within an organization.

Type: Effective

NDS Operational: Yes

LDAP Name

organizationalRole

ASN.1 ID

2.5.6.8

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Organizational Role

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Organizational Role
	Organization (page 202)	Organizational Role
	Organizational Unit (page 215)	Organizational Role
Named By	CN (Common Name) (page 400)	Organizational Role

Mandatory Attributes

Organizational Role	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

Organizational Role	
Description (page 410)	Postal Address (page 629)
EMail Address (page 419)	Postal Code (page 631)
Facsimile Telephone Number (page 427)	Postal Office Box (page 633)
L (Locality Name) (page 458)	Role Occupant (page 660)
Mailbox ID (page 516)	S (State or Province Name) (page 663)
Mailbox Location (page 518)	SA (Street Address) (page 665)
OU (Organizational Unit Name) (page 606)	See Also (page 675)
Physical Delivery Office Name (page 627)	Telephone Number (page 696)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Normally, an organizational role is thought to be performed by a particular organizational person. Over its lifetime, however, an organizational role can be filled by a succession of different organizational people. In general, an organizational role can be filled by a person or a nonhuman entity.

The LDAP server maps requests for the organizationalRole class to this class.

Organizational Unit

Defines objects representing subdivisions of organizations.

Type: Effective

NDS Operational: Yes

LDAP Name

organizationalUnit

ASN.1 ID

2.5.6.5

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	ndsLoginProperties (page 159)
	ndsLoginProperties (page 159)	Organizational Unit

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Organizational Unit
	Locality (page 141)	Organizational Unit
	Organization (page 202)	Organizational Unit
	Organizational Unit (page 215)	Organizational Unit
Named By	OU (Organizational Unit Name) (page 606)	Organizational Unit

Mandatory Attributes

Organizational Unit	Inherited from Top	Inherited from ndsLoginProperties
OU (Organizational Unit Name) (page 606)	Object Class (page 601)	

Optional Attributes

Organizational Unit	
Description (page 410)	NDPS Printer Install List (page 775)
Detect Intruder (page 413)	NDPS Printer Install Timestamp (page 776)
E-Mail Address (page 419)	NDPS Public Printer Install List (page 779)
Entrust:AttributeCertificate (page 741)	NDPS Replace All Client Printers (page 780)
Facsimile Telephone Number (page 427)	Network Address (page 556)
Intruder Attempt Reset Interval (page 454)	NNS Domain (page 562)
Intruder Lockout Reset Interval (page 456)	Physical Delivery Office Name (page 627)
L (Locality Name) (page 458)	Postal Address (page 629)
Last Login Time (page 461)	Postal Code (page 631)
Lockout After Detection (page 500)	Postal Office Box (page 633)
Login Intruder Limit (page 508)	Print Job Configuration (page 638)
Login Time (page 513)	Printer Control (page 642)
Login Script (page 511)	Private Key (page 643)
Mailbox ID (page 516)	Public Key (page 648)
Mailbox Location (page 518)	S (State or Province Name) (page 663)
NDPS Control Flags (page 766)	SA (Street Address) (page 665)
NDPS Default Printer (page 770)	See Also (page 675)
NDPS Default Public Printer (page 771)	Telephone Number (page 696)
Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from ndsLoginProperites

Account Balance (page 363)	Minimum Account Balance (page 534)
Allow Unlimited Credit (page 365)	Network Address (page 556)
Group Membership (page 436)	Network Address Restriction (page 558)
Higher Privileges (page 440)	Password Allow Change (page 616)
Language (page 460)	Password Expiration Interval (page 617)
Last Login Time (page 461)	Password Expiration Time (page 618)
Locked By Intruder (page 499)	Password Minimum Length (page 621)
Login Allowed Time Map (page 501)	Password Required (page 622)
Login Disabled (page 502)	Password Unique Required (page 623)
Login Expiration Time (page 503)	Passwords Used (page 624)
Login Grace Limit (page 504)	Private Key (page 643)
Login Grace Remaining (page 505)	Profile (page 645)
Login Intruder Address (page 506)	Profile Membership (page 646)
Login Intruder Attempts (page 507)	Public Key (page 648)
Login Intruder Reset Time (page 509)	Security Equals (page 673)
Login Maximum Simultaneous (page 510)	Security Flags (page 674)
Login Script (page 511)	Server Holds (page 679)
Login Time (page 513)	Timezone (page 700)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Self]	Read	Print Job Configuration	Organizational Unit
[Self]	Read	Login Script	Organizational Unit

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Partition

Encapsulates the information required to maintain the synchronization and connectivity of NDS distributed operation.

Type: Auxiliary

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.15

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	On
Ambiguous Naming	On
Ambiguous Container	On
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Partition
Containment	(none)	Partition and Top
Named By	(none)	Partition and Top

Mandatory Attributes

Partition	Inherited from Top
(None)	Object Class (page 601)

Optional Attributes

Partition	
Authority Revocation (page 378)	Partition Control (page 612)
CA Private Key (page 393)	Partition Creation Time (page 614)
CA Public Key (page 394)	Partition Status (page 615)
Certificate Revocation (page 397)	Purge Vector (page 650)
Convergence (page 403)	Received Up To (page 653)
Cross Certificate Pair (page 405)	Replica (page 656)
High Convergence Sync Interval (page 439)	Replica Up To (page 657)
Inherited ACL (page 450)	SAS:Security DN (page 668)
Low Convergence Reset Time (page 514)	Synchronization Tolerance (page 693)
Low Convergence Sync Interval (page 515)	Synchronized Up To (page 694)
masvLabel (page 523)	Transitive Vector (page 703)
Obituary Notify (page 599)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

NDS automatically adds this class (and its required attributes) to the Root object of a partition. (This class is added in addition to the base class of the object, but does not change that base class.) Most of the partition attributes are operational, so NDS can supply initial values for these attributes automatically.

The optional attributes, Authority Revocation, CA Private Key, CA Public Key, Certification Revocation, Convergence, and Cross Certificate Pair, were originally only in the Partition class. However, in the next NDS release, they were also added to the Top object class. Thus, they are now redundant in the Partition class because they are inherited from Top.

NDS 8 defines the Partition class as an auxiliary class. Previous versions of NDS define it as an effective class.

Person

Represents the common elements of organizational and residential persons.

Type: Effective

NDS Operational: Yes

LDAP Name

person

ASN.1 ID

2.5.6.6

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	ndsLoginProperties
	ndsLoginProperties	Person

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Person
	Organization (page 202)	Person
	Organizational Unit (page 215)	Person
Named By	CN (Common Name) (page 400)	Person
	uniqueID (page 707)	Person

Mandatory Attributes

Person	Inherited from Top	Inherited from ndsLoginProperties
CN (Common Name) (page 400)	Object Class (page 601)	(None)
Surname (page 691)		

Optional Attributes

Person
Description (page 410) Full Name (page 429) Generational Qualifier (page 431) Given Name (page 434) Initials (page 451) See Also (page 675) Telephone Number (page 696) uniqueID (page 707)

Added to Person in NDS eDirectory 8.5
departmentNumber (page 409) employeeType (page 422) manager (page 519) mobile (page 535) pager (page 611) roomNumber (page 661) S (State or Province Name) (page 663)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from ndsLoginProperities

Account Balance (page 363)	Minimum Account Balance (page 534)
Allow Unlimited Credit (page 365)	Network Address (page 556)
Group Membership (page 436)	Network Address Restriction (page 558)
Higher Privileges (page 440)	Password Allow Change (page 616)
Language (page 460)	Password Expiration Interval (page 617)
Last Login Time (page 461)	Password Expiration Time (page 618)
Locked By Intruder (page 499)	Password Minimum Length (page 621)
Login Allowed Time Map (page 501)	Password Required (page 622)
Login Disabled (page 502)	Password Unique Required (page 623)
Login Expiration Time (page 503)	Passwords Used (page 624)
Login Grace Limit (page 504)	Private Key (page 643)
Login Grace Remaining (page 505)	Profile (page 645)
Login Intruder Address (page 506)	Profile Membership (page 646)
Login Intruder Attempts (page 507)	Public Key (page 648)
Login Intruder Reset Time (page 509)	Security Equals (page 673)
Login Maximum Simultaneous (page 510)	Security Flags (page 674)
Login Script (page 511)	Server Holds (page 679)
Login Time (page 513)	Timezone (page 700)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The X.500 standard defines two subclasses of person: Organizational Person and Residential Person. The schema defined by this document does not include Residential Person, but the division of Person from Organizational Person has been maintained for future compatibility with X.500.

pkiCA

Defines a certificate authority (CA) for the NDS tree.

Type: Auxiliary Class

LDAP Name

pkiCA

ASN.1 ID

2.5.6.22

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	(None)	pkiCA
Containment	(None)	pkiCA
Named By	(None)	pkiCA

Mandatory Attributes

pkiCA
(None)

Optional Attributes

pkiCA	
attributeCertificate (page 367)	Last Login Time (page 461)
authorityRevocationList (page 377)	Login Time (page 513)
cACertificate (page 392)	Network Address (page 556)
certificateRevocationList (page 396)	Private Key (page 643)
crossCertificatePair (page 399)	Public Key (page 648)

Default ACL Template

None

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Certificate authorities issue public/private key pairs and public key certificates.

pkiUser

Defines a PKI user.

Type: Auxiliary Class

LDAP Name

pkiUser

ASN.1 ID

2.5.6.21

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	(None)	pkiUser
Containment	(None)	pkiUser
Named By	(None)	pkiUser

Mandatory Attributes

pkiUser

(None)

Optional Attributes

pkiUser

userCertificate (page 715)

Default ACL Template

None

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Print Server

Represents NetWare print servers.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.18

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	Print Server
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

Print Server	Inherited from Top	Inherited from Server
(None)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

Print Server	
Operator (page 603) Printer (page 640)	SAP Name (page 667)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Server

Account Balance (page 363)	Public Key (page 648)
Allow Unlimited Credit (page 365)	Resource (page 658)
Description (page 410)	Security Equals (page 673)
Full Name (page 429)	Security Flags (page 674)
Host Device (page 447)	See Also (page 675)
L (Locality Name) (page 458)	Status (page 683)
Minimum Account Balance (page 534)	SvcInfo (page 903)
Network Address (page 556)	SvcType (page 904)
O (Organization Name) (page 595)	SvcTypeID (page 905)
OU (Organizational Unit Name) (page 606)	User (page 714)
Private Key (page 643)	Version (page 719)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)
[Root]	Read	[All Attributes]	Print Server

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The Print Server class differs from the NCP Server class in that print servers use the SPX protocol for communications rather than NCP.

The Print Server class is used for both bindery-based and NDS-based print servers. Bindery-based print servers do not have the Public Key or Private Key attributes. The Private Key and Public Key attributes are present if the server is a client of NDS Authentication Services.

The Version attribute (inherited from Server) indicates whether the server is bindery-based or NDS-based.

The Operator and User attributes are used by the print server as access control lists. (The User attribute is inherited from Server.) The Operator attribute identifies those individuals who are authorized to act as print server operators. The User attribute identifies individuals authorized to use the print server.

The Host Device attribute identifies the device that hosts the server. This is usually a computer, but it might be some other device. For example, a printer could host a built-in print server.

The Resource attribute contains a list of resources managed by this service.

The User attribute contains a list of objects that are authorized to use this server. The server must determine if the user list is to be maintained by an administrator, or if the list is automatically generated by the server. If the user list is used by the server as an access control list, the administrator will usually maintain the list. If the user list is purely informational, reflecting access control information stored elsewhere, the server usually maintains the list.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

Printer

Represents printers in the NDS tree. A printer object points to the queues to which it is attached.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.17

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Device (page 113)
	Device (page 113)	Printer
Containment	domain (page 119)	Device (page 113)
	Organization (page 202)	Device (page 113)
	Organizational Unit (page 215)	Device (page 113)

Rule	Class/Attribute	Defined For
Named By	CN (Common Name) (page 400)	Device (page 113)

Mandatory Attributes

Printer	Inherited from Top	Inherited from Device
(None)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

Printer	
Cartridge (page 395)	Page Description Language (page 610)
Default Queue (page 407)	Print Server (page 639)
Host Device (page 447)	Printer Configuration (page 641)
Memory (page 527)	Queue (page 651)
Network Address Restriction (page 558)	Status (page 683)
Notify (page 563)	Supported Typefaces (page 690)
Operator (page 603)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Device

Description (page 410)	See Also (page 675)
L (Locality Name) (page 458)	Serial Number (page 677)
Network Address (page 556)	SvcInfo (page 903)
O (Organization Name) (page 595)	SvcType (page 904)
OU (Organizational Unit Name) (page 606)	SvcTypeID (page 905)
Owner (page 608)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The Host Device attribute is used in this class to denote the computer (or other device) to which the printer is attached.

The Print Server attribute identifies the print servers that manage this device.

The Supported Typefaces and Page Description Language attributes are included to aid a search for printers with a particular set of capabilities. The contents of these attributes are statically maintained by an administrator, rather than being dynamically updated from printer feedback.

The Queue attribute identifies the associated queues through which this printer can be accessed.

The L (Locality Name) attribute can be used to identify the physical location of a device. For example, if the device were a printer, the locality might be "Building D, Section 24, by Ed Bender's desk."

The O (Organization Name) and the OU (Organizational Unit Name) attributes may already be present in the device's distinguished name. They are repeated here to aid searching when an organization spans multiple subtrees in the NDS tree. However, NDS does not automatically add these attributes

even though they may be present in the device's distinguished name. Additional values for the organization name or organizational unit name can be useful when a device is "co-owned" by multiple organizations.

Profile

Specifies a shared login configuration.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.19

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class	Defined For
Super Classes	Top (page 263)	Profile
Containment	domain (page 119)	Profile
	Organization (page 202)	Profile
	Organizational Unit (page 215)	Profile
Named By	CN (Common Name) (page 400)	Profile

Mandatory Attributes

Profile	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)
Login Script (page 511)	

Optional Attributes

Profile	
Description (page 410)	O (Organization Name) (page 595)
Full Name (page 429)	OU (Organizational Unit Name) (page 606)
L (Locality Name) (page 458)	See Also (page 675)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

A Profile class has an associated Login Script attribute that contains the bulk of the configuration information. This login script is executed after the container login script, but before the user login script. The login script for the Profile class should contain group-related conditional statements.

Queue

Represents batch processing queues available in the NetWare NCP environment.

Type: Effective

NDS Operational: Yes

ASN.1 ID

2.16.840.1.113719.1.1.6.1.20

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	Queue
Containment	domain (page 119)	Resource (page 245)
	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)

Rule	Class	Defined For
Named By	CN (Common Name) (page 400)	Resource (page 245)

Mandatory Attributes

Queue	Inherited from Top	Inherited from Resource
Queue Directory (page 652)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

Queue	
Device (page 414)	Server (page 678)
Host Server (page 449)	User (page 714)
Network Address (page 556)	Volume (page 720)
Operator (page 603)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Resource	
Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Root]	Read	[All Attributes]	Queue

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The Host Server attribute identifies the server that provides the Queue Management Service (QMS) for this queue. Requests to manipulate a particular resource must usually be directed to the host server.

The Queue Directory attribute names the subdirectory where this queue's files are stored. The Host Resource Name attribute is used when the host's local identification for a resource differs from the global resource identification. For example, a server might recognize SYS as the local name for a volume with the NDS name:

```
Project X.Engineering.Acme.US
```

The L (Locality Name), O (Organization Name), and OU (Organizational Unit Name) attributes are useful when a resource is used by multiple localities, organizations, or organizational units. If these attributes contain appropriate values, a search can be initiated for resources associated with a particular locality or organization.

The Network Address attribute acts as a cache for the server's network address. The user can contact the server without having to dereference the Host Server attribute.

The Operator, Server, and User attributes are used by QMS for Access Control Lists. The Operator attribute identifies the users or groups that have operator privileges. The Server attribute identifies print servers allowed to service this queue.

The User attribute contains a list of objects that are authorized to use this queue. The server that controls the resource must determine if the user list is maintained by an administrator or if the list is automatically generated by the server. If the user list is used by the server as an access control list, the administrator will usually maintain the list. If the user list is purely informational, reflecting access control information stored elsewhere, the server usually maintains the list.

Resource

Identifies logical resources available on the network.

Type: Noneffective

NDS Operational: Yes

LDAP Name

resource

ASN.1 ID

2.16.840.1.113719.1.1.6.1.21

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class	Defined For
Super Classes	Top (page 263)	Resource

Rule	Class	Defined For
Containment	domain (page 119)	Resource
	Organization (page 202)	Resource
	Organizational Unit (page 215)	Resource
Named By	CN (Common Name) (page 400)	Resource

Mandatory Attributes

Resource	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

Resource	
Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The Resource class differs from the Device class in that a device is a physical unit such as a printer, and a resource is some nonphysical, logical unit such as a queue or volume.

The Host Resource Name attribute is used when the host's local identification for a resource differs from the global resource identification. For example, a server might recognize SYS as the local name for a volume with the NDS name:

```
Project X.Engineering.Acme.US
```

The L (Locality Name), O (Organization Name), and OU (Organizational Unit Name) attributes are useful when a resource is used by multiple localities, organizations, or organizational units. If these attributes contain appropriate values, a search can be initiated for resources associated with a particular locality or organization.

SAS:Security

Identifies a container object that holds tree-wide security configuration information

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.39.6.1.1

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	SAS:Security
Containment	Top (page 263)	SAS:Security
	Tree Root (page 266)	SAS:Security
Named By	CN (Common Name) (page 400)	SAS:Security

Mandatory Attributes

SAS:Security	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

SAS:Security
NDSPKI:Tree CA DN (page 553)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

In NetWare 4.x, this object is placed under Top, which is the virtual tree root in NetWare 4.x.

In NetWare 5.x, this object is placed under the Tree Root object.

In a WAN environment, the replica containing this object needs to be widely distributed.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

SAS:Service

Identifies the NDSPKI:Key Material object (the server certificates) that belongs to the server.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.39.6.1.2

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	SAS:Service
Containment	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)
Named By	CN (Common Name) (page 400)	SAS:Service

Mandatory Attributes

SAS:Service	Inherited from Top	Inherited from Resource
	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

SAS:Service	
Host Server (page 449) NDSPKI:Key Material DN (page 539)	Private Key (page 643) Public Key (page 648)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Resource	
Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For each server on which SAS (Secure Authentication Services) is installed, a SAS:Service object exists for that server. This object contains attributes that identify the NCP Server object (the Host Server attribute) and the NDSPKI:Key Material object (the NDSPKI:Key Material DN attribute).

The Private Key and Public Key attributes are not currently used.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

Server

Identifies entities that manage one or more resources and provide access to those resources through a communications protocol.

Type: Noneffective

NDS Operational: Yes

LDAP Name

server

ASN.1 ID

2.16.840.1.113719.1.1.6.1.22

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Server
	Organization (page 202)	Server
	Organizational Unit (page 215)	Server
Named By	CN (Common Name) (page 400)	Server

Mandatory Attributes

Server	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

Server	
Account Balance (page 363)	Private Key (page 643)
Allow Unlimited Credit (page 365)	Public Key (page 648)
Description (page 410)	Resource (page 658)
Full Name (page 429)	Security Equals (page 673)
Host Device (page 447)	Security Flags (page 674)
L (Locality Name) (page 458)	See Also (page 675)
Minimum Account Balance (page 534)	Status (page 683)
Network Address (page 556)	Timezone (page 700)
O (Organization Name) (page 595)	User (page 714)
OU (Organizational Unit Name) (page 606)	Version (page 719)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server
[Self]	Supervisor	[Entry Rights]	Server

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The Host Device attribute identifies the device that hosts the server. This is usually a computer, but it might be some other device. For example, a printer could host a built-in print server.

The Private Key and Public Key attributes are present if the server is a client of NDS Authentication Services. The Resource attribute contains a list of resources managed by this service.

The User attribute contains a list of objects that are authorized to use this server. The server must determine if the user list is to be maintained by an administrator, or if the list is automatically generated by the server. If the user

list is used by the server as an access control list, the administrator will usually maintain the list. If the user list is purely informational, reflecting access control information stored elsewhere, the server usually maintains the list.

strongAuthenticationUser

Defines a Netscape user who can use certificates for authentication.

Type: Auxiliary Class

LDAP Name

strongAuthenticationUser

ASN.1 ID

2.5.6.15

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	(None)	strongAuthenticationUser
Containment	(None)	strongAuthenticationUser
Named By	(None)	strongAuthenticationUser

Mandatory Attributes

strongAuthenticationUser

(None)

Optional Attributes

strongAuthenticationUser

userCertificate (page 715)

Default ACL Template

None

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Template

Identifies an object that can be used to create User objects with default values for file system and NDS rights.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.64.6.1.1

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Template
Containment	Organization (page 202)	Template
	Organizational Unit (page 215)	Template
Named By	CN (Common Name) (page 400)	Template

Mandatory Attributes

Template	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

Template	
Account Balance (page 363)	New Object's FS Rights (page 560)
Allow Unlimited Credit (page 365)	New Object's Self Rights (page 561)
Description (page 410)	OU (Organizational Unit Name) (page 606)
EMail Address (page 419)	Password Allow Change (page 616)
Facsimile Telephone Number (page 427)	Password Expiration Interval (page 617)
Group Membership (page 436)	Password Expiration Time (page 618)
Higher Privileges (page 440)	Password Minimum Length (page 621)
Home Directory (page 441)	Password Required (page 622)
Home Directory Rights (page 443)	Password Unique Required (page 623)
L (Locality Name) (page 458)	Physical Delivery Office Name (page 627)
Language (page 460)	Postal Address (page 629)
Login Allowed Time Map (page 501)	Postal Code (page 631)
Login Disabled (page 502)	Postal Office Box (page 633)
Login Expiration Time (page 503)	Profile (page 645)
Login Grace Limit (page 504)	Run Setup Script (page 662)
Login Maximum Simultaneous (page 510)	S (State or Province Name) (page 663)
Login Script (page 511)	SA (Street Address) (page 665)
Mailbox ID (page 516)	Security Equals (page 673)
Mailbox Location (page 518)	Security Flags (page 674)
Member (page 525)	See Also (page 675)
Members Of Template (page 526)	Set Password After Create (page 681)
Message Server (page 529)	Setup Script (page 682)
Minimum Account Balance (page 534)	Telephone Number (page 696)
Network Address Restriction (page 558)	Title (page 701)
New Object's DS Rights (page 559)	Trustees Of New Object (page 704)
	Volume Space Restrictions (page 721)

Added to Template in NDS eDirectory 8.5

departmentNumber (page 409)	manager (page 519)
employeeType (page 422)	S (State or Province Name) (page 663)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Top

All classes are a subclass of Top. This class mandates that all objects contain an Object Class attribute. Although Top is an effective class, it is a special case in that no objects can be constructed from this class by the user.

Type: Effective

NDS Operational: Yes

LDAP Name

top

ASN.1 ID

2.5.6.0

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	On
Ambiguous Naming	On
Ambiguous Container	On
Auxiliary Class	Off

Class Structure

Rule	Class	Defined For
Super Classes	(none)	Top

Rule	Class	Defined For
Containment	(none)	Top
Named By	(none)	Top

Mandatory Attributes

Top
Object Class (page 601)

Optional Attributes

Top	
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424)	GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

This class mandates that all objects will contain an Object Class attribute, that all classes can have values for the attributes needed for authentication and synchronization, and that all objects are supervised by the object that created them. Although Top is an effective class, it is a special case in that the user cannot construct objects from this class.

In NetWare 4.x, the NDS tree root has a base class of Top. In NetWare 5.x, the NDS tree root can use a new object, Tree Root, as its base class.

The optional attributes, Authority Revocation, CA Private Key, CA Public Key, Certificate Revocation, and Cross Certificate Pair, were originally only in the Partition class. However, in the next NDS release, they were also added to the Top object class. Thus they are now redundant in the Partition class because they are inherited from Top.

The LDAP server maps requests for the top class to this class.

Tree Root

Defines the NDS tree's [Root] object.

Type: Effective

NDS Operational: Yes

LDAP Name

treeRoot

ASN.1 ID

2.16.840.1.113719.1.1.6.1.32

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class	Defined For
Super Classes	Top (page 263)	Tree Root
Containment	(Nothing)	Tree Root
Named By	T (Tree Name) (page 695)	Tree Root

Mandatory Attributes

Tree Root	Inherited from Top
T (Tree Name) (page 695)	Object Class (page 601)

Optional Attributes

Tree Root
SAS:Security DN (page 668)

Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

This class object is new to NetWare 5.x, and in NetWare 5.x, the Tree Root class is one of three objects that can be the topmost object in the tree. The Country class and Organization class are the other two.

NetWare 4.x supports either Country class or Organization class as the topmost object in the tree.

Unknown

Represents any object created by the server to restore an object whose base class is no longer defined by the schema.

Type: Effective

NDS Operational: Yes

LDAP Name

unknown

ASN.1 ID

2.16.840.1.113719.1.1.6.1.24

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class	Defined For
Super Classes	Top (page 263)	Unknown
Containment	(none)	Unknown and Top

Rule	Class	Defined For
Named By	(none)	Unknown and Top

Mandatory Attributes

Unknown	Inherited from Top
	Object Class (page 601)

Optional Attributes

Unknown
(Special)

Inherited from Top
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424) GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

Any attribute is valid as an optional attribute for this class. An NDS client cannot create objects of class Unknown.

The LDAP server maps requests for the unknown class to this class.

User

Represents users of network services.

Type: Effective

NDS Operational: Yes

LDAP Name

inetOrgPerson

ASN.1 ID

2.16.840.1.113730.3.2.2

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class	Defined For
Super Classes	Top (page 263)	ndsLoginProperties (page 159)
	ndsLoginProperties (page 159)	Person (page 222)
	Person (page 222)	Organizational Person (page 207)
	Organizational Person (page 207)	User
Containment	domain (page 119)	Organizational Person (page 207)
	Organization (page 202)	Organizational Person (page 207)
	Organizational Unit (page 215)	Organizational Person (page 207)
Named By	CN (Common Name) (page 400)	Organizational Person (page 207)
	uniqueID (page 707)	Organizational Person (page 207)
	OU (Organizational Unit Name) (page 606)	Organizational Person (page 207)

Mandatory Attributes

User	Inherited from Top	Inherited from ndsLoginProperties	Inherited from Person	Inherited from Organizational Person
(None)	Object Class (page 601)	(None)	CN (Common Name) (page 400) Surname (page 691)	(None)

Optional Attributes Defined for User

User

Account Balance (page 363)	Login Intruder Reset Time (page 509)
Allow Unlimited Credit (page 365)	Login Maximum Simultaneous (page 510)
audio (page 730)	Login Script (page 511)
businessCategory (page 389)	Login Time (page 513)
carLicense (page 731)	manager (page 519)
departmentNumber (page 409)	Message Server (page 529)
displayName (page 739)	Minimum Account Balance (page 534)
Employee ID (page 740)	mobile (page 535)
employeeType (page 422)	NDSPKI:Keystore (page 540)
Entrust:User (page 742)	NDSPKI:userCertificateInfo (page 555)
Group Membership (page 436)	Network Address (page 556)
Higher Privileges (page 440)	Network Address Restriction (page 558)
Home Directory (page 441)	NRD:Registry Data (page 847)
homePhone (page 444)	NRD:Registry Index (page 848)
homePostalAddress (page 445)	NSCP:employeeNumber (page 579)
Internet EMail Address (page 453)	NSCP:mailAccessDomain (page 849)
jpegPhoto (page 747)	NSCP:mailAlternateAddress (page 850)
labeledUri (page 748)	NSCP:mailAutoReplyMode (page 851)
Language (page 460)	NSCP:mailAutoReplyText (page 852)
Last Login Time (page 461)	NSCP:mailDeliveryOption (page 853)
LDAP:ARL (page 752)	NSCP:mailForwardingAddress (page 854)
LDAP:caCertificate (page 753)	NSCP:mailHost (page 855)
LDAP:CRL (page 754)	NSCP:mailMessageStore (page 856)
LDAP:crossCertificatePair (page 755)	NSCP:mailProgramDeliveryInfo (page 857)
ldapPhoto (page 750)	NSCP:mailQuota (page 858)
LDAPUserCertificate (page 751)	NSCP:nsLicenseEndTime (page 860)
Locked By Intruder (page 499)	NSCP:nsLicensedFor (page 861)
Login Allowed Time Map (page 501)	NSCP:nsLicenseStartTime (page 862)
Login Disabled (page 502)	pager (page 611)
Login Expiration Time (page 503)	Password Allow Change (page 616)
Login Grace Limit (page 504)	Password Expiration Interval (page 617)
Login Grace Remaining (page 505)	Password Expiration Time (page 618)
Login Intruder Address (page 506)	Password Minimum Length (page 621)
Login Intruder Attempts (page 507)	Password Required (page 622)

User	
Password Unique Required (page 623)	secretary (page 884)
Passwords Used (page 624)	Security Equals (page 673)
preferredLanguage (page 864)	Security Flags (page 674)
Print Job Configuration (page 638)	Server Holds (page 679)
Printer Control (page 642)	Timezone (page 700)
Private Key (page 643)	Type Creator Map (page 705)
Profile (page 645)	UID (User ID) (page 706)
Profile Membership (page 646)	userCertificate (page 715)
Public Key (page 648)	userSMIMECertificate (page 906)
roomNumber (page 661)	x500UniqueIdentifier (page 728)

Optional Attributes from inetOrgPerson*	
audio (page 730)	ldapPhoto (page 750)
businessCategory (page 389)	manager (page 519)
carLicense (page 731)	mobile (page 535)
departmentNumber (page 409)	NSCP:employeeNumber (page 579)
displayName (page 739)	pager (page 611)
employeeType (page 422)	preferredLanguage (page 864)
Given Name (page 434)	roomNumber (page 661)
homePhone (page 444)	secretary (page 884)
homePostalAddress (page 445)	uniqueID (page 707)
Initials (page 451)	userCertificate (page 715)
Internet EMail Address (page 453)	userSMIMECertificate (page 906)
jpegPhoto (page 747)	x500UniqueIdentifier (page 728)
labeledUri (page 748)	

*The inetOrgPerson attributes are not added automatically to User. See Remarks for more information.

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from ndsLoginProperties

Account Balance (page 363)	Minimum Account Balance (page 534)
Allow Unlimited Credit (page 365)	Network Address (page 556)
Group Membership (page 436)	Network Address Restriction (page 558)
Higher Privileges (page 440)	Password Allow Change (page 616)
Language (page 460)	Password Expiration Interval (page 617)
Last Login Time (page 461)	Password Expiration Time (page 618)
Locked By Intruder (page 499)	Password Minimum Length (page 621)
Login Allowed Time Map (page 501)	Password Required (page 622)
Login Disabled (page 502)	Password Unique Required (page 623)
Login Expiration Time (page 503)	Passwords Used (page 624)
Login Grace Limit (page 504)	Private Key (page 643)
Login Grace Remaining (page 505)	Profile (page 645)
Login Intruder Address (page 506)	Profile Membership (page 646)
Login Intruder Attempts (page 507)	Public Key (page 648)
Login Intruder Reset Time (page 509)	Security Equals (page 673)
Login Maximum Simultaneous (page 510)	Security Flags (page 674)
Login Script (page 511)	Server Holds (page 679)
Login Time (page 513)	Timezone (page 700)

Inherited from Person	
Description (page 410)	Initials (page 451)
Full Name (page 429)	See Also (page 675)
Generational Qualifier (page 431)	Telephone Number (page 696)
Given Name (page 434)	

Inherited from Organizational Person	
destinationIndicator (page 412)	Postal Code (page 631)
EEmail Address (page 419)	Postal Office Box (page 633)
Facsimile Telephone Number (page 427)	preferredDeliveryMethod (page 636)
internationaliSDNNumber (page 452)	registeredAddress (page 655)
L (Locality Name) (page 458)	S (State or Province Name) (page 663)
Mailbox ID (page 516)	SA (Street Address) (page 665)
Mailbox Location (page 518)	telexNumber (page 698)
OU (Organizational Unit Name) (page 606)	telexTerminalIdentifier (page 699)
Physical Delivery Office Name (page 627)	Title (page 701)
Postal Address (page 629)	x121Address (page 727)

Default ACL Template Defined for User

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top
[Public]	Read	Message Server	User
[Root]	Browse	[Entry Rights]	User
[Root]	Read	Group Membership	User
[Root]	Read	Network Address	User
[Self]	Read	[All Attributes]	User
[Self]	Read/Write	Login Script	User
[Self]	Read/Write	Print Job Configuration	User

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

The User class includes both clients and service providers. In this context, Private Key stores the object's private key encrypted by the object's password.

The X.500 standard defines two subclasses of person: Organizational Person and Residential Person. The operational schema does not defined Residential Person, but the division of Person from Organizational Person has been maintained for compatibility with X.500. An optional file, rperson.sch, that ships with the LDAP server defines the Residential Person class. See residentialPerson (page 332) for more information.

In NDS 8, the User class inherits the NDS login attributes from the ndsLoginProperties class. For backwards compatibility with previous versions of NDS, they have not been removed from the list of optional attributes in the User class definition.

The LDAP server ships with an optional nov_inet.sch file. This file, when executed, adds inetOrgPerson attributes to the User class. To add these attributes and allow LDAP applications to use the User class as if it were the inetOrgPerson class, run nwconfig.nlm and run the sys:system\schema\nov_inet.sch file. For information on setting up a separate class for inetOrgPerson, see inetOrgPerson (page 291).

userSecurityInformation

Defines a list of supported algorithms.

Type: Auxiliary Class

LDAP Name

userSecurityInformation

ASN.1 ID

2.5.6.18

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	On

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	(None)	userSecurityInformation
Containment	(None)	userSecurityInformation
Named By	(None)	userSecurityInformation

Mandatory Attributes

userSecurityInformation
(None)

Optional Attributes

userSecurityInformation
supportedAlgorithms (page 685)

Default ACL Template

None

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For more information, see RFC 2256.

Volume

Represents NetWare file system volumes.

Type: Effective

NDS Operational: Yes

LDAP Name

volume

ASN.1 ID

2.16.840.1.113719.1.1.6.1.26

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	On
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	Volume

Rule	Class/Attribute	Defined For
Containment	domain (page 119)	Resource (page 245)
	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)
Named By	CN (Common Name) (page 400)	Resource (page 245)

Mandatory Attributes

Volume	Inherited from Top	Inherited from Resource
Host Server (page 449)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

Volume
Status (page 683)

Inherited from Top
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424) GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Inherited from Resource	
Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Root]	Read	Host Resource Name	Volume
[Root]	Read	Host Server	Volume

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

This subclass exists primarily to allow Volume objects to be distinguished from other types of Resource objects.

The Host Server attribute identifies the server that owns and services the resource. Requests to manipulate a particular resource must usually be directed to the host server.

When present, the Host Resource Name attribute is used to contain the local volume name that corresponds to the volume name on the server. This attribute is used when the host's local identification for a resource differs from the global resource identification. For example, a server might recognize SYS as the local name for a volume with the NDS name:

```
Project X.Engineering.Acme.US
```

If the attribute is not present, the local volume name "SYS" can be assumed.

The L (Locality Name), O (Organization Name), and OU (Organizational Unit Name) attributes are useful when a resource is used by multiple localities,

organizations, or organizational units. If these attributes contain appropriate values, a search can be initiated for resources associated with a particular locality or organization.

WANMAN:LAN Area

Contains the information on the group of servers that belong to a LAN area and can share a single WAN policy.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.38.6.1.4

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	WANMAN:LAN Area
Containment	Country (page 105)	WANMAN:LAN Area
	Locality (page 141)	WANMAN:LAN Area
	Organization (page 202)	WANMAN:LAN Area
	Organizational Unit (page 215)	WANMAN:LAN Area
Named By	CN (Common Name) (page 400)	WANMAN:LAN Area

Mandatory Attributes

WANMAN:LAN Area	Inherited from Top
CN (Common Name) (page 400)	Object Class (page 601)

Optional Attributes

WANMAN:LAN Area	
Description (page 410)	Owner (page 608)
L (Locality Name) (page 458)	See Also (page 675)
Member (page 525)	WANMAN:Cost (page 722)
O (Organization Name) (page 595)	WANMAN:Default Cost (page 724)
OU (Organizational Unit Name) (page 606)	WANMAN:WAN Policy (page 726)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

The Member attribute contains the list of servers that belong to the LAN area.

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

3

Novell Object Class Extensions

This chapter lists alphabetically the object classes that extend the schema in a default NetWare installation and from the downloadable schema files (<http://www.novell.com/products/nds/schema/index.html>). It also includes the obsolete object classes that have extended the schema in NDS releases previous to NDS eDirectory 8.5.

- ♦ For object class definitions that are in the base schema, see “Base Object Class Definitions” on page 73.
- ♦ For an explanation of the types of information included about each class, see “Reading Class Definitions” on page 32.

Entrust:CRLDistributionPoint

Replaced by cRLDistributionPoint (page 108).

inetOrgPerson

Defines an LDAP user-type object that can log in to the network and contains information about that object's relationship to an organization.

Type: Effective

LDAP Name

inetOrgPerson

ASN.1 ID

2.16.840.1.113730.3.2.2

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	ndsLoginProperties (page 159)
	ndsLoginProperties (page 159)	Person (page 222)
	Person (page 222)	Organizational Person (page 207)
	Organizational Person (page 207)	inetOrgPerson
Containment	Country (page 105)	inetOrgPerson
	domain (page 119)	inetOrgPerson
	Locality (page 141)	inetOrgPerson
	Organization (page 202)	inetOrgPerson
	Organizational Unit (page 215)	inetOrgPerson
Named By	CN (Common Name) (page 400)	inetOrgPerson
	uniqueID (page 707)	inetOrgPerson
	Given Name (page 434)	inetOrgPerson
	Internet EMail Address (page 453)	inetOrgPerson
	Surname (page 691)	inetOrgPerson

Mandatory Attributes

inetOrgPerson	Inherited from Top	Inherited from Person
	Object Class (page 601)	CN (Common Name) (page 400)
		Surname (page 691)

Optional Attributes

inetOrgPerson

audio (page 730)	ldapPhoto (page 750)
businessCategory (page 389)	manager (page 519)
carLicense (page 731)	mobile (page 535)
departmentNumber (page 409)	NSCP:employeeNumber (page 579)
displayName (page 739)	pager (page 611)
employeeType (page 422)	preferredLanguage (page 864)
Given Name (page 434)	roomNumber (page 661)
homePhone (page 444)	secretary (page 884)
homePostalAddress (page 445)	uniqueID (page 707)
Initials (page 451)	userCertificate (page 715)
Internet EMail Address (page 453)	userSMIMECertificate (page 906)
jpegPhoto (page 747)	x500UniqueIdentifier (page 728)
labeledUri (page 748)	

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from ndsLoginProperties

Account Balance (page 363)	Minimum Account Balance (page 534)
Allow Unlimited Credit (page 365)	Network Address (page 556)
Group Membership (page 436)	Network Address Restriction (page 558)
Higher Privileges (page 440)	Password Allow Change (page 616)
Language (page 460)	Password Expiration Interval (page 617)
Last Login Time (page 461)	Password Expiration Time (page 618)
Locked By Intruder (page 499)	Password Minimum Length (page 621)
Login Allowed Time Map (page 501)	Password Required (page 622)
Login Disabled (page 502)	Password Unique Required (page 623)
Login Expiration Time (page 503)	Passwords Used (page 624)
Login Grace Limit (page 504)	Private Key (page 643)
Login Grace Remaining (page 505)	Profile (page 645)
Login Intruder Address (page 506)	Profile Membership (page 646)
Login Intruder Attempts (page 507)	Public Key (page 648)
Login Intruder Reset Time (page 509)	Security Equals (page 673)
Login Maximum Simultaneous (page 510)	Security Flags (page 674)
Login Script (page 511)	Server Holds (page 679)
Login Time (page 513)	Timezone (page 700)

Inherited from Person

Description (page 410)	Initials (page 451)
Full Name (page 429)	See Also (page 675)
Generational Qualifier (page 431)	Telephone Number (page 696)
Given Name (page 434)	

Inherited from Organizational Person	
destinationIndicator (page 412)	Postal Code (page 631)
EMail Address (page 419)	Postal Office Box (page 633)
Facsimile Telephone Number (page 427)	preferredDeliveryMethod (page 636)
internationaliSDNNumber (page 452)	registeredAddress (page 655)
L (Locality Name) (page 458)	S (State or Province Name) (page 663)
Mailbox ID (page 516)	SA (Street Address) (page 665)
Mailbox Location (page 518)	telexNumber (page 698)
OU (Organizational Unit Name) (page 606)	telexTerminalIdentifier (page 699)
Physical Delivery Office Name (page 627)	Title (page 701)
Postal Address (page 629)	x121Address (page 727)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

This is an optional class that is not installed automatically with the LDAP server. The administrator must use the nwconfig.nlm to execute the Sys:system\schema\iperson.sch file. The administrator must then manually reconfigure the LDAP mapping so that inetOrgPerson points to inetOrgPerson rather than to User.

The User class can be modified to support the additional inetOrgPerson attributes by executing the nov_inet.sch file. This is the preferred method because most NDS applications assume that User is the only class used to define user-type objects in the tree.

NDPS Broker

Defines the NDPS broker that manages the following three services:
service registry, notification, and resource management

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	NDPS Broker
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

NDPS Broker	Inherited from Top	Inherited from Server
	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

NDPS Broker	
Delivery Methods Installed (page 738)	Registry Advertising Name (page 874)
GW API Gateway Directory Path (page 743)	Registry Service Enabled (page 875)
GW API Gateway Directory Volume (page 744)	Registry Srv Net Addr (page 876)
MHS Send Directory Path (page 763)	Registry Srv Net Address (page 877)
MHS Send Directory Volume (page 764)	Resource Mgmt Service Enabled (page 881)
NDPS SMTP Server (page 781)	Resource Mgmt Srv Net Addr (page 879)
Notification Service Enabled (page 844)	Resource Mgmt Srv Net Address (page 880)
Notification Srv Net Addr (page 845)	Resource Mgr Database Path (page 882)
Notification Srv Net Address (page 846)	Resource Mgr Database Volume (page 883)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Server

Account Balance (page 363)	Private Key (page 643)
Allow Unlimited Credit (page 365)	Public Key (page 648)
Description (page 410)	Resource (page 658)
Full Name (page 429)	Security Equals (page 673)
Host Device (page 447)	Security Flags (page 674)
L (Locality Name) (page 458)	See Also (page 675)
masvAuthorizedRange (page 520)	Status (page 683)
masvDefaultRange (page 521)	SvcInfo (page 903)
masvProposedLabel (page 524)	SvcType (page 904)
Minimum Account Balance (page 534)	SvcTypeID (page 905)
Network Address (page 556)	User (page 714)
O (Organization Name) (page 595)	Version (page 719)
OU (Organizational Unit Name) (page 606)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDPS Manager

Defines the NDPS Printer Manager that manages NDPS Printer objects and Public Access Printer Agents.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	NDPS Manager
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

NDPS Manager	Inherited from Top	Inherited from Server
Database Dir Path (page 734)	Object Class (page 601)	CN (Common Name) (page 400)
Database Volume Name (page 735)		
NDPS Manager Status (page 773)		

Optional Attributes

NDPS Manager	
Datapool Locations (page 737)	NDPS Database Saved Timestamp (page 767)
NDPS Database Saved Data Image (page 768)	Printer to PA ID Mappings (page 872)
NDPS Database Saved Index Image (page 769)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Server

Account Balance (page 363)	Private Key (page 643)
Allow Unlimited Credit (page 365)	Public Key (page 648)
Description (page 410)	Resource (page 658)
Full Name (page 429)	Security Equals (page 673)
Host Device (page 447)	Security Flags (page 674)
L (Locality Name) (page 458)	See Also (page 675)
masvAuthorizedRange (page 520)	Status (page 683)
masvDefaultRange (page 521)	SvcInfo (page 903)
masvProposedLabel (page 524)	SvcType (page 904)
Minimum Account Balance (page 534)	SvcTypeID (page 905)
Network Address (page 556)	User (page 714)
O (Organization Name) (page 595)	Version (page 719)
OU (Organizational Unit Name) (page 606)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDPS Printer

Identifies a printer agent and contains information about a printer's features and about the printer's authorized users and operators.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Device (page 113)
	Device (page 113)	NDPS Printer
Containment	domain (page 119)	Device (page 113)
	Organization (page 202)	Device (page 113)
	Organizational Unit (page 215)	Device (page 113)
Named By	CN (Common Name) (page 400)	Device (page 113)

Mandatory Attributes

NDPS Printer	Inherited from Top	Inherited from Device (Class)
NDPS Operator Role (page 774)	Object Class (page 601)	CN (Common Name) (page 400)
NDPS User Role (page 782)		
Printer Status (page 871)		

Optional Attributes

NDPS Printer	
Client Install Candidate (page 732)	NDPS Printer Queue List (page 777)
Color Supported (page 733)	NDPS Printer Siblings (page 778)
Datapool Location (page 736)	Notification Consumers (page 842)
Host Device (page 447)	Notification Profile (page 843)
IPP URI (page 745)	Page Description Languages (page 863)
IPP URI Security Scheme (page 746)	Primary Notification Service (page 865)
Maximum Speed (page 761)	Primary Resource Service (page 866)
Maximum Speed Units (page 762)	Printer Agent Name (page 867)
NDPS Accountant Role (page 765)	Printer Manufacturer (page 868)
NDPS Job Configuration (page 772)	

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Device (Class)

Description (page 410)	See Also (page 675)
L (Locality Name) (page 458)	Serial Number (page 677)
Network Address (page 556)	SvcInfo (page 903)
O (Organization Name) (page 595)	SvcType (page 904)
OU (Organizational Unit Name) (page 606)	SvcTypeID (page 905)
Owner (page 608)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDSCat:Catalog

Identifies logical attributes that catalog objects may need.

Type: Noneffective

ASN.1 ID

2.16.840.1.113719.1.1.6.1.42

Class Flags

Class Flags	Setting
Container	Off
Effective	Off
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	NDSCat: Catalog
Containment	domain (page 119)	Resource (page 245)
	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)
Named By	CN (Common Name) (page 400)	Resource (page 245)

Mandatory Attributes

NDSCat: Catalog	Inherited from Top	Inherited from Resource
	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

NDSCat: Catalog	
NDSCat:Actual All Attributes (page 783)	NDSCat:Actual Scope (page 792)
NDSCat:Actual Attribute Count (page 784)	NDSCat:Actual Search Aliases (page 793)
NDSCat:Actual Attributes (page 785)	NDSCat:Actual Start Time (page 794)
NDSCat:Actual Base Object (page 786)	NDSCat:Actual Value Count (page 795)
NDSCat:Actual Catalog Size (page 787)	NDSCat:AttrDefTbl (page 797)
NDSCat:Actual End Time (page 788)	NDSCat:CatalogDB (page 801)
NDSCat:Actual Filter (page 789)	NDSCat:IndexDefTbl (page 805)
NDSCat:Actual Object Count (page 790)	NDSCat:Label (page 807)
NDSCat:Actual Return Code (page 791)	NDSCat:Log (page 808)
Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Resource	
Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDSCat:Master Catalog

Defines and contains the catalog database.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.1.6.1.43

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	NDSCat:Catalog (page 305)
	NDSCat:Catalog (page 305)	NDSCat: Master Catalog
Containment	domain (page 119)	Resource (page 245)
	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)
Named By	CN (Common Name) (page 400)	Resource (page 245)

Mandatory Attributes

NDSCat: Master Catalog	Inherited from Top	Inherited from Resource	Inherited from NDSCat: Catalog
	Object Class (page 601)	CN (Common Name) (page 400)	

Optional Attributes

NDSCat: Master Catalog	
Host Server (page 449)	NDSCat:Max Retries (page 811)
NDSCat:All Attributes (page 796)	NDSCat:Retry Interval (page 813)
NDSCat:Attributes (page 798)	NDSCat:Scope (page 814)
NDSCat:Auto Dredge (page 799)	NDSCat:Search Aliases (page 815)
NDSCat:Base Object (page 800)	NDSCat:Slave Catalog List (page 816)
NDSCat:Dredge Interval (page 803)	NDSCat:Start Time (page 817)
NDSCat:Filter (page 804)	Private Key (page 643)
NDSCat:Indexes (page 806)	Public Key (page 648)
NDSCat:Max Log Size (page 810)	Security Equals (page 673)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Resource

Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Inherited from NDSCat: Catalog

NDSCat:Actual All Attributes (page 783)	NDSCat:Actual Scope (page 792)
NDSCat:Actual Attribute Count (page 784)	NDSCat:Actual Search Aliases (page 793)
NDSCat:Actual Attributes (page 785)	NDSCat:Actual Start Time (page 794)
NDSCat:Actual Base Object (page 786)	NDSCat:Actual Value Count (page 795)
NDSCat:Actual Catalog Size (page 787)	NDSCat:AttrDefTbl (page 797)
NDSCat:Actual End Time (page 788)	NDSCat:CatalogDB (page 801)
NDSCat:Actual Filter (page 789)	NDSCat:IndexDefTbl (page 805)
NDSCat:Actual Object Count (page 790)	NDSCat:Label (page 807)
NDSCat:Actual Return Code (page 791)	NDSCat:Log (page 808)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NDSCat:Slave Catalog

Identifies a catalog that is an exact duplicate of a master catalog object.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.1.6.1.44

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	NDSCat:Catalog (page 305)
	NDSCat:Catalog (page 305)	NDSCat:Slave Catalog
Containment	domain (page 119)	Resource (page 245)
	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)
Named By	CN (Common Name) (page 400)	Resource (page 245)

Mandatory Attributes

NDSCat:Slave Catalog	Inherited from Top	Inherited from Resource	Inherited from NDSCat: Catalog
(None)	Object Class (page 601)	CN (Common Name) (page 400)	(None)

Optional Attributes

NDSCat:Slave Catalog	
NDSCat:Master Catalog (page 809)	
Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	
Inherited from Resource	
Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Inherited from NDSCat: Catalog	
NDSCat:Actual All Attributes (page 783)	NDSCat:Actual Scope (page 792)
NDSCat:Actual Attribute Count (page 784)	NDSCat:Actual Search Aliases (page 793)
NDSCat:Actual Attributes (page 785)	NDSCat:Actual Start Time (page 794)
NDSCat:Actual Base Object (page 786)	NDSCat:Actual Value Count (page 795)
NDSCat:Actual Catalog Size (page 787)	NDSCat:AttrDefTbl (page 797)
NDSCat:Actual End Time (page 788)	NDSCat:CatalogDB (page 801)
NDSCat:Actual Filter (page 789)	NDSCat:IndexDefTbl (page 805)
NDSCat:Actual Object Count (page 790)	NDSCat:Label (page 807)
NDSCat:Actual Return Code (page 791)	NDSCat:Log (page 808)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

After the dredger writes the master catalog, it writes any specified slave catalogs. Slave catalogs can be placed in any location in the NDS tree, including in local partitions for performance enhancements. If a new dredge results in an updated master catalog, each slave catalog is also replaced with an updated copy.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NetSvc

Identifies a Winsock 2 service on the network.

Type: Effective

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	NetSvc
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

NetSvc	Inherited from Top	Inherited from Server
SvcTypeID (page 905)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

NetSvc
(None)

Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Inherited from Server

Account Balance (page 363)	Public Key (page 648)
Allow Unlimited Credit (page 365)	Resource (page 658)
Description (page 410)	Security Equals (page 673)
Full Name (page 429)	Security Flags (page 674)
Host Device (page 447)	See Also (page 675)
L (Locality Name) (page 458)	SvcInfo (page 903)
Minimum Account Balance (page 534)	SvcType (page 904)
Network Address (page 556)	SvcTypeID (page 905)
O (Organization Name) (page 595)	Status (page 683)
OU (Organizational Unit Name) (page 606)	User (page 714)
Private Key (page 643)	Version (page 719)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

NLS:License Certificate

Stores the license certificate, usage information, and the object that is assigned to it.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.51.6.1.1

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NLS:License Certificate
Containment	NLS:Product Container (page 323)	NLS:License Certificate
Named By	NLS:License ID (page 826)	NLS:License Certificate

Mandatory Attributes

NLS:License Certificate	Inherited from Top
NLS:Common Certificate (page 819)	Object Class (page 601)
NLS:License ID (page 826)	
NLS:Revision (page 834)	

Optional Attributes

NLS:License Certificate
NLS:Owner (page 829)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NLS:License Server

Stores the configuration information for the licensing server (the nlsfsp nlm).

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.51.6.1.2

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Server (page 254)
	Server (page 254)	NLS:License Server
Containment	domain (page 119)	Server (page 254)
	Organization (page 202)	Server (page 254)
	Organizational Unit (page 215)	Server (page 254)
Named By	CN (Common Name) (page 400)	Server (page 254)

Mandatory Attributes

NLS:License Server	Inherited from Top	Inherited from Server
Host Server (page 449)	Object Class (page 601)	CN (Common Name) (page 400)
NLS:License Database (page 825)		
NLS:LSP Revision (page 828)		
NLS:Transaction Database (page 838)		

Optional Attributes

NLS:License Server	
NLS:Transaction Log Name (page 839)	NLS:Search Type (page 835)
NLS:Transaction Log Size (page 840)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from Server

Account Balance (page 363)	Public Key (page 648)
Allow Unlimited Credit (page 365)	Resource (page 658)
Description (page 410)	Security Equals (page 673)
Full Name (page 429)	Security Flags (page 674)
Host Device (page 447)	See Also (page 675)
L (Locality Name) (page 458)	Status (page 683)
Minimum Account Balance (page 534)	SvcInfo (page 903)
Network Address (page 556)	SvcType (page 904)
O (Organization Name) (page 595)	SvcTypeID (page 905)
OU (Organizational Unit Name) (page 606)	User (page 714)
Private Key (page 643)	Version (page 719)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)
[Public]	Read	Network Address	Server (page 254)
[Self]	Supervisor	[Entry Rights]	Server (page 254)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NLS:Product Container

Contains the license certificates for a product and stores the summary information.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.51.6.1.0

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	NLS:Product Container
Containment	Organization (page 202)	NLS:Product Container
	Organizational Unit (page 215)	NLS:Product Container
Named By	NLS:Publisher (page 833)	NLS:Product Container
	NLS:Product (page 832)	NLS:Product Container
	NLS:Version (page 841)	NLS:Product Container

Mandatory Attributes

NLS:Product Container	Inherited from Top
NLS:Product (page 832)	Object Class (page 601)
NLS:Publisher (page 833)	
NLS:Revision (page 834)	
NLS:Version (page 841)	

Optional Attributes

NLS:Product Container	
NLS:Current Installed (page 820)	NLS:Peak Installed Data (page 830)
NLS:Current Peak Installed (page 821)	NLS:Peak Used Data (page 831)
NLS:Current Peak Used (page 822)	NLS:Summary Update Time (page 836)
NLS:Current Used (page 823)	NLS:Summary Version (page 837)
NLS:Hourly Data Size (page 824)	

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NSCP:groupOfUniqueNames5

Represents a Netscape group.

Type: Effective

ASN.1 ID

2.5.6.17

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	List (page 138)
	List (page 138)	NSCP:mailGroup5 (page 329)
	NSCP:mailGroup5 (page 329)	NSCP:groupOfUniqueNames5
Containment	Country (page 105)	NSCP:mailGroup5 (page 329)
	Locality (page 141)	NSCP:mailGroup5 (page 329)
	Organization (page 202)	NSCP:mailGroup5 (page 329)
	Organizational Unit (page 215)	NSCP:mailGroup5 (page 329)

Rule	Class/Attribute	Defined For
Named By	CN (Common Name) (page 400)	List (page 138)

Mandatory Attributes

NSCP:groupOfUniqueNames5	Inherited from Top	Inherited from List	Inherited from NSCP:mailGroup5
(None)	Object Class (page 601)	CN (Common Name) (page 400)	(None)

Optional Attributes

NSCP:groupOfUniqueNames5
(None)

Inherited from Top
ACL (page 364) Audit:File Link (page 372) Authority Revocation (page 378) Back Link (page 382) Bindery Property (page 384) CA Private Key (page 393) CA Public Key (page 394) Certificate Revocation (page 397) Certificate Validity Interval (page 398) Cross Certificate Pair (page 405) Equivalent To Me (page 424) GUID (page 437) Last Referenced Time (page 462) masvAuthorizedRange (page 520) masvDefaultRange (page 521) masvProposedLabel (page 524) Obituary (page 597) Other GUID (page 604) Reference (page 654) Revision (page 659) Used By (page 712)

Inherited from List

Description (page 410)	Member (page 525)
E-Mail Address (page 419)	O (Organization Name) (page 595)
Full Name (page 429)	OU (Organizational Unit Name) (page 606)
L (Locality Name) (page 458)	Owner (page 608)
Mailbox ID (page 516)	See Also (page 675)
Mailbox Location (page 518)	

Inherited from NSCP:mailGroup5

Internet E-Mail Address (page 453)	NSCP:mailHost (page 855)
NSCP:mailAlternateAddress (page 850)	NSCP:mgrpRFC822mailmember (page 583)
NSCP:mailForwardingAddress (page 854)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

LDAP added this class for Netscape support. For more information, see the documentation for the Netscape schema.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

NSCP:mailGroup5

Replaced by NSCP:mailGroup1 (page 183).

NSCP:Nginfo

Replaced by NSCP:nginfo3 (page 197).

NSCP:Nginfo2

Replaced by NSCP:nginfo3 (page 197).

residentialPerson

Defines an LDAP user-type object that can log in to the network and contains residential information about the object.

Type: Effective

LDAP Name

residentialPerson

ASN.1 ID

2.5.6.10

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	ndsLoginProperties (page 159)
	ndsLoginProperties (page 159)	Person (page 222)
	Person (page 222)	residentialPerson

Rule	Class/Attribute	Defined For
Containment	Country (page 105)	residentialPerson
	domain (page 119)	residentialPerson
	Locality (page 141)	residentialPerson
	Organization (page 202)	residentialPerson
	Organizational Unit (page 215)	residentialPerson
Named By	CN (Common Name) (page 400)	residentialPerson
	uniqueID (page 707)	residentialPerson
	Given Name (page 434)	residentialPerson
	Surname (page 691)	residentialPerson

Mandatory Attributes

residentialPerson	Inherited from Top
	Object Class (page 601)

Optional Attributes

residentialPerson	
businessCategory (page 389)	Postal Office Box (page 633)
destinationIndicator (page 412)	preferredDeliveryMethod (page 636)
Facsimile Telephone Number (page 427)	registeredAddress (page 655)
internationaliSDNNumber (page 452)	S (State or Province Name) (page 663)
L (Locality Name) (page 458)	SA (Street Address) (page 665)
Physical Delivery Office Name (page 627)	Telephone Number (page 696)
Postal Address (page 629)	telexNumber (page 698)
Postal Code (page 631)	telexTerminalIdentifier (page 699)
	x121Address (page 727)

Inherited from Top

ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Inherited from ndsLoginProperties

Account Balance (page 363)	Minimum Account Balance (page 534)
Allow Unlimited Credit (page 365)	Network Address (page 556)
Group Membership (page 436)	Network Address Restriction (page 558)
Higher Privileges (page 440)	Password Allow Change (page 616)
Language (page 460)	Password Expiration Interval (page 617)
Last Login Time (page 461)	Password Expiration Time (page 618)
Locked By Intruder (page 499)	Password Minimum Length (page 621)
Login Allowed Time Map (page 501)	Password Required (page 622)
Login Disabled (page 502)	Password Unique Required (page 623)
Login Expiration Time (page 503)	Passwords Used (page 624)
Login Grace Limit (page 504)	Private Key (page 643)
Login Grace Remaining (page 505)	Profile (page 645)
Login Intruder Address (page 506)	Profile Membership (page 646)
Login Intruder Attempts (page 507)	Public Key (page 648)
Login Intruder Reset Time (page 509)	Security Equals (page 673)
Login Maximum Simultaneous (page 510)	Security Flags (page 674)
Login Script (page 511)	Server Holds (page 679)
Login Time (page 513)	Timezone (page 700)

Inherited from Person	
Description (page 410)	Initials (page 451)
Full Name (page 429)	See Also (page 675)
Generational Qualifier (page 431)	Telephone Number (page 696)
Given Name (page 434)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

This is an optional class that is not installed automatically with the LDAP server. The administrator must use the `nwconfig.nlm` to execute the `Sys:system\schema\person.sch` file. The administrator must then manually refresh the LDAP configuration (at the console prompt type “ldap refresh immediate”).

SLP Scope Unit

Contains SLP Service objects.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.49.6.1.1

Class Flags

Class Flags	Setting
Container	On
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	SLP Scope Unit
Containment	Country (page 105)	SLP Scope Unit
	Locality (page 141)	SLP Scope Unit
	Organization (page 202)	SLP Scope Unit
	Organizational Unit (page 215)	SLP Scope Unit
Named By	SU (page 902)	SLP Scope Unit

Mandatory Attributes

SLP Scope Unit	Inherited from Top
(None)	Object Class (page 601)

Optional Attributes

SLP Scope Unit	
SLP Scope Name (page 892)	SLP SU Type (page 897)
SLP SU Back Link (page 896)	SU (page 902)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

Each SLP Scope Unit object contains all the SLP Service objects found in the specified scope or scopes. Since this is an NDS container object, it can become its own partition, which can then be replicated and placed on a server in any location that needs service information.

In a typical NDS tree, the tree will contain one SLP Scope Unit object with a scope for each geographically separate location (for example, a scope for Provo and a scope for San Jose). The object would be its own partition, and a replica would be placed in Provo and San Jose. The San Jose replica would keep the San Jose services current, and NDS would synchronize the For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.information with the Provo replica. The Provo services would be kept current in Provo, and then synchronized with San Jose. Both sites would contain a complete list of available services.

The SLP Directory Agent object obtains its rights to SLP Service objects from the SLP Scope Unit object. The SLP Scope Unit object should have Read, Write, and Browse rights to the objects in its container. The agent control over the SLP Service objects can be restricted by modifying these rights.

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

SLP Directory Agent

Identifies an instance of a Directory Agent and contains its configuration information.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.49.6.1.2

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	SLP Directory Agent
Containment	Country (page 105)	SLP Directory Agent
	Locality (page 141)	SLP Directory Agent
	Organization (page 202)	SLP Directory Agent
	Organizational Unit (page 215)	SLP Directory Agent
Named By	CN (Common Name) (page 400)	SLP Directory Agent

Mandatory Attributes

SLP Directory Agent	Inherited from Top
(None)	Object Class (page 601)

Optional Attributes

SLP Directory Agent	
CN (Common Name) (page 400)	SLP DA Back Link (page 888)
Private Key (page 643)	SLP Scope Unit DN (page 893)
Public Key (page 648)	SLP Start Purge Hour (page 894)
SLP Cache Limit (page 887)	SLP Status (page 895)

Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

Each NetWare server that is running a SLP Directory Agent requires an SLP Directory Agent object.

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

SLP Service

Identifies a network resource that has registered its service with the SLP Directory Agent.

Type: Effective

ASN.1 ID

2.16.840.1.113719.1.49.6.1.3

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	SLP Service
Containment	SLP Scope Unit (page 336)	SLP Service
Named By	CN (Common Name) (page 400)	SLP Service

Mandatory Attributes

SLP Service	Inherited from Top
SLP Language (page 890)	Object Class (page 601)
SLP Lifetime (page 891)	
SLP Type (page 898)	
SLP URL (page 899)	

Optional Attributes

SLP Directory Agent
CN (Common Name) (page 400)
SLP Attribute (page 886)

Inherited from Top
ACL (page 364)
Audit:File Link (page 372)
Authority Revocation (page 378)
Back Link (page 382)
Bindery Property (page 384)
CA Private Key (page 393)
CA Public Key (page 394)
Certificate Revocation (page 397)
Certificate Validity Interval (page 398)
Cross Certificate Pair (page 405)
Equivalent To Me (page 424)
GUID (page 437)
Last Referenced Time (page 462)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvProposedLabel (page 524)
Obituary (page 597)
Other GUID (page 604)
Reference (page 654)
Revision (page 659)
Used By (page 712)

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

When the SLP Directory Agent first loads, it creates an SLP Service object for each network service that it discovers. The object contains all the information a client requires to discover the service type and its location.

Network administrators can create SLP Service objects with a snap-in to the NetWare Administrator utility. This is the recommended procedure, for with the utility, the administrator can place the SLP Service objects in an SLP Scope Unit container.

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

SMS SMDR Class

Provides for SMDR (Storage Management Data Requester) location and name resolution on an IP network.

Type: Effective

Class Flags

Class Flags	Setting
Container	Off
Effective	On
Nonremovable	Off
Ambiguous Naming	Off
Ambiguous Container	Off
Auxiliary Class	Off

Class Structure

Rule	Class/Attribute	Defined For
Super Classes	Top (page 263)	Resource (page 245)
	Resource (page 245)	SMS SMDR Class
Containment	domain (page 119)	Resource (page 245)
	Organization (page 202)	Resource (page 245)
	Organizational Unit (page 215)	Resource (page 245)
Named By	CN (Common Name) (page 400)	Resource (page 245)

Mandatory Attributes

SMS SMDR Class	Inherited from Top	Inherited from Resource
(None)	Object Class (page 601)	CN (Common Name) (page 400)

Optional Attributes

SMS SMDR Class	
SAP Name (page 667)	Status (page 683)
SMS Protocol Address (page 900)	Version (page 719)
SMS Registered Service (page 901)	
Inherited from Top	
ACL (page 364)	GUID (page 437)
Audit:File Link (page 372)	Last Referenced Time (page 462)
Authority Revocation (page 378)	masvAuthorizedRange (page 520)
Back Link (page 382)	masvDefaultRange (page 521)
Bindery Property (page 384)	masvProposedLabel (page 524)
CA Private Key (page 393)	Obituary (page 597)
CA Public Key (page 394)	Other GUID (page 604)
Certificate Revocation (page 397)	Reference (page 654)
Certificate Validity Interval (page 398)	Revision (page 659)
Cross Certificate Pair (page 405)	Used By (page 712)
Equivalent To Me (page 424)	
Inherited from Resource	
Description (page 410)	OU (Organizational Unit Name) (page 606)
Host Resource Name (page 448)	See Also (page 675)
L (Locality Name) (page 458)	Uses (page 718)
O (Organization Name) (page 595)	

Default ACL Template

Object Name	Default Rights	Affected Attributes	Class Defined For
[Creator]	Supervisor	[Entry Rights]	Top (page 263)

Remarks

When the SMDR starts the first time, it creates an SMS SMDR Class object in its configured context. This object is made a member of a container object which holds objects for all the SMDRs on the network. When an SMDR receives a service or name resolution request on an IP network, the SMDR uses the objects in the container to find the requested information.

For help in understanding the class definition template, see “Reading Class Definitions” on page 32.

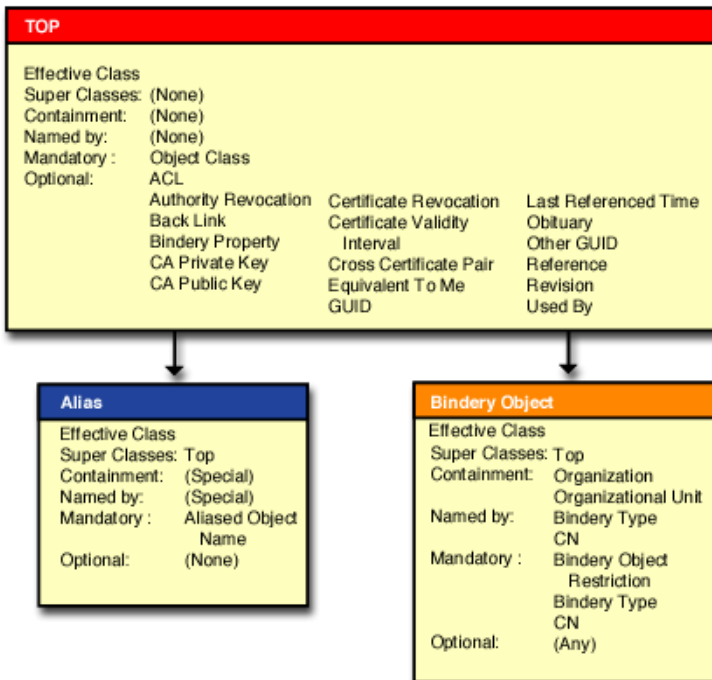
For LDAP clients to access this class, the LDAP server must map this class to a name that contains no spaces or colons.

4

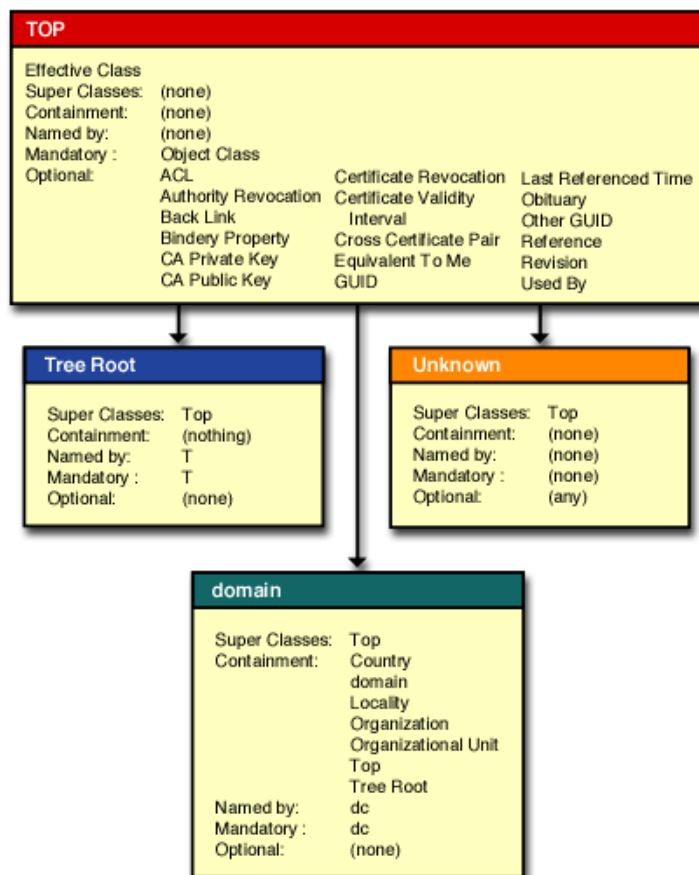
Graphical View of Object Class Inheritance

This section displays the inheritance for all the object classes defined in the NDS operational schema. For more information, see “Graphical View Explanation” on page 46.

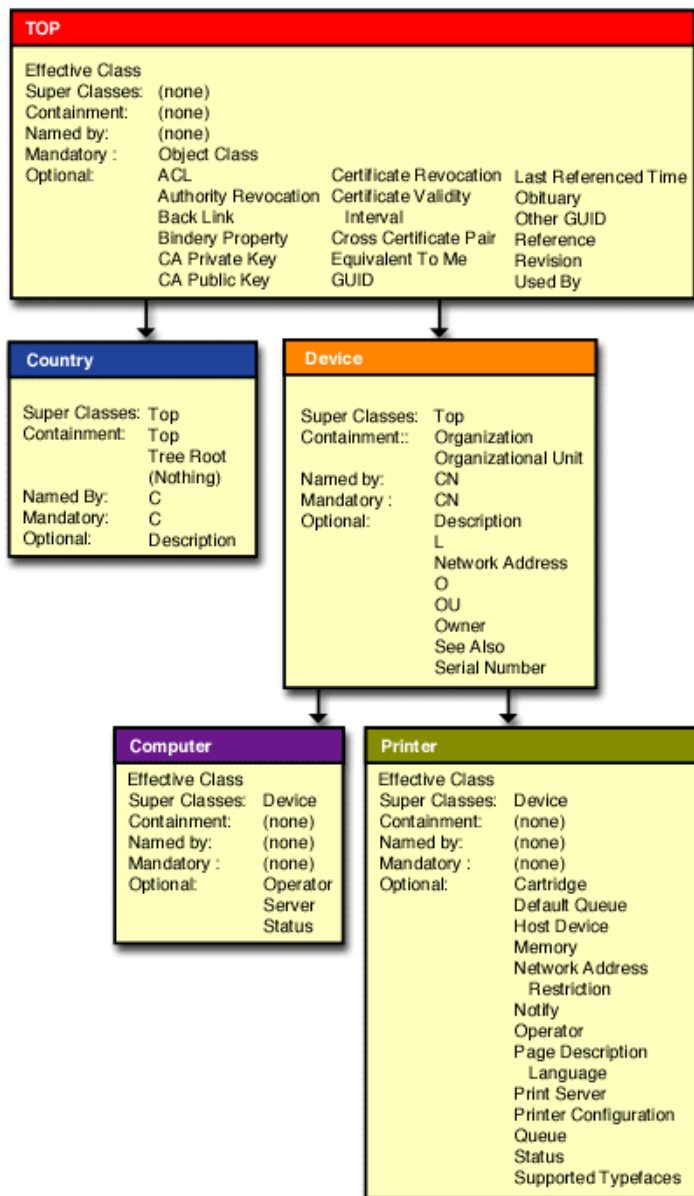
Alias and Bindery Object Classes



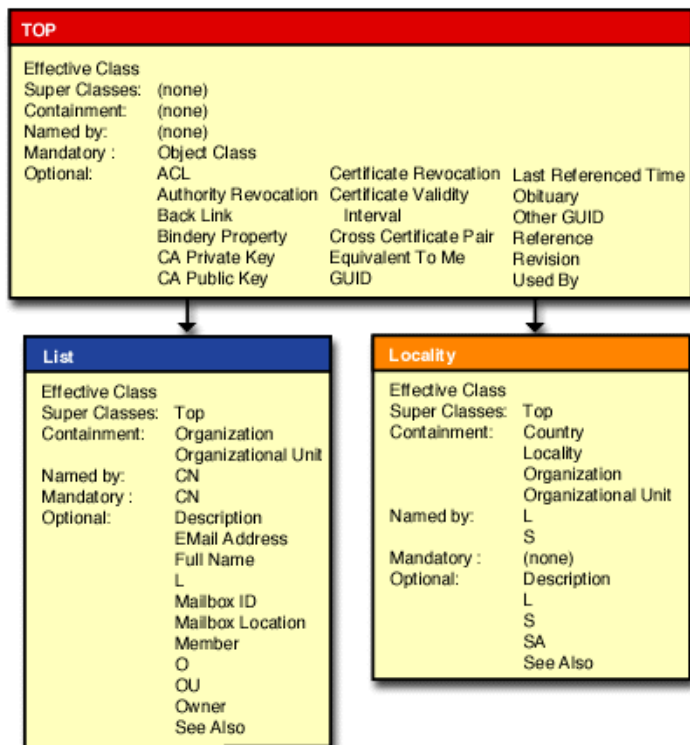
Tree Root, domain, and Unknown



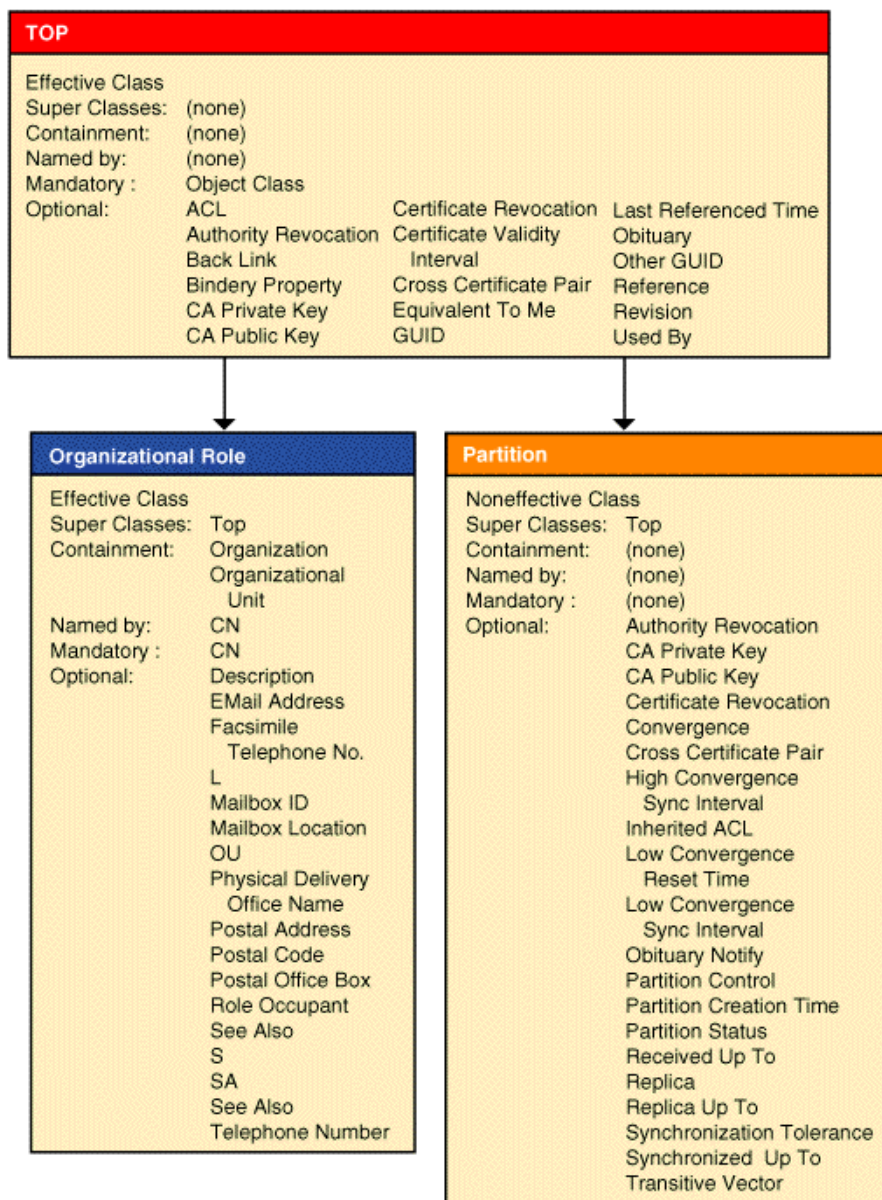
Computer, Country, Device, and Printer



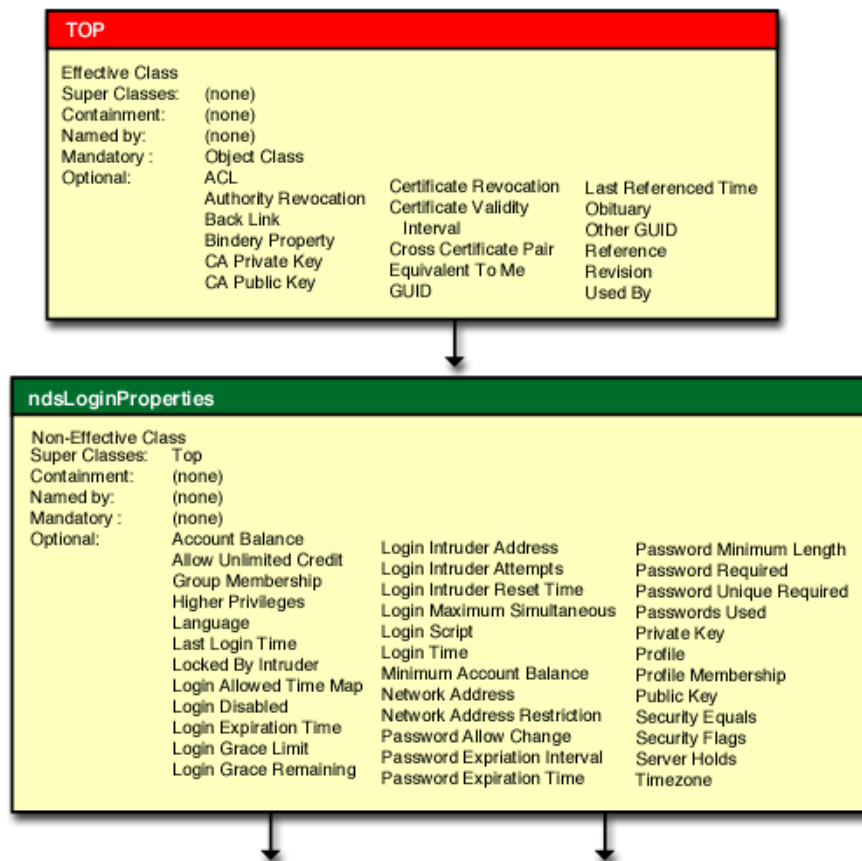
List and Locality



Organizational Role and Partition



ndsLoginProperties, Organization, and Organizational Unit



Organization	
Effective Class	
Super Classes:	Top
Containment:	Country
	Locality
	Top
	Tree Root
	(Nothing)
Named by:	O
Mandatory :	O
Optional:	Description
	Detect Intruder
	Email Address
	Facsimile
	Telephone No.
	Intruder Attempt
	Reset Interval
	Intruder Lockout
	Reset Interval
	L
	Lockout After
	Detection
	Login Intruder
	Limit
	Login Script
	Mailbox ID
	Mailbox Location
	NNS Domain
	Physical Delivery
	Office Name
	Postal Address
	Postal Code
	Postal Office Box
	Print Job
	Configuration
	Printer Control
	S
	SA
	See Also
	Telephone Number

Organizational Unit	
Effective Class	
Super Classes:	Top
Containment:	Locality
	Organization
	Organizational Unit
Named by:	OU
Mandatory :	OU
Optional:	Description
	Detect Intruder
	Email Address
	Facsimile
	Telephone No.
	Intruder Attempt
	Reset Interval
	Intruder Lockout
	Reset Interval
	L
	Lockout After
	Detection
	Login Intruder
	Limit
	Login Script
	Mailbox ID
	Mailbox Location
	NNS Domain
	Physical Delivery
	Office Name
	Postal Address
	Postal Code
	Postal Office Box
	Print Job
	Configuration
	Printer Control
	S
	SA
	See Also
	Telephone Number

ndsLoginProperties, Person, Organizational Person, and User

TOP			
Effective Class			
Super Classes: (none)			
Containment: (none)			
Named by: (none)			
Mandatory : Object Class			
Optional:			
	ACL	Certificate Revocation	Last Referenced Time
	Authority Revocation	Certificate Validity	Obituary
	Back Link	Interval	Other GUID
	Bindery Property	Cross Certificate Pair	Reference
	CA Private Key	Equivalent To Me	Revision
	CA Public Key	GUID	Used By



ndsLoginProperties			
Non-Effective Class			
Super Classes: Top			
Containment: (none)			
Named by: (none)			
Mandatory : (none)			
Optional:			
	Account Balance	Login Intruder Address	Password Minimum Length
	Allow Unlimited Credit	Login Intruder Attempts	Password Required
	Group Membership	Login Intruder Reset Time	Password Unique Required
	Higher Privileges	Login Maximum Simultaneous	Passwords Used
	Language	Login Script	Private Key
	Last Login Time	Login Time	Profile
	Locked By Intruder	Minimum Account Balance	Profile Membership
	Login Allowed Time Map	Network Address	Public Key
	Login Disabled	Network Address Restriction	Security Equals
	Login Expiration Time	Password Allow Change	Security Flags
	Login Grace Limit	Password Expiration Interval	Server Holds
	Login Grace Remaining	Password Expiration Time	Timezone



Person			
Effective Class			
Super Classes: Top			
Containment: (none)			
Named by: (none)			
Mandatory : CN			
Optional:			
	Surname	Given Name	See Also
	Description	Initials	Telephone Number
	Full Name		
	Generational Qualifier		

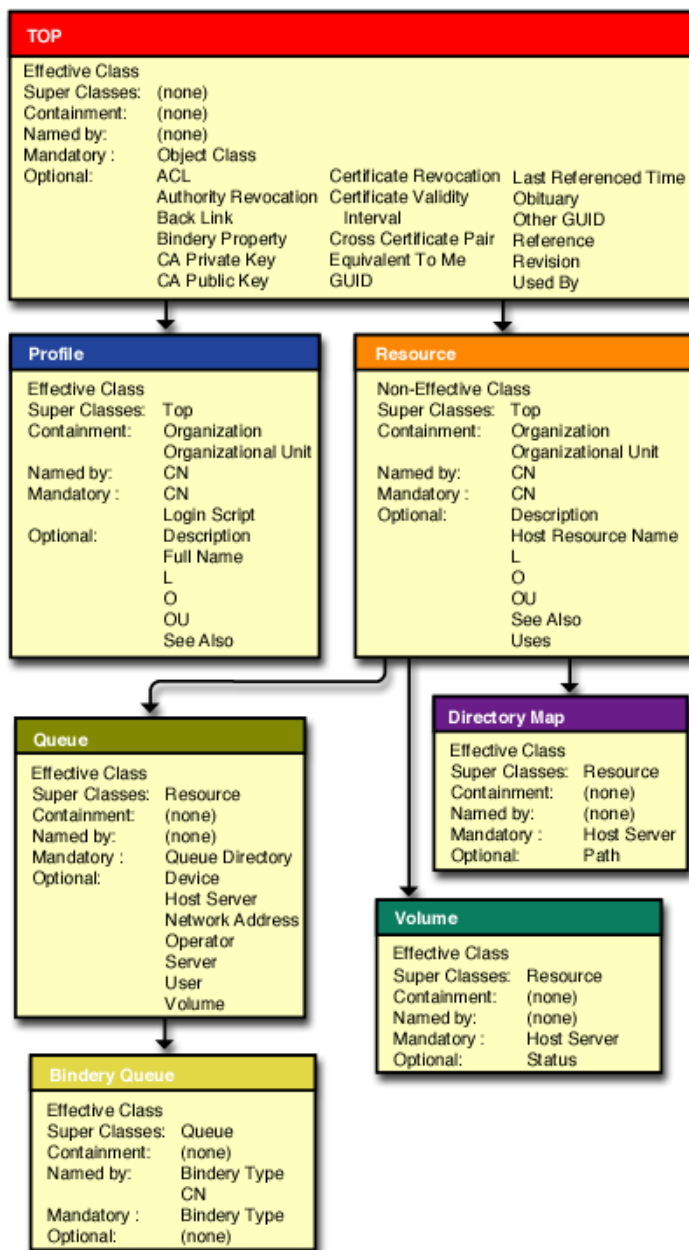


Organizational Person			
Effective Class			
Super Classes:	Person		
Containment:	Organization Organizational Unit		
Named by:	CN OU		
Mandatory :	(none)		
Optional:	E Mail Address	OU	Postal Office Box
	Facsimile Telephone No.	Physical Delivery	S
	L	Office Name	SA
	Mailbox ID	Postal Address	Title
	Mailbox Location	Postal Code	

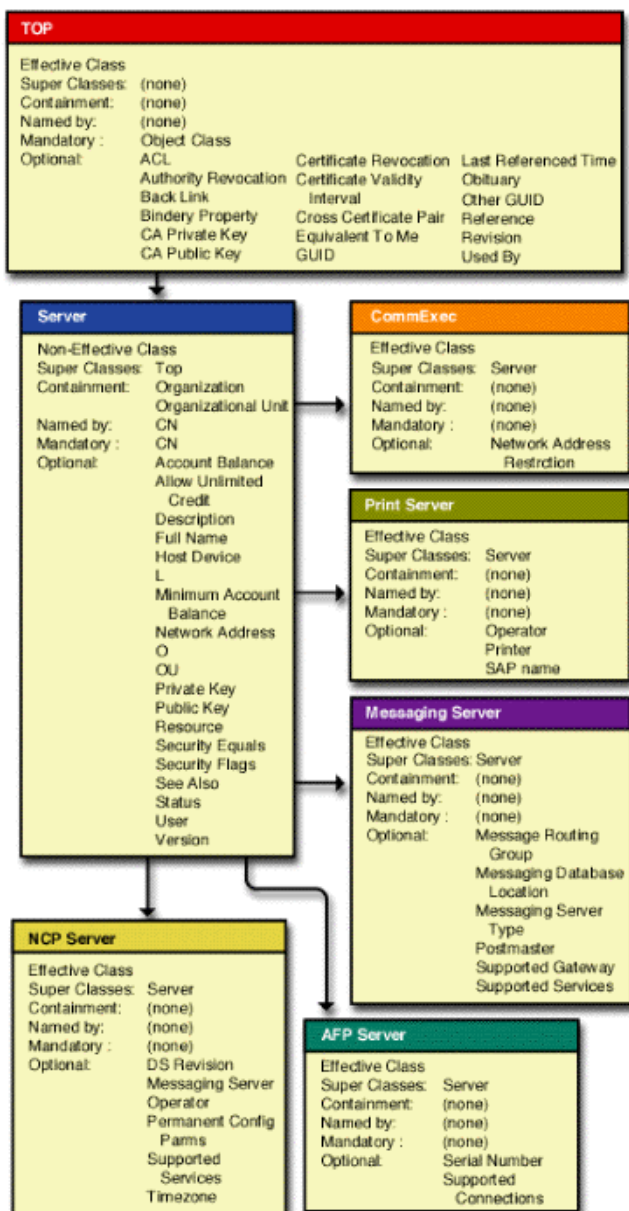


User			
Effective Class			
Super Classes:	Organizational Person		
Containment:	(none)		
Named by:	(none)		
Mandatory :	(none)		
Optional:	Account Balance	Login Intruder Attempts	Password Unique Required
	Allow Unlimited Credit	Login Intruder Reset Time	Passwords Used
	Group Membership	Login Maximum Simultaneous	Print Job Configuration
	Higher Privileges	Login Script	Printer Control
	Home Directory	Login Time	Private Key
	Language	Message Server	Profile
	Last Login Time	Minimum Account Balance	Profile Membership
	Locked By Intruder	Network Address	Public Key
	Login Allowed Time Map	Network Address Restriction	Security Equals
	Login Disabled	Password Allow Change	Security Flags
	Login Expiration Time	Password Expiration Interval	Server Holds
	Login Grace Limit	Password Expiration Time	Type Creator Map
	Login Grace Remaining	Password Minimum Length	UID
	Login Intruder Address	Password Required	

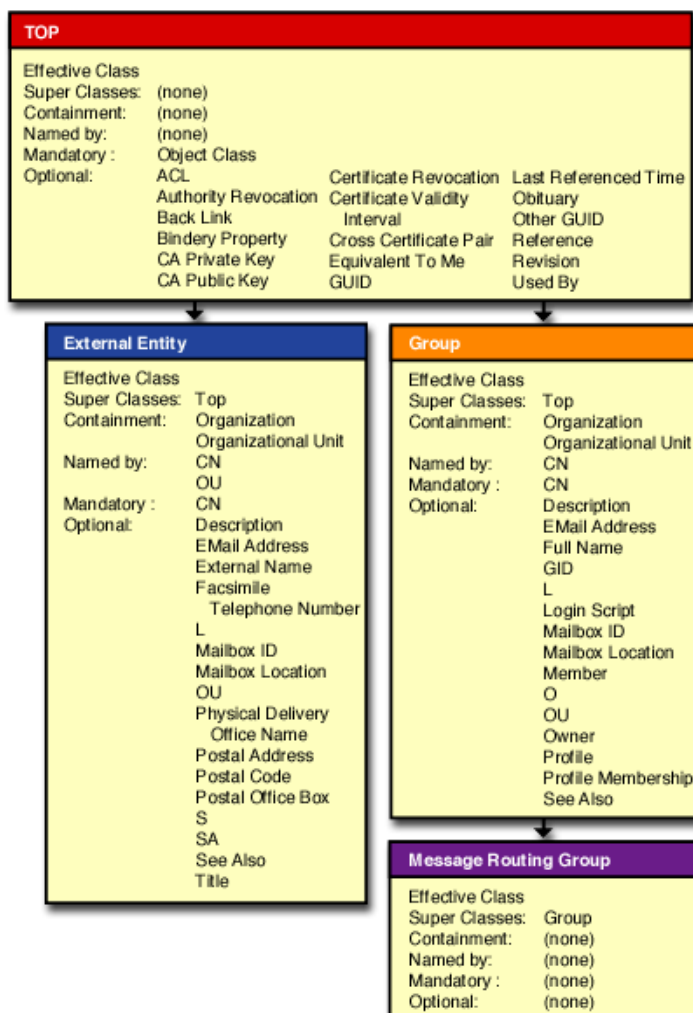
Directory Map, Profile, Queues, Resource, and Volume



Servers (AFP, Messaging, NCP, Print) and CommExec



External Entity, Group, and Message Routing Group



5

Base Attribute Definitions

This chapter lists alphabetically all the attribute definitions that NDS installs with NDS eDirectory 8.5 on all platforms : NetWare, NT, Solaris, and Linux..

- ♦ For the attribute definitions which are added during a default NetWare installation or with the downloadable schema files, see “Novell Attribute Extensions” on page 729.
- ♦ For LDAP operational attributes, see “LDAP Operational Attributes” on page 907.
- ♦ For an explanation of the types of information included about each attribute, see “Reading NDS Attribute Type Definitions” on page 63.

Aliased Object Name

Assigned to alias objects in the NDS tree. The aliased object is the object to which the alias points.

NDS Operational: Yes

LDAP Name

aliasedObjectName

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.1

Used In

Alias (page 77)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For Alias class objects, this attribute is mandatory.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Account Balance

Specifies the amount of credit (or money) the user has to spend on the purchase of network services. If the server has accounting rules activated, services are refused to the user when the user's account balance drops below a specified minimum balance.

NDS Operational: Yes

Syntax

Counter (page 936)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.1

Used In

Server (page 254)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Counter syntax.

ACL

Contains access control information for the object and its attributes.

NDS Operational: Yes

Syntax

Object ACL (page 957)

Constraints

DS_NONREMOVABLE_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.2

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The LDAP server does not currently support the Object ACL syntax. LDAP clients cannot access this attribute.

Allow Unlimited Credit

Allows the user to access and use all network services he or she has rights to, without maintaining a minimum account balance.

NDS Operational: Yes

Syntax

Boolean (page 922)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.4

Used In

Server (page 254)
User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

associatedName

Holds the distinguished name of the entry's associated object.

LDAP Name

associatedName

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.38

Used In

domain (page 119)

attributeCertificate

Specifies an attribute certificate.

LDAP Name

attributeCertificate

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.58

Used In

pkiCA (page 226)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used by Certificate Authorities (Novell and third party).

Audit:A Encryption Key

Not currently used.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.1.0

Used In

Audit:File Object (page 86)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Audit:B Encryption Key

Not currently used.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.2.0

Used In

Audit:File Object (page 86)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Audit:Contents

Contains no useful data.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.3.0

Used In

Audit:File Object (page 86)

Remarks

NDS rights to this attribute determine the rights to the physical audit file. For example, the Write right to the attribute allows the submitting of audit records; the Read right allows the reading of audit records.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Audit:Current Encryption Key

Not currently used.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.5.0

Used In

Audit:File Object (page 86)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Audit:File Link

Contains the link to the Audit:File Object object associated with this audited object.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.6.0

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Audit:Link List

Contains a list of objects that can submit audit records.

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.7.0

Used In

Audit:File Object (page 86)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Audit:Path

Specifies the volume on which an audit file is stored.

Syntax

Path (page 966)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.8.0

Used In

Audit:File Object (page 86)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Audit:Policy

Stores all the policy related information for an audited object.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.9.0

Used In

Audit:File Object (page 86)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Audit:Type

Specifies the type of audit records an Audit:File Object is currently accepting.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.12.4.4.0

Used In

Audit:File Object (page 86)

Remarks

Audit:File Objects can accept the following types of audit records:

Value	Type
0	Volume
1	Container
2	External

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

authorityRevocationList

Specifies an authentication revocation list for certificate management.

LDAP Name

authorityRevocationList
authorityRevocationList;binary

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.38

Used In

certificationAuthority (page 96)
cRLDistributionPoint (page 108)
pkiCA (page 226)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used by Certificate Authorities (Novell and third party).

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Authority Revocation

Contains a time-stamped list of revoked public keys of all Certificate Authorities known and certified by the Certificate Authority.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_READ_ONLY_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.5

Used In

Partition (page 219)

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the octet string syntax.

AuxClass Object Class Backup

Allows auxiliary class information to be stored on an NDS server not running NDS 8.

NDS Operational: Yes

Syntax

Class Name (page 934)

Constraints

DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.507

Used In

(Special)

Remarks

The name has a maximum size of 32 characters.

When an NDS 8 server must send object information that includes auxiliary class attributes to a server running an earlier version of NDS, the NDS 8 server copies the values in the Object Class attribute to the AuxClass Object Class Backup attribute and changes the object's class to Unknown.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

For help in understanding the attribute definition template, see "Reading NDS Attribute Type Definitions" on page 63.

Auxiliary Class Flag

Indicates that the class is an auxiliary class.

NDS Operational: Yes

Syntax

Unknown (page 989)

Constraints

DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.503

Used In

(Special)

Remarks

When the schema definition for an auxiliary class is sent to an NDS server that is running a version of NDS previous to NDS 8, the NDS 8 server adds this attribute to the auxiliary class, removes the auxiliary class flag, and then sends the definition to the server. The NDS server running a version of NDS previous to NDS 8 can then store the definition as a standard class definition.

If an NDS 8 server receives a class definition with this attribute, it knows that the class is an auxiliary class and adds the definition to its schema as an auxiliary class.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Back Link

In NetWare 4.x, attached to any object for which an external reference is required by a remote server.

NDS Operational: Yes

Syntax

Back Link (page 920)

Constraints

DS_NONREMOVABLE_ATTR
DS_READ_ONLY_ATTR
DS_SERVER_READ
DS_SPARSE_REQUIRED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.6

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute contains the set of servers that store an external reference to an associated object. The attribute is used to notify such servers of changes in the status of the object. For NetWare 5.x, see the Used By attribute.

LDAP clients cannot access this attribute because the LDAP server does not currently support the Back Link syntax.

Bindery Object Restriction

Contains a single-valued integer attribute used by bindery objects. It consists of an error code that indicates the reason the bindery object cannot be represented as an NDS object.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.7

Used In

Bindery Object (page 89)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For the Bindery Object class, this attribute is mandatory.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Bindery Property

Emulates bindery properties that cannot be represented with other attribute types.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_READ_ONLY_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.8

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In the bindery, properties of any name and data structure can be attached to objects. This is not the case with NDS. Bindery Property attributes hold the information stored in bindery properties through the bindery API and the Bindery Services.

Any object can have a bindery property through inheritance from Top, but only Users, Groups, Queues, Print Servers, Profiles, and Bindery Objects can be accessed through bindery service calls. Bindery Property attributes can only be added through bindery service calls. So even though it may be legal in the schema, in practice, no other objects will ever have bindery properties.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Bindery Restriction Level

Restricts the rights granted to objects that are security equivalent to the bindery SUPERVISOR.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR

DS_OPERATIONAL

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.179

Used In

(None)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is new in NetWare 5.x and must be added to the class definition of Organization or Organizational Unit. Once added to the class, the attribute can be stored on a container that is part of the bindery context path of a server.

If the attribute is present and the value is nonzero, objects which are equivalent to the server object are restricted to the rights the objects obtain through normal NDS rights inheritance. If this attribute is set to zero or not present, objects which are security equivalent the server object and to the bindery SUPERVISOR are granted all rights to the server object.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Bindery Type

Associates a bindery object type with an object of class Bindery Object.

NDS Operational: Yes

Syntax

Numeric String (page 955)

Constraints

DS_NONREMOVABLE_ATTR

DS_READ_ONLY_ATTR

DS_SINGLE_VALUED_ATTR

DS_STRING_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.9

Used In

Bindery Object (page 89)

Bindery Queue (page 92)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For the Bindery Object and Bindery Queue classes, this attribute is mandatory.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

businessCategory

Specifies the kind of business performed by an organization.

LDAP Name

businessCategory

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (1..128)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.15

Used In

inetOrgPerson (page 291)

NSCP:groupOfCertificates (page 180)

Organizational Person (page 207)

residentialPerson (page 332)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extensions to return the NDS attribute constraints.

C (Country Name)

Specifies the name of a country.

NDS Operational: Yes

LDAP Name

c
countryName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (2, 2)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.6

Used In

Country (page 105)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For the Country class, this attribute is mandatory.

When used as a component of an NDS name, a country name identifies the country in which the named object is either physically located or associated

with in some other important way. An attribute value for country name is a string chosen from ISO 3166, which is a list of 2-character country abbreviations for all the countries in the world.

A non-extended LDAP schema query returns only the single-valued constraint.

cACertificate

Specifies a certificate authority certificate for certificate management.

LDAP Name

cACertificate
cACertificate;binary

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.37

Used In

certificationAuthority (page 96)
cRLDistributionPoint (page 108)
pkiCA (page 226)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used by Certificate Authorities (Novell and third party).

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

CA Private Key

Contains the Certificate Authority private key.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_HIDDEN_ATTR
DS_NONREMOVABLE_ATTR
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.11

Used In

Partition (page 219)
Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Applications cannot access this attribute since it is a hidden attribute.

This attribute signs public keys that are produced for objects. The attribute contains the private key encrypted with the Certificate Authority's password.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

CA Public Key

Contains the Certificate Authority public key.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_PUBLIC_READ

DS_READ_ONLY_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.12

Used In

Partition (page 219)

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute verifies public keys that are produced for objects subordinate to the Certificate Authority. The attribute contains the public key along with the certificate information.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Cartridge

Contains a list of font cartridges present on the printer.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.10

Used In

Printer (page 234)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

certificateRevocationList

Specifies a certificate revocation list for certificate management.

LDAP Name

certificateRevocationList
certificateRevocationList;binary

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.39

Used In

certificationAuthority (page 96)
cRLDistributionPoint (page 108)
pkiCA (page 226)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used by Certificate Authorities (Novell and third party).

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Certificate Revocation

Contains a time-stamped list of all public keys revoked by the Certificate Authority.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_READ_ONLY_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.13

Used In

Partition (page 219)

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Certificate Validity Interval

Specifies the amount of time that a certificate is valid.

NDS Operational: Yes

Syntax

Interval (page 951)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (60..-1)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.123

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The length of time is specified when the certificate is made.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

crossCertificatePair

Specifies a cross certificate pair for certificate management.

LDAP Name

crossCertificatePair
crossCertificatePair;binary

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.40

Used In

certificationAuthority (page 96)
cRLDistributionPoint (page 108)
pkiCA (page 226)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used by Certificate Authorities (Novell and third party).

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

CN (Common Name)

Specifies an identifier of an object. A common name is not a complete NDS name; it is a name by which an object is commonly known in a particular context, such as within an organization.

NDS Operational: Yes

LDAP Name

cn
commonName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (1..64)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.3

Used In

applicationEntity (page 80)
applicationProcess (page 83)
Bindery Object (page 89)
Bindery Queue (page 92)
cRLDistributionPoint (page 108)
Entrust:CRLDistributionPoint (page 290)
External Entity (page 125)
Group (page 128)

inetOrgPerson (page 291)
LDAP Server (page 135)
List (page 138)
NSCP:groupOfCertificates (page 180)
Organizational Role (page 212)
NSCP:Nginfo (page 330)
NSCP:Nginfo2 (page 331)
Person (page 222)
Profile (page 238)
Resource (page 245)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For all the object classes that this attribute is defined for, this attribute is mandatory. Through inheritance, this attribute becomes mandatory for all classes subordinated to these classes.

The user, typically the administrator, who creates the object chooses the value for this attribute, subject to the guidelines the organization defines for naming new objects. By default, an object does not have rights to rename itself; thus, an object has no control over its own name. A common name is not necessarily unique in an NDS tree and can admit ambiguity within its limited scope.

Common names are usually derived from the object's complete name. For example, a typical complete name for a person in an English-speaking country may comprise a personal title (Mr., Ms., Dr., Professor, Sir, Lord, and so on), a first name, middle names, a last name, a generational qualifier (Jr., Sr., and so on) and decorations and awards. This type of person would have a complete name that looks like the following:

Mr. Robin Lachlan Smith BSc (Hons) CEng MIEE

Often the first initial of the first name and the last name are used to create the person's common name. Using this convention, Mr. Robin Smith might have the common name of rsmith.

Common names are leaf object names, are usually short, and conform to the naming conventions of the country or culture with which they are associated.

The User object class has attributes for the components of the complete name. See the following attributes: Full Name, Generational Qualifier, Given Name, Initials, Surname, Title.

Variant names are associated with a named object as separate and alternative attribute values. Common variant names should be made available (for example, the use of a middle name as a preferred first name or the use of “Bill” in place of “William”).

An LDAP schema query requires extensions to return the NDS attribute constraints.

Convergence

Indicates how persistent a partition should be in attempting to keep its replicas up to date.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0,1)
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.15

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The possible values of this attribute and their meanings are:

0	Low	Do not propagate updates as they come in. Rather, wait for a periodic synchronization to update any replicas. Synchronizations can be done at a low frequency to save resources. Partitions with low convergence and pending updates should be synchronized at least once every 24 hours. See the Low Convergence Sync Interval and Low Convergence Reset Time attributes for details on setting the convergence cycle.
---	-----	---

1	High	Make one attempt to propagate an update to all replicas when it comes in. If this fails, schedule a synchronization for the next interval time.
---	------	---

High is the default for this attribute.

Cross Certificate Pair

Contains a pair of public keys that allow public key verification to circumvent the normal certification hierarchy. This provides a shorter certification path.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.17

Used In

Partition (page 219)
Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

dc

Contains the domain name for an object.

NDS Operational: Yes

LDAP Name

dc

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED (1..64)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.25

Used In

domain (page 119)
dcObject (page 111)

Remarks

The attribute holds one component of a domain name. For more information, see RFC 2247.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Default Queue

Specifies a queue where jobs submitted to the specified printer go unless a different queue is specified.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR
DS_SERVER_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.18

Used In

Printer (page 234)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

deltaRevocationList

Specifies a delta revocation list for certificate management.

LDAP Name

deltaRevocationList;binary
deltaRevocationList

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.53

Used In

cRLDistributionPoint (page 108)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used by Certificate Authorities (Novell and third party).

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

departmentNumber

Contains the object's department number.

LDAP Name

departmentNumber

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.2

Used In

inetOrgPerson (page 291)

Person (page 222)

Template (page 260)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

Description

Specifies text that describes the associated object.

NDS Operational: Yes

LDAP Name

description

multiLineDescription

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SIZED_ATTR (1..1024)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.13

Used In

applicationEntity (page 80)

applicationProcess (page 83)

Country (page 105)

Device (page 113)

External Entity (page 125)

Group (page 128)

List (page 138)

Locality (page 141)

NSCP:groupOfCertificates (page 180)

Organization (page 202)

Organizational Role (page 212)
Organizational Unit (page 215)
Person (page 222)
Profile (page 238)
Resource (page 245)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An attribute value for the Description attribute is a string in human-readable format. For example, the object “Standards Interest” might have the associated description “distribution list for exchange of information about intracompany standards development.”

An LDAP schema query requires extensions to return the NDS attribute constraints.

destinationIndicator

Specifies an attribute used in telegram service.

LDAP Name

destinationIndicator

Syntax

Printable String (page 971)

Constraints

DS_SIZED_ATTR (1..128)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.27

Used In

Organizational Person (page 207)

residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Detect Intruder

Indicates a desire to identify suspicious login attempts.

NDS Operational: Yes

Syntax

Boolean (page 922)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.20

Used In

Organization (page 202)

Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Device

Contains a list of all printers that service the specified queue.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.21

Used In

Queue (page 241)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

dmdName

Specifies a directory management domain (DMD).

LDAP Name

dmdName

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (1..32768)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.54

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The DMD is the administrative authority which operates the directory server. For more information, see RFC 2256.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

dn

Specifies a distinguished name.

LDAP Name

distinguishedName

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.49

Used In

None

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for general LDAP support.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

dnQualifier

Specifies information which makes an object unique when information is being merged from multiple sources and objects could have the same RDNs.

LDAP Name

dnQualifier

Syntax

Printable String (page 971)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.46

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Each source should use the same value for the dnQualifier for all objects. For more information, see RFC 2256.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

DS Revision

Contains the internal revision number of the NDS agent that is running on a NetWare server.

NDS Operational: Yes

LDAP Name

dsRevision

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.172

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Email Address

Contains the e-mail address of the user. The name must conform to the established conventions for e-mail names.

NDS Operational: Yes

LDAP Name

mail

Syntax

Email Address (page 942)

Constraints

DS_NONREMOVABLE_ATTR

DS_PUBLIC_READ

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.3

Used In

External Entity (page 125)

Group (page 128)

Organization (page 202)

Organizational Person (page 207)

Organizational Role (page 212)

Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Mail Applications

The information stored in this attribute is application specific. The syntax supports a *type* field and an *address* field. The mail application defines the type and the format of the address. NDS does not verify the information; it only stores the information.

MHS Mail Applications

For MHS applications, the e-mail address for a non-MHS user is stored in the EMail Address attribute. The e-mail address for an MHS user is stored in the Mailbox Location and Mailbox ID attributes.

EMail Address is used to specify a user's mailbox that resides in a non-MHS E-mail system. For example, an NDS user can choose to have his or her e-mail delivered to a UNIX machine that supports the SMTP messaging protocol. This user's SMTP address in the UNIX machine is placed in the EMail Address attribute. (In this case, the user's Mailbox Location and Mailbox ID attributes are not used.)

EMail Address can also specify a user's e-mail alias as known in a foreign messaging system. A non-MHS messaging system can use the alias to send mail to an MHS user. For example, an MHS user (whose mailbox is in an MHS messaging server) can have an X400 alias so that X400 users can use this alias to send mail to the MHS user.

A user can have more than one e-mail alias, one for each non-MHS system. The EMail Address attribute information is stored in an EMail_Address_T structure. The convention for storing non-MHS E-mail addresses and aliases in the structure is as follows:

- ♦ **Non-MHS E-mail Address.** If the type field of the structure is set to zero, the data structure contains an E-mail address, in the following form:

non-MHS_Email_Protocol:non-MHS_Email_Address

The *non_MHS_Email_Protocol* is a 1-8 character string, and the *non-MHS_Email_Address* is a string for the actual address value.

Example: SMTP:JohnD@Novell.Com

- ♦ **Alias.** If the type field of the structure is set to one, the data structure contains an e-mail alias, in the following form:

non-MHS_Email_Protocol:non-MHS_Email_Alias.

The *non_MHS_Email_Protocol* is a 1-8 character string, and the *non-MHS_Email_Address* is a string for the actual alias value.

Example: SMTP:JohnD@Novell.Com

employeeType

Identifies what type of employee the object is, for example, full time or part time.

LDAP Name

employeeType

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.4

Used In

inetOrgPerson (page 291)

Person (page 222)

Template (page 260)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

enhancedSearchGuide

Specifies a search filter used by X.500 users.

LDAP Name

enhancedSearchGuide

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.47

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

Equivalent To Me

Specifies a list of objects that are security equivalent to the object containing the attribute.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR
DS_SERVER_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.174

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

External Name

Specifies the name of the external entity in the form used by that service.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.164

Used In

External Entity (page 125)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The External Name attribute is used to hold names that will not be interpreted by NDS. These names should not be translated.

MHS uses External Name to include users from non-NDS directories, in order to provide an integrated address book for sending mail.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

External Synchronizer

Is reserved for future use.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.124

Used In

(None)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Facsimile Telephone Number

Specifies the telephone number and, optionally, the parameters for a facsimile terminal associated with an object.

NDS Operational: Yes

LDAP Name

facsimileTelephoneNumber

Syntax

Facsimile Telephone Number (page 944)

Constraints

DS_NONREMOVABLE_ATTR

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.23

Used In

External Entity (page 125)

Organization (page 202)

Organizational Person (page 207)

Organizational Role (page 212)

Organizational Unit (page 215)

residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An attribute value for the facsimile telephone number is a string that complies with the internationally agreed format for showing international telephone numbers, E.123, (for example "+81 3 347 7418") and an optional bit string (formatted according to Recommendation T.30).

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Full Name

Specifies the full name of an object.

NDS Operational: Yes

LDAP Name

fullName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SIZED_ATTR (0..127)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.120

Used In

Group (page 128)

List (page 138)

Person (page 222)

Profile (page 238)

Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Full names conform to the naming conventions of the country or culture they are associated with. For example a typical full name for a person in an English speaking country might comprise a personal title, a first name, a middle name, a last name, generational qualifier, and decorations and awards.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Generational Qualifier

Specifies the generation of an object.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (1..8)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.170

Used In

Person (page 222)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Junior, *Jr.*, *Senior*, *Sr.*, and *II* are examples of generational qualifiers.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

generationQualifier

Specifies the generation of the object, which is usually the suffix.

LDAP Name

generationQualifier

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (1..32768)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.44

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Examples of generation qualifiers are Junior, Jr., Senior, Sr., and III. For more information, see RFC 2256.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

GID (Group ID)

Specifies a unique group ID for use by UNIX[®] clients.

NDS Operational: Yes

LDAP Name

groupID

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.24

Used In

Group (page 128)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The LDAP server maps requests for the groupOfNames and group attributes to this attribute.

Given Name

Specifies the given name of an object.

NDS Operational: Yes

LDAP Name

givenName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_SIZED_ATTR (1..32)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.42

Used In

inetOrgPerson (page 291)
Person (page 222)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In the American culture, Given Name is the first name of a person. It does not include the family name.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Group Membership

Contains a list of the groups to which the object belongs.

NDS Operational: Yes

LDAP Name

groupMembership

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

DS_WRITE_MANAGED

ASN.1 ID

2.16.840.1.113719.1.1.4.1.25

Used In

External Entity (page 125)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

GUID

Contains a guaranteed unique value for any object in the NDS tree.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_OPERATIONAL

DS_PUBLIC_READ

DS_READ_ONLY_ATTR

DS_SINGLE_VALUED

DS_SIZED_ATTR (16, 16)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.501

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is new in NetWare 5.x. The internal format of the globally unique ID (GUID) is defined by the SGUID NLM of the NetWare operating system. To generate the GUIDs, this NLM uses the policies described for UUIDs and GUIDs done by the Network Working Group.

GUID has been added as an optional attribute to the Top class so that all objects can (and eventually will) have a GUID.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

High Convergence Sync Interval

Contains the interval at which a partition synchronization will occur if no events have caused synchronization to occur.

NDS Operational: Yes

Syntax

Interval (page 951)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.117

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When high convergence is in effect, synchronization of partition information is event-driven. When a partition's information is changed, an immediate synchronization with other replicas is initiated. If no update activity occurs within the high convergence synchronization interval, synchronization will be initiated. The default for this interval is 60 minutes.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Higher Privileges

Specifies an alternative set of security access privileges.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SERVER_READ

DS_SYNC_IMMEDIATE

DS_WRITE_MANAGED

ASN.1 ID

2.16.840.1.113719.1.1.4.1.116

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is not currently implemented. When implemented, it will allow a person to activate a certain set of privileges, then deactivate them when desired. This avoids the need to log in as user Admin to perform certain functions. You can activate the higher privileges, perform the desired function, and then deactivate the privileges.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Home Directory

Contains the initial value for a client's current working directory.

NDS Operational: Yes

LDAP Name

homeDirectory

Syntax

Path (page 966)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (1..255)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.26

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The Home Directory attribute is intended for use primarily by UNIX clients. UNIX clients maintain their configuration information in the client's home directory. The Home Directory attribute should contain a legal name in the UNIX name space of the Host Server defined for the client.

For DOS clients logging in using bindery mode, this attribute sets a default directory location.

DOS and OS/2 clients use login scripts to provide their initial configuration information, and MAC clients have other means for storing configuration information.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Path syntax.

Home Directory Rights

Specifies the rights the User objects will have to their home directories when the Users are created with this Template object.

Syntax

Integer (page 948)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.62.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

homePhone

Contains the object's home phone number.

LDAP Name

homePhone

Syntax

Telephone Number (page 979)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.20

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

homePostalAddress

Contains the object's home address for mail.

LDAP Name

homePostalAddress

Syntax

Postal Address (page 969)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.39

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

houseIdentifier

Identifies a building within a location.

LDAP Name

houseIdentifier

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (1..32768)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.51

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

Host Device

Contains the distinguished name of the device with which the object is associated.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.27

Used In

Printer (page 234)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Host Resource Name

Contains the name by which the resource is known on the local host.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.28

Used In

Resource (page 245)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used when a resource is locally known by a different name on the host server. For example, a volume's global name might be:

```
'Tools.Development.Acme.US'
```

That volume's local name on the server might be SYS.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Host Server

Identifies the server with which an object is associated.

NDS Operational: Yes

LDAP Name

hostServer

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.29

Used In

Directory Map (page 116)

Queue (page 241)

Volume (page 281)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For Volume class objects, this attribute is mandatory.

Inherited ACL

Contains a summary of access control information inherited from a superior partition.

NDS Operational: Yes

Syntax

Object ACL (page 957)

Constraints

DS_NONREMOVABLE_ATTR
DS_READ_ONLY_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.30

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The information in this attribute is used to optimize effective rights calculations within a partition. The attribute lists all access control information inherited from objects superior to the partition object.

The LDAP server does not currently support the Object ACL syntax. LDAP clients cannot access this attribute.

Initials

Specifies the initials of an object.

NDS Operational: Yes

LDAP Name

initials

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_SIZED_ATTR (1..8)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.43

Used In

inetOrgPerson (page 291)
Person (page 222)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For example, the initials for John Steven Doe are JSD.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

internationaliSDNNumber

Specifies an international ISDN (Integrated Services Digital Network) number used in voice, video, and data transmissions.

LDAP Name

internationaliSDNNumber

Syntax

Numeric String (page 955)

Constraints

DS_SIZED_ATTR (1..16)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.25

Used In

Organizational Person (page 207)

residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

Internet EMail Address

Specifies an Internet e-mail address.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.216

Used In

NSCP:mailGroup5 (page 329)

NSCP:mailGroup5 (page 329)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP Services for NDS uses this attribute to store an Internet e-mail address. A value is not required, but if supplied, the address is delivered to the LDAP client when a user's e-mail address is requested.

LDAP added this attribute for Netscape support. LDAP maps requests for the mail attribute to this attribute.

Intruder Attempt Reset Interval

Designates the time frame in which to monitor consecutive failed login attempts.

NDS Operational: Yes

Syntax

Interval (page 951)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.31

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

If the number of consecutive failed attempts to log in exceeds the Login Intruder Limit, the user's account is locked to prevent further login attempts. The Intruder Attempt Reset Interval is the amount of time in which consecutive login attempts must fall. For example, assume the interval is 30 minutes and the limit is 3 login attempts. If a user attempted to log in every 20 minutes, the user would never be locked out. If the user attempted to log in every minute, a lock would be implemented after the third attempt.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Intruder Lockout Reset Interval

Identifies the amount of time a user remains locked out once an intruder detection lock has been applied.

NDS Operational: Yes

Syntax

Interval (page 951)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.32

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

knowledgeInformation

Specifies a standard LDAP attribute.

LDAP Name

knowledgeInformation

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (0..32768)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.2

Used In

dSA (page 122)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for backwards compatibility. LDAP no longer uses this attribute. For more information, see RFC 2256.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

L (Locality Name)

Specifies a physical or geographical location.

NDS Operational: Yes

LDAP Name

l
localityName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (1..128)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.7

Used In

applicationEntity (page 80)
applicationProcess (page 83)
Device (page 113)
External Entity (page 125)
Group (page 128)
List (page 138)
Locality (page 141)
Organization (page 202)
Organizational Person (page 207)
Organizational Role (page 212)

Organizational Unit (page 215)
Profile (page 238)
residentialPerson (page 332)
Resource (page 245)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When used as a component of an NDS name, a locality name identifies a geographical area or locality in which the named object is physically located or with which it is associated in some other important way. An attribute value for L is a string, (for example, L = Edinburgh).

An LDAP schema query requires extensions to return the NDS attribute constraints.

Language

Contains an ordered list of languages.

NDS Operational: Yes

Syntax

Case Ignore List (page 926)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.34

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is intended for use by applications to determine which language display the object desires.

Last Login Time

Contains the login time of the session previous to the current session.

NDS Operational: Yes

Syntax

Time (page 981)

Constraints

DS_NONREMOVABLE_ATTR

DS_READ_ONLY_ATTR

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.35

Used In

Organizational Unit (page 215)

pkiCA (page 226)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For the time of the current login session, see the Login Time (page 513) attribute.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Last Referenced Time

Specifies the last time the object was referenced.

NDS Operational: Yes

Syntax

Timestamp (page 983)

Constraints

DS_NONREMOVABLE_ATTR

DS_PER_REPLICA

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.167

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The LDAP server does not currently support the Timestamp syntax.

LDAP ACL v11

Defines access control restrictions for LDAP clients.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.210

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. LDAP ACLs allow an additional layer of security between LDAP clients and NDS.

LDAP Allow Clear Text Password

Specifies whether LDAP clients can use clear text passwords to authenticate to NDS.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.213

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

The default is FALSE, which forces all LDAP clients with passwords to use encryption over an SSL connection. Anonymous binds and proxy users can still authenticate or bind because they do not use passwords.

LDAP Anonymous Identity

Specifies which NDS user account is used for LDAP anonymous binds.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.214

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

If this attribute has no value, LDAP uses the NDS [Public] object for anonymous binds.

If an NDS user account is specified as the value, the account must be configured so that it does not require a password or require password changes. It also should not allow the user to create or change the password. LDAP anonymous binds do not have passwords, so any bind that includes both a user and a password is treated as an NDS user bind.

LDAP Attribute Map v11

Defines how LDAP attribute names are mapped to NDS attribute names.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.211

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. To avoid corrupting the data in this attribute, use the option in NetWare Administrator to change any mappings.

This attribute allows the LDAP server to expose NDS attributes in LDAP format. When an LDAP client requests an LDAP attribute from the LDAP server, the server returns the corresponding NDS attribute.

LDAP Backup Log Filename

Specifies the path and name of the file that is used when the LDAP log file reaches the maximum size.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.205

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. The backup log file is stored in the SYS:\ETC directory.

This attribute works in conjunction with the LDAP Log Filename attribute. If a log filename is configured and the log file reaches the maximum size, the log file is renamed to the name specified in this attribute. All new log data continues to be stored in the log file.

If no backup log filename is specified, old data is lost whenever the log file reaches the maximum file size.

LDAP Class Map v11

Defines how LDAP class names are mapped to NDS class names.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.212

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. To avoid corrupting the data in this attribute, use the option in NetWare Administrator to change any mappings.

This attribute allows the LDAP server to expose NDS classes in LDAP format. When an LDAP client requests an LDAP class from the LDAP server, the server returns the corresponding NDS class.

LDAP Enable SSL

Is reserved.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.200

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP Enable TCP

Is reserved.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.198

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP Enable UDP

Is reserved.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.199

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

TRUE indicates that the LDAP server is configured for UDP.

For the LDAP server to function, this attribute must always be TRUE.

LDAP Group

Specifies the LDAP group object that contains the configuration settings for this LDAP server.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.191

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

LDAP Host Server

Specifies the name of the NetWare server that is hosting the LDAP Services for NDS.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.190

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. The host server is the NetWare server that is running the NLDAP NLM.

LDAP Log Filename

Specifies the name of the LDAP log file.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.204

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

The default is no log file. When a name is specified, logging is enabled and the log file is stored in the SYS:\ETC directory on the LDAP server.

LDAP Log Level

Specifies the type of messages that are recorded in the LDAP log file.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0..32768)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.1.192

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. LDAP Services for NDS uses the following values for message levels.

Value	Description
1	Trace function calls
4	LDAP requests
8	Connection information
16	BER decoding
32	Search filter processing

Value	Description
64	Configuration processing
128	LDAP ACL processing
256	Client request summary
512	Search response summary
4096	Error messages from all options
8192	Server console messages generated by LDAP
65535	All messages

The values, except for the last one (65535), can be added together to select more than one kind of message. The default value is 8192.

LDAP Log Size Limit

Specifies the maximum size, in bytes, of the LDAP log file.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (2048..4294967295)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.206

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

When the log file reaches the maximum size, it is renamed to the name specified in the LDAP Backup Log Filename attribute. All new data continues to be stored in the file specified by the LDAP Log Filename attribute.

A log size of zero specifies an unlimited file size. The default is 100000.

LDAP Referral

Specifies the URL of an alternate LDAP server that is to handle any requests for objects that are not found within any of the subtrees known to the current LDAP server.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.208

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

Referrals use the following format:

ldap://hostname

To limit subtrees known to an LDAP server, see the LDAP Suffix (page 488) attribute.

LDAP Screen Level

Specifies the type of messages that are sent to the console of an LDAP server.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0..32768)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.193

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. LDAP Services for NDS uses the following values for message levels.

Value	Description
1	Trace function calls
4	LDAP requests
8	Connection information
16	BER decoding

Value	Description
32	Search filter processing
64	Configuration processing
128	LDAP ACL processing
256	Client request summary
512	Search response summary
4096	Error messages from all options
8192	Server console messages generated by LDAP
65535	All messages

The values, except for the last one (65535), can be added together to select more than one kind of message. The default value is 8192.

LDAP Search Size Limit

Specifies the maximum number of objects about which an LDAP server will return data.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..2147483647)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.194

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

LDAP Services for NDS returns object data until the object number exceeds the value of this attribute. The LDAP server then stops sending object data and considers the request complete. The default value is 500.

LDAP Search Time Limit

Specifies the maximum amount of time, in milliseconds, that the LDAP server can use to return data.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..2147483647
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.195

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. The default is 3600 seconds.

The LDAP server continues processing the request until the search time has expired. It then returns an error reporting that the search time was exceeded. The LDAP server considers the request complete.

LDAP Server

Specifies the name of the LDAP server that is running on the NetWare server.

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.215

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

LDAP Server Bind Limit

Specifies the maximum number of simultaneous LDAP binds or connections the LDAP server allows.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0..4294967295)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.196

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

If the bind limit is set to zero, there is no limit on the number of binds. This is the default value.

If the bind limit is set to a number other than zero, the LDAP server accepts binds until the bind limit is reached. It then rejects additional binds until the total number of binds drops below the bind limit.

LDAP Server Idle Timeout

Specifies the maximum amount of time, in seconds, that an LDAP connection can be inactive before the connection is destroyed.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0..4294967295)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.197

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

The LDAP server requires that this attribute's value be set to zero or a multiple of 10 (for example, 10, 150, or 900).

If the value is set to zero, there is no time limit on idle connections. This is the default.

If the value is set to a number other than zero, the LDAP server disconnects inactive connections after the specified number of seconds have elapsed.

LDAP Server List

Specifies which LDAP servers can use the configuration information stored in the LDAP group object.

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.209

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

LDAP SSL Port

Specifies the SSL port for LDAP Services for NDS.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (0..65535)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.203

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. The default value is 636, a well-known SSL port for LDAP services.

LDAP Suffix

Defines the subtrees that LDAP Services for NDS makes available to LDAP clients.

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.207

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

If this attribute has no value, the LDAP server makes all subtrees from the root of the NDS tree available to LDAP clients. Otherwise, only objects that are contained below the suffixes can be accessed.

LDAP TCP Port

Specifies the port number that the LDAP server uses for TCP packets.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (0..65535)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.201

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. The default value is 389, a well-known port address for LDAP services.

LDAP UDP Port

Specifies the port number that the LDAP server uses for UDP packets.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (0..65535)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.202

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. The default value is 389, a well-known port address for LDAP services.

LDAP:bindCatalog

Is reserved for future use.

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.238

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP:bindCatalogUsage

Is reserved for future use.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.237

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP:keyMaterialName

Specifies the Key Material object for storing the SSL certificate used in SSL connections for this LDAP server.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.291

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. A value specified in this attribute enables SSL connections.

LDAP:otherReferralUsage

Not currently used.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.294

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP:searchCatalog

Specifies the catalog that the LDAP server can use for searches.

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.236

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter.

LDAP:searchCatalogUsage

Specifies the type of searches which the LDAP server can use the catalog for.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.235

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. The LDAP server uses the following values.

Value	Description
First Bit	The LDAP server doesn't search the catalog but uses the NDS database for all searches.
Second Bit	The LDAP server doesn't search the NDS database but uses the catalog for all searches.

Value	Description
First and Second Bit	If the attribute type is not in the catalog, the LDAP server searches the NDS database. This is the default at installation.

LDAP:searchReferralUsage

Specifies the LDAP search referral option.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.293

Used In

inetOrgPerson (page 291)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP Group object can be configured to handle referrals in the following ways:

- ◆ Always traverse the NDS tree (default option). This causes the request to use the referrals to search out information on other NDS servers.
- ◆ Always refer. This causes the request to return referrals and allows the caller to determine whether or not to traverse the NDS tree for the information.
- ◆ Refer if possible. This allows referrals to NDS 8 servers to be returned. This option is used in trees which contain a mix of NDS 8 and previous versions of NDS. Previous versions of the LDAP server do not register in the referral list, and therefore cannot be returned.

Locked By Intruder

Specifies that an object has been disabled due to intruder detection.

NDS Operational: Yes

Syntax

Boolean (page 922)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.37

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In the case of a User object, if TRUE this attribute indicates the user cannot log in to the network because too many sequential attempts to log in have been denied. This usually happens when a person does not know the correct password and tries to guess it too many times.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Lockout After Detection

Indicates that users should be kept from attempting to log in once they are identified as an intruder.

NDS Operational: Yes

Syntax

Boolean (page 922)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.38

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Monitoring attempts of users to log in can occur without actually enforcing any restrictions on logging in. If this attribute is set to TRUE, it indicates that users should be locked out once the limits of tolerance have been exceeded. Those tolerable limits are designated using other attribute values.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Login Allowed Time Map

Specifies the allowed login time periods for an account for each day of the week to a precision of one-half hour.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (42, 42)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.39

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Login Disabled

Notifies the user that the account has been disabled.

NDS Operational: Yes

Syntax

Boolean (page 922)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.40

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When this attribute is set to True, it disables the user account. This is used for explicit permanent disabling of an account. This attribute can only be manually set and cleared.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Login Expiration Time

Specifies a date and time after which a client cannot log in and authenticate as an object.

NDS Operational: Yes

Syntax

Time (page 981)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.41

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Login Grace Limit

Specifies the total number of times an old password can be used (after the old password has expired) to access the account.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.42

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Login Grace Remaining

Specifies how many grace logins are left before the account is locked.

NDS Operational: Yes

Syntax

Counter (page 936)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.43

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Counter syntax.

Login Intruder Address

Specifies the address of the node that caused the intruder detection lockout.

NDS Operational: Yes

LDAP Name

loginIntruderAddress

Syntax

Net Address (page 953)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.44

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Login Intruder Attempts

Specifies the number of failed login attempts that have occurred in the current interval.

NDS Operational: Yes

Syntax

Counter (page 936)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.45

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Counter syntax.

Login Intruder Limit

Specifies the number of failed login attempts allowed before an account is locked due to intruder detection.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.46

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Login Intruder Reset Time

Specifies the next time that the intruder attempts variable will be reset.

NDS Operational: Yes

Syntax

Time (page 981)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.47

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Time syntax.

Login Maximum Simultaneous

Specifies the number of authenticated login sessions an object can initiate simultaneously.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.48

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Login Script

Contains the object's login script.

NDS Operational: Yes

Syntax

Stream (page 977)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.49

Used In

Group (page 128)

Organization (page 202)

Organizational Unit (page 215)

Profile (page 238)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For the Profile class, the Login Script attribute is mandatory.

The Login Script attribute contains the login script for Organization and Organization Unit containers. When a user logs in, the LOGIN program searches one level above (to either the Organization or Organizational Unit) and runs its script. The LOGIN program then runs the specified profile script (if any) and the user's login script (if any).

If a login script has been created for a group, the group login script is executed only if it is called from either the container or the profile login script.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Stream syntax.

Login Time

Specifies the login time of the current session.

NDS Operational: Yes

Syntax

Time (page 981)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.50

Used In

Organizational Unit (page 215)

pkiCA (page 226)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Time syntax.

Low Convergence Reset Time

Specifies the time at which to start the low convergence synchronization cycle.

NDS Operational: Yes

Syntax

Time (page 981)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.118

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This reset time is implemented only if the Convergence attribute of the partition specifies low convergence. The time stored in this attribute is the time of day at which a synchronization occurs and the interval counter is reset (or initialized). The next synchronization will occur after the low synchronization interval has lapsed.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Time syntax.

Low Convergence Sync Interval

Specifies the amount of time (in seconds) that must pass from the start of one partition synchronization to the next.

NDS Operational: Yes

Syntax

Interval (page 951)

Constraints

DS_NONREMOVABLE_ATTR
DS_SINGLE_VALUED_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.104

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This synchronization interval is implemented only if the Convergence attribute specifies low convergence.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Mailbox ID

Contains a unique ID associated with the object's mailbox and messaging server.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..8)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.163

Used In

External Entity (page 125)
Group (page 128)
List (page 138)
Organization (page 202)
Organizational Person (page 207)
Organizational Role (page 212)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Mailbox Location

Contains the name of the Messaging Server that services the object's mailbox.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.162

Used In

External Entity (page 125)

Group (page 128)

List (page 138)

Organization (page 202)

Organizational Person (page 207)

Organizational Role (page 212)

Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

manager

Contains the name of the object's manager.

LDAP Name

manager

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.10

Used In

inetOrgPerson (page 291)

Person (page 222)

Template (page 260)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

masvAuthorizedRange

Contains a list of clearances that the object is authorized to use during login and authentication.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.2.6

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The session clearance requested during log in and authentication must be within the list of clearances defined by this attribute.

masvDefaultRange

Defines the session clearance to assign if one is not specified during login and authentication.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.2.5

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This value must be within the list of authorized clearances as specified in the MASV:Authorized Range attribute.

masvDomainPolicy

Defines security policy information which is global to an NDS tree.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.2.7

Used In

MASV:Security Policy (page 144)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The global policy information includes default classifications for resources and default clearances for logged-in objects. These default values are used only if classifications and clearances are not explicitly defined for an object.

masvLabel

Defines the classification of all NDS objects contained on the partition.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.2.3

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Once a label is explicitly assigned, the partition label cannot be modified.

masvProposedLabel

Defines the classification to be assigned to a partition that is to be created by splitting an existing partition.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.2.4

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A proposed label is not a label, but rather a value that will be assigned when the partition is split to create a new subordinate partition. If a proposed label is not assigned when a partition is created, the label of the new partition is set to the parent partition’s label. The MASV:Proposed Label attribute may be modified up until the new partition is created.

Member

Lists objects associated with a group or list.

NDS Operational: Yes

LDAP Name

member

uniqueMember

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.31 (member)

2.5.4.50 (uniqueMember)

Used In

Group (page 128)

List (page 138)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Members Of Template

Specifies the names of the User objects that have been created with this Template object.

Syntax

Distinguished Name (page 938)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.59.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Memory

Specifies the amount of printer memory (in kilobytes).

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.52

Used In

Printer (page 234)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Currently the Memory attribute is used only for printers, but it is not limited to printers. In the future, it could be used for other devices.

Message Routing Group

Specifies the name of the Message Routing Group objects to which a messaging server can belong.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.126

Used In

Messaging Server (page 150)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Message Server

Specifies the name of a server object that stores and forwards broadcast-type messages.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.127

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Messaging Database Location

Contains the location of the messaging file-directory structure.

NDS Operational: Yes

LDAP Name

messagingDatabaseLocation

Syntax

Path (page 966)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.125

Used In

Messaging Server (page 150)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An example for a messaging directory is "\mhs".

MHS messaging servers use a file-system subtree to (1) receive messages from applications or other messaging servers and gateways, (2) store messages while they are being routed, (3) store internal control files, and (4) extract files.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Path syntax.

Messaging Server

Identifies a Messaging Server object that is running on the NCP Server object.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.53

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Messaging Server Type

Identifies the type of a Messaging Server.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (1..32)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.166

Used In

Messaging Server (page 150)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Examples of messaging server types are MHS, GMHS, and X400.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Minimum Account Balance

Specifies the minimum amount of credit (or money) a user must have in his or her account to access specified services.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.54

Used In

Server (page 254)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

mobile

Contains the phone number of the object's cellular or mobile phone.

LDAP Name

mobile

Syntax

Telephone Number (page 979)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.41

Used In

inetOrgPerson (page 291)

Person (page 222)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

NDSPKI:Certificate Chain

Defines a multi-valued attribute that contains all the public key certificates that chain back to the root authority.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.4

Used In

NDSPKI:Certificate Authority (page 162)
NDSPKI:Key Material (page 165)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Two bytes of chaining data preceding each certificate.

- ♦ If the first byte is nonzero, the certificate is the root certificate. If it is zero, the certificate is nested in the chain.
- ♦ The second byte contains an index that specifies where the current certificate belongs in the chain. One indicates the first certificate; two, the second; etc.

NDSPKI:Given Name

Contains the distinguished name of the NDSPKI:Key Material object when it is created.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.11

Used In

NDSPKI:Key Material (page 165)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The syntax for this attribute is a case ignore string, which means that this name does not change when the NDSPKI:Key Material object is renamed. Such an object should never be renamed, and this attribute allows the original name to be preserved.

NDSPKI:Key File

Contains all the key material in Terisa format.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.7

Used In

NDSPKI:Key Material (page 165)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute contains all the information that would normally be found in a Terisa system key file. Only Terisa software can process the information.

NDSPKI:Key Material DN

Contains the distinguished names of the NDSPKI:Key Material objects that belong to the server that owns the SAS:Service object.

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.9

Used In

SAS:Service (page 251)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:Keystore

Contains wrapped private keys.

Syntax

Octet String (page 962)

Constraints

DS_HIDDEN_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.105

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:Not After

Contains a validity period for a certificate. The certificate cannot be used after this period.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.15

Used In

NDSPKI:Trusted Root Object (page 177)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:Not Before

Contains a validity period for a certificate. The certificate cannot be used before this period.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.14

Used In

NDSPKI:Trusted Root Object (page 177)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:Parent CA

Indicates who is the parent of the NDSPKI:Certificate Authority object.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.5

Used In

NDSPKI:Certificate Authority (page 162)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Currently, the parent is a hard-coded string because only one NDSPKI:Certificate Authority object can exist in an NDS tree.

NDSPKI:Parent CA DN

Not currently used.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.6

Used In

NDSPKI:Certificate Authority (page 162)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:Private Key

Contains the private key associated with the object.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.2

Used In

NDSPKI:Certificate Authority (page 162)

NDSPKI:Key Material (page 165)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Private keys are encrypted by the server’s Key Storage Key. Only the server that created the public/private key pair can use the key.

NDSPKI:Public Key

Contains the public key associated with the object.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.1

Used In

NDSPKI:Certificate Authority (page 162)
NDSPKI:Key Material (page 165)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Public keys are not encrypted and can be read by anyone.

NDSPKI:Public Key Certificate

Holds the public key certificate associated with the object.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.3

Used In

NDSPKI:Certificate Authority (page 162)
NDSPKI:Key Material (page 165)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A public key certificate binds a public key to a given entity, making that entity the owner of the public key.

NDSPKI:SD Key Cert

Not currently used.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.103

Used In

NDSPKI:SD Key List (page 171)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:SD Key ID

Not currently used.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.104

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:SD Key Server DN

Contains the distinguished name of the server that can be contacted to get a copy of a security domain key.

Syntax

Distinguished Name (page 938)

Constraints

DS_SERVER_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.101

Used In

NDSPKI:SD Key List (page 171)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:SD Key Struct

Not currently used.

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.102

Used In

NCP Server (page 155)

NDSPKI:SD Key List (page 171)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:Subject Name

Contains the subject name of the owner of the associated key pair and public key certificate.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.8

Used In

NDSPKI:Key Material (page 165)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The NDSPKI:Subject Name attribute specifies who owns the key. Usually, this is the Internet name service (DNS) name of the server.

NDSPKI:Tree CA DN

Contains the distinguished name of the Certificate Authority for the NDS tree.

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.10

Used In

SAS:Security (page 248)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Each NDS tree contains only one CA, a NDSPKI:Certificate Authority object.

NDSPKI:Trusted Root Certificate

Contains a certificate from a Certificate Authority.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.13

Used In

NDSPKI:Trusted Root Object (page 177)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSPKI:userCertificateInfo

Contains information that links a certificate in a user certificate attribute to a private key in the NDSPKI:Keystore attribute.

Syntax

Path (page 966)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.48.4.1.12

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Besides the link information, this attribute can contain additional information about the certificate.

Network Address

Contains one or more network addresses of the associated object.

NDS Operational: Yes

LDAP Name

networkAddress

Syntax

Net Address (page 953)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.55

Used In

Device (page 113)

Organizational Unit (page 215)

pkiCA (page 226)

Queue (page 241)

Server (page 254)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute specifies the type of the underlying transport. The attribute can have multiple values.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Network Address Restriction

Restricts objects to specific network or node addresses.

NDS Operational: Yes

LDAP Name

networkAddressRestriction

Syntax

Net Address (page 953)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.56

Used In

CommExec (page 98)

Printer (page 234)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute specifies the type of the underlying transport. The attribute can have multiple values.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

New Object's DS Rights

Specifies which NDS objects the User objects are granted NDS rights to when the User objects are created with this Template object.

Syntax

Object ACL (page 957)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.55.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

New Object's FS Rights

Specifies the file, directory, or volume path and the rights to which the User objects are granted when created with this Template object.

Syntax

Path (page 966)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.56.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

New Object's Self Rights

Specifies the NDS rights that a User object is granted to its own object when the User object is created with this Template object.

Syntax

Object ACL (page 957)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.63.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NNS Domain

Specifies the name of the NetWare Name Service (NNS) Domain that has been upgraded from NetWare 3.x into the container.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SIZED_ATTR (1.128)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.119

Used In

Organization (page 202)

Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Notify

Specifies a list of objects that are to be notified when a specified event occurs.

NDS Operational: Yes

LDAP Name

notify

Syntax

Typed Name (page 986)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.57

Used In

Printer (page 234)

Remarks

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NSCP:administratorContactInfo

Specifies Netscape administrator information.

LDAP Name

administratorContactInfo

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.74

Used In

NSCP:NetscapeServer5 (page 193)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:adminURL

Specifies a Netscape administrator URL.

LDAP Name

adminURL

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.75

Used In

NSCP:NetscapeServer5 (page 193)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:AmailAccessDomain

Specifies a Netscape mail access domain.

LDAP Name

mailAccessDomain

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.12

Used In

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AmailAlternateAddress

Specifies an alternate mail address used by Netscape.

LDAP Name

mailAlternateAddress

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.13

Used In

NSCP:mailGroup5 (page 329)

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AmailAutoReplyMode

Specifies a Netscape auto reply mode for mail.

LDAP Name

mailAutoReplyMode

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.14

Used In

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:MailAutoReplyText

Specifies a text message used by Netscape for auto reply.

LDAP Name

mailAutoReplyText

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.15

Used In

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AmalDeliveryOption

Specifies a delivery option used by Netscape.

LDAP Name

mailDeliveryOption

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.16

Used In

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:MailForwardingAddress

Specifies a forwarding address used by Netscape.

LDAP Name

mailForwardingAddress

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.17

Used In

NSCP:mailGroup5 (page 329)

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AmailHost

Specifies a mail host used by Netscape.

LDAP Name

mailHost

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.18

Used In

NSCP:mailGroup5 (page 329)

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AmailMessageStore

Specifies a message store used by Netscape.

LDAP Name

mailMessageStore

Syntax

Case Exact String (page 924)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.19

Used In

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AmalProgramDeliveryInfo

Specifies delivery information used by Netscape.

LDAP Name

mailProgramDeliveryInfo

Syntax

Case Exact String (page 924)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.20

Used In

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AmailQuota

Specifies a mail quota used by Netscape.

LDAP Name

mailQuota

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.21

Used In

NSCP:mailRecipient (page 187)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AnsLicenseEndTime

Specifies an end time for a Netscape license.

LDAP Name

nsLicenseEndTime

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.38

Used In

NSCP:nsLicenseUser (page 200)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AnsLicensedFor

Specifies a Netscape license.

LDAP Name

nsLicensedFor

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.36

Used In

NSCP:nsLicenseUser (page 200)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:AnsLicenseStartTime

Specifies a start time for a Netscape license.

LDAP Name

nsLicenseStartTime

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.37

Used In

NSCP:nsLicenseUser (page 200)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for auxiliary class support. For more information, see the documentation for the Netscape schema.

NSCP:employeeNumber

Specifies an ID that has been assigned to an employee.

LDAP Name

employeeNumber

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.3

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used by Netscape products.

NSCP:installationTimeStamp

Specifies a Netscape installation time stamp.

LDAP Name

installationTimeStamp

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.73

Used In

NSCP:NetscapeServer5 (page 193)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:mailRoutingAddress

Specifies a mail routing address used by Netscape.

LDAP Name

mailRoutingAddress

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.47

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The LDAP server adds this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:memberCertificateDesc

Specifies a member certificate description for Netscape products.

LDAP Name

memberCertificateDescription

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.199

Used In

NSCP:groupOfCertificates (page 180)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDS supports only 32 characters in an attribute name. This attribute maps to Netscape’s memberCertificateDescription attribute.

NSCP:mgrpRFC822mailmember

Specifies mail members used by Netscape.

LDAP Name

mgrpRFC822mailmember

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.30

Used In

NSCP:mailGroup5 (page 329)

NSCP:mailGroup5 (page 329)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

In NDS 8 and above, this attribute has a syntax of case ignore string. In some previous versions of NDS, this attribute is defined with a syntax of case exact string.

NSCP:ngcomponentCIS

Specifies a Netscape component.

LDAP Name

ngcomponent

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.196

Used In

NSCP:Nginfo2 (page 331)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 8, LDAP adds this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

The attribute has been redefined in NDS 8 because the NSCP:ngComponent attribute defined in NetWare 5 uses the wrong syntax.

NSCP:nsaclrole

Specifies an ACL role used by Netscape.

LDAP Name

nsaclrole

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.192

Used In

NSCP:Nginfo (page 330)

NSCP:Nginfo2 (page 331)

NSCP:Nginfo2 (page 331)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:nscreator

Specifies a Netscape creator.

LDAP Name

nscreator

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.195

Used In

NSCP:Nginfo (page 330)
NSCP:Nginfo2 (page 331)
NSCP:Nginfo2 (page 331)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. The LDAP server maps requests for the nscreator attribute to this attribute. For more information, see the documentation for the Netscape SuiteSpot schema.

NSCP:nsflags

Specifies Netscape flags.

LDAP Name

nsflags

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.194

Used In

NSCP:Nginfo (page 330)

NSCP:Nginfo2 (page 331)

NSCP:Nginfo2 (page 331)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:nsnewsACL

Specifies an ACL for Netscape news.

LDAP Name

nsnewsACL

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.191

Used In

NSCP:Nginfo (page 330)
NSCP:Nginfo2 (page 331)
NSCP:Nginfo2 (page 331)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:nsprettyname

Specifies a Netscape name.

LDAP Name

nsprettyname

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.193

Used In

NSCP:Nginfo (page 330)

NSCP:Nginfo2 (page 331)

NSCP:Nginfo2 (page 331)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:serverHostName

Specifies a host name server used by Netscape.

LDAP Name

serverHostName

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.76

Used In

NSCP:NetscapeServer5 (page 193)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:serverProductName

Specifies a server product name used by Netscape.

LDAP Name

serverProductName

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.71

Used In

NSCP:NetscapeServer5 (page 193)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:serverRoot

Specifies a server root used by Netscape.

LDAP Name

serverRoot

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.70

Used In

NSCP:NetscapeServer5 (page 193)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:serverVersionNumber

Specifies a server version number used by Netscape.

LDAP Name

serverVersionNumber

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.72

Used In

NSCP:NetscapeServer5 (page 193)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

NSCP:subtreeACI

Specifies a subtree ACI used by Netscape.

LDAP Name

subtreeACI

Syntax

Case Ignore String (page 929)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.69

Used In

NSCP:Nginfo (page 330)
NSCP:Nginfo2 (page 331)
NSCP:Nginfo2 (page 331)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Netscape support. For more information, see the documentation for the Netscape schema.

O (Organization Name)

Specifies the name of an organization.

NDS Operational: Yes

LDAP Name

o
organizationName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (1..64)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.10

Used In

applicationEntity (page 80)
Device (page 113)
Group (page 128)
List (page 138)
NSCP:groupOfCertificates (page 180)
Organization (page 202)
Profile (page 238)
Resource (page 245)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For the Organization class, this attribute is mandatory.

When used as a component of an NDS name, an organization name identifies an organization with which the named object is affiliated. An attribute value for O is a string chosen by the organization (for example, O=Scottish Telecommunications plc). Any variant's name should be associated with the named organization as separate and alternative attribute values.

An LDAP schema query requires extensions to return NDS attribute constraints.

Obituary

Used to avoid name collisions during certain operations.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR
DS_READ_ONLY_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.114

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used internally by NDS. The DS_READ_ONLY_ATTR constraint restricts access to the NDS server. Obituaries provide a unique identification, other than the object name, that may be required in order to resolve certain functions.

In a distributed database, each server receives updated information through synchronization. Because the servers do not receive updates simultaneously, the server might not hold the same information at a given time. For this reason, each server holds on to the old information until all the other servers receive updates. NDS uses obituaries to keep track of such information.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Obituary Notify

Added to a partition root when an object, which is referenced in the partition but not resident on the partition, has its RDN modified or is renamed, deleted, moved, or restored.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR
DS_READ_ONLY_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.500

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The attribute is new in NetWare 5.x and works in conjunction with the Used By attribute. For example, if object A has a Used By attribute referencing partition root B, then deleting object A would cause an Obituary Notify attribute to be placed on B. This will cause partition B to clean up all references to object A in its partition.

The format of the octet string is internal to NDS.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Object Class

Contains an unordered list of object classes. These classes are the fully expanded set of super classes for the object to which this attribute is assigned.

NDS Operational: Yes

LDAP Name

objectClass

Syntax

Class Name (page 934)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.0

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For the Top class and all of its subordinate classes, this attribute is mandatory.

When an object is created, a single initial value for object class must be specified. When the server creates the object, it expands the value set of the

object class attribute to include all of the super classes of the initially specified class.

Auxiliary classes are added to or removed from an object class by modifying the Object Class attribute.

An LDAP schema query requires extensions to return the NDS attribute constraints.

Operator

Specifies an object with operator privileges.

NDS Operational: Yes

LDAP Name

operator

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SERVER_READ

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.59

Used In

Computer (page 101)

NCP Server (page 155)

Print Server (page 230)

Printer (page 234)

Queue (page 241)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Other GUID

Contains recovered GUIDs.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_OPERATIONAL

DS_PUBLIC_READ

DS_SIZED (16, 16)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.502

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used to manage the case where an object loses its GUID, has another one generated, and then the old GUID is recovered or restored. This attribute will store the recovered GUIDs so that an object can be accessed by the new one (stored in the GUID attribute) and the recovered one (stored in this attribute).

The octet string is sized at 16 octets and has the same format as the GUID attribute.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

OU (Organizational Unit Name)

Specifies the name of an organizational unit.

NDS Operational: Yes

LDAP Name

ou

organizationalUnitName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SIZED_ATTR (1..64)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.11

Used In

applicationEntity (page 80)

applicationProcess (page 83)

Device (page 113)

External Entity (page 125)

Group (page 128)

List (page 138)

NSCP:groupOfCertificates (page 180)

Organizational Person (page 207)

Organizational Role (page 212)

Organizational Unit (page 215)

Profile (page 238)
Resource (page 245)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For the Organizational Unit class, this attribute is mandatory.

When used as a component of an NDS name, an organizational unit name identifies a unit of an organization with which the named object is affiliated. The unit is understood to be part of an organization designated by an O (Organization Name) attribute assigned to the same object. It follows that if an OU (Organizational Unit Name) attribute is used in an NDS name, it must be associated with an O (Organization Name) attribute.

An attribute value for OU (Organizational Unit Name) is a string chosen by the organization to which the unit belongs. An example of an organizational unit name is

`OU=Technology Division`

An LDAP schema query requires extensions to return NDS attribute constraints.

Owner

Specifies the name of an object that has some responsibility for the associated object.

NDS Operational: Yes

LDAP Name

owner

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.32

Used In

Device (page 113)

Group (page 128)

List (page 138)

NSCP:groupOfCertificates (page 180)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An attribute value for this attribute is a distinguished name, which could represent a group. This attribute can have multiple values.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Page Description Language

Identifies the page description languages (PDLs) supported by a printer. Multiple PDLs should be represented as multiple values.

NDS Operational: Yes

Syntax

Printable String (page 971)

Constraints

DS_NONREMOVABLE_ATTR

DS_SIZED_ATTR (1..64)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.63

Used In

Printer (page 234)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An example of an attribute value for this attribute is

Page Description Language=PostScript

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

pager

Contains the phone number of the object's pager.

LDAP Name

pager

Syntax

Telephone Number (page 979)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.42

Used In

inetOrgPerson (page 291)

Person (page 222)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

Partition Control

Contains the states of split and join operations.

NDS Operational: Yes

LDAP Name

partitionControl

Syntax

Typed Name (page 986)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.121

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The Typed Name syntax has three fields: entry ID, level, and interval. The partition operation (PC_IDLE, PC_JOINING_UP, PC_JOINING_DOWN, PC_MOVE_SUBTREE_DEST, PC_MOVE_SUBTREE_SCR, etc.) is stored in the level field. The partition state which uses the same values defined for

replica states is stored the interval field (for possible values, see Replica States).

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Partition Creation Time

Identifies the particular incarnation of the list of replicas of a partition.

NDS Operational: Yes

Syntax

Timestamp (page 983)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.64

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Partition Creation Time is also known as the partition's "epoch." An epoch is defined by explicit management intervention to recover from the hard failure of a replica that prevents synchronizations from completing.

The LDAP server does not currently support the Timestamp syntax. LDAP clients cannot access this attribute.

Partition Status

Provides information on the success or failure of the last synchronization cycle.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_PER_REPLICA
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SPARSE_REQUIRED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.176

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Password Allow Change

Determines whether the person logged in under an account can change the password for that account.

NDS Operational: Yes

Syntax

Boolean (page 922)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.66

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Password Expiration Interval

Specifies the time interval a password can remain active. When the interval expires, the current password is marked expired.

NDS Operational: Yes

Syntax

Interval (page 951)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.67

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Password Expiration Interval uses a syntax definition (SYN_INTERVAL) that is a 32-bit integer. The time is stored in seconds. NWAdmin divides the value by 86400 (seconds per day) and displays it in days.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Password Expiration Time

Specifies when the password will expire.

NDS Operational: Yes

Syntax

Time (page 981)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.68

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Password Management

Allows an object to be granted password management rights for all objects in a container or subtree without having full supervisor rights.

NDS Operational: Yes

Syntax

Unknown (page 989)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.185

Used In

(None)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is new in NetWare 5.x and has been defined specifically for password management, and as such, can never be added to any class definition. It is a special attribute to be used in ACLs. When referenced in an ACL, it grants rights to modify the password of the object the ACL belongs to, or if inheritance is set on the ACL, to all subordinate objects.

To grant permission to change the password, NDS now looks for either write rights on the ACL attribute (original behavior) or write rights to the Password Management attribute (new behavior).

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Password Minimum Length

Establishes the minimum length for an object's clear-text password.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.69

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Password Required

Establishes that a password is required for the object to log in.

NDS Operational: Yes

Syntax

Boolean (page 922)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.70

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Password Unique Required

Establishes that when an object's password is changed, it must be different from those in the Passwords Used attribute.

NDS Operational: Yes

Syntax

Boolean (page 922)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.71

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Passwords Used

Specifies old (previously used) passwords.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_HIDDEN_ATTR

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.65

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Applications cannot access this attribute since it is a hidden attribute.

The Passwords Used attribute is maintained internally if Unique Passwords Required is set to True. When a password is changed, the old password value is stored in this attribute. Before a new password is accepted, this list is searched to see if the new password is already in the list. Old passwords are purged from the list when the password timestamp is older than 8 times the password expiration interval.

Path

Specifies the physical location of a file system directory.

NDS Operational: Yes

LDAP Name

path

Syntax

Path (page 966)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.72

Used In

Directory Map (page 116)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Permanent Config Parm

Holds the permanent configuration parameters.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_OPERATIONAL

DS_PUBLIC_READ

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.177

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Physical Delivery Office Name

Specifies the name of the city, village, or place, where a physical delivery office is located.

NDS Operational: Yes

LDAP Name

physicalDeliveryOfficeName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (1..128)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.19

Used In

External Entity (page 125)
Organization (page 202)
Organizational Person (page 207)
Organizational Role (page 212)
Organizational Unit (page 215)
residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Postal Address

Specifies the address information required for the physical delivery of postal messages to a named object.

NDS Operational: Yes

LDAP Name

postalAddress

Syntax

Postal Address (page 969)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.16

Used In

External Entity (page 125)

Organization (page 202)

Organizational Person (page 207)

Organizational Role (page 212)

Organizational Unit (page 215)

residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A value for this attribute is typically composed of selected attributes from the MHS Unformatted Postal O/R Address version 1, according to Recommendation F.401. The address is limited to 6 lines of 30 characters each, including a Postal Country Name.

Normally the information contained in such an address could include an addressee's name, street address, city, state or province, postal code and possibly a Post Office Box number, depending on the specific requirements of the named object.

NetWare administrative utilities will link the postal address value to the following attributes: Physical Delivery Office Name, Postal Code, Postal Office Box, and Street Address.

The postal address uses the following as default values:

Line 1: The object's RDN

Line 2: Street Address or Post Office Box Number

Line 3: (no default value)

Line 4: Physical Delivery Office Name, State or Province Name

Line 5: Postal Code

Line 6: Country Name (from the object's DN)

The second line of the postal address will default to the post office box, if present, or otherwise to the street address. The third line of the postal address should be presented in the utility as the second line of both the post office box and street address. Each line of the postal address is limited to 30 characters. Attribute values are truncated to 30 characters when used in the postal address.

Developers are encouraged to follow these guidelines.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Postal Code

Specifies the postal code of the named object. If this attribute value is present, it is part of the object's postal address.

NDS Operational: Yes

LDAP Name

postalCode

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (0..40)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.17

Used In

External Entity (page 125)
Organization (page 202)
Organizational Person (page 207)
Organizational Role (page 212)
Organizational Unit (page 215)
residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Postal Office Box

Specifies the post office box at which the object receives physical postal delivery.

NDS Operational: Yes

LDAP Name

postOfficeBox

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (0..40)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.18

Used In

External Entity (page 125)
Organization (page 202)
Organizational Person (page 207)
Organizational Role (page 212)
Organizational Unit (page 215)
residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Postmaster

Specifies one or more users who have the privileges to manage a messaging server.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.128

Used In

Messaging Server (page 150)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Postmasters have the required rights to remove mailboxes. They also receive messages about special events in the messaging server, such as messages not being processed.

preferredDeliveryMethod

Specifies the preferred delivery method for a message.

LDAP Name

preferredDeliveryMethod

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.28

Used In

Organizational Person (page 207)
residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A non-extended LDAP schema query returns only the single-valued constraint.

For more information, see RFC 2256.

presentationAddress

Specifies an OSI presentation layer address.

LDAP Name

presentationAddress

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.29

Used In

applicationEntity (page 80)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A non-extended LDAP schema query returns only the single-valued constraint.

For more information, see RFC 2256.

Print Job Configuration

Contains information on the specified print job configuration.

NDS Operational: Yes

Syntax

Stream (page 977)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.80

Used In

Organization (page 202)

Organizational Unit (page 215)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Print Server

Designates an object as the host server for a specific printer.

NDS Operational: Yes

LDAP Name

printServer

Syntax

Typed Name (page 986)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.81

Used In

Printer (page 234)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Printer

Contains a list of printer object names that are serviced by the print server.

NDS Operational: Yes

LDAP Name

printer

Syntax

Typed Name (page 986)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.77

Used In

Print Server (page 230)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Printer Configuration

Contains information on the specified printer configuration.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.78

Used In

Printer (page 234)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Printer Control

Contains the NDS counterpart of the DOS printer definition file, NET\$PRN.DAT.

NDS Operational: Yes

Syntax

Stream (page 977)

Constraints

DS_NONREMOVABLE_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.79

Used In

Organization (page 202)
Organizational Unit (page 215)
User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Stream syntax.

Private Key

Contains an RSA private key.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_HIDDEN_ATTR
DS_NONREMOVABLE_ATTR
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.82

Used In

Organizational Unit (page 215)
pkiCA (page 226)
Server (page 254)
User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Your applications cannot access this attribute since it is a hidden attribute.

This attribute is used to produce signed authentication information that can be used to verify the identity of an object.

The Private Key attribute contains an RSA private key encrypted with the password of the object.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Profile

Identifies the login profile to be used if the user doesn't specify one at login time.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.83

Used In

Group (page 128)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Profile Membership

Contains a list of profiles that the object can use.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

DS_WRITE_MANAGED

ASN.1 ID

2.16.840.1.113719.1.1.4.1.171

Used In

Group (page 128)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute allows objects to have multiple profiles. This is useful for NNS users who are brought into the NDS tree, since NNS allows users to have multiple domains.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

protocolInformation

Specifies information which is used with the presentationAddress attribute for the OSI network service layer.

LDAP Name

protocolInformation

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.48

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

Public Key

Contains a certified RSA public key.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.84

Used In

Organizational Unit (page 215)
pkiCA (page 226)
Server (page 254)
User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute verifies authentication information produced with the corresponding private key.

The attribute contains the public key, along with information that certifies the integrity of the public/private key pair. Certification information is necessary to prevent legitimate key pairs from being replaced by illicit key pairs.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Purge Vector

Contains a publicly available attribute that can be used by NDS management utilities (such as NDS Manager) to determine the synchronization status of a given partition.

NDS Operational: Yes

Syntax

Timestamp (page 983)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_PER_REPLICA
DS_SCHEDULE_SYNC_NEVER
DS_SPARSE_REQUIRED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.183

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is new in NetWare 5.x. It gives the time since the last successful synchronization cycle that allowed NDS to purge old information because it has now been seen by all other interested servers.

The LDAP server does not currently support the Timestamp syntax.

Queue

Contains the distinguished name of the queue with which the object is associated.

NDS Operational: Yes

LDAP Name

queue

Syntax

Typed Name (page 986)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.85

Used In

Printer (page 234)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The attribute contains the queue's Distinguished Name and the queue's priority.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Queue Directory

Contains the name of the subdirectory where the queue's files are stored.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SERVER_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..255)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.86

Used In

Queue (page 241)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The subdirectory name should include the local volume designation along with the subdirectory name. The subdirectory name should be a valid DOS name.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Received Up To

Specifies the last time the replica has received any updates.

NDS Operational: Yes

Syntax

Timestamp (page 983)

Constraints

DS_NONREMOVABLE_ATTR

DS_PUBLIC_READ

DS_READ_ONLY_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.87

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The LDAP server does not currently support the Timestamp syntax. LDAP clients cannot access this attribute.

Reference

Contains a list of objects referenced by the owner of this attribute.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_HIDDEN_ATTR
DS_NONREMOVABLE_ATTR
DS_READ_ONLY_ATTR
DS_PER_REPLICA
DS_SPARSE_REQUIRED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.115

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Applications cannot access to this attribute since it is a hidden attribute.

This attribute is kept only on the replica in which it originated. It is not skulked or updated to other replicas. It is used to make it easier to find the objects that have a reference to a certain object. If objects A, B, C, and D all have privileges granted to object X (through use of the ACL attribute), the Reference attribute of object X will contain the object names of A, B, C, and D.

This attribute is also used to list bindery object relations in Bindery Services.

registeredAddress

Specifies a postal address for the delivery of telegrams or expedited documents which require the receiver to accept delivery.

LDAP Name

registeredAddress

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.26

Used In

Organizational Person (page 207)
residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

Replica

Identifies the NDS servers that store replicas of a partition.

NDS Operational: Yes

Syntax

Replica Pointer (page 974)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.88

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is multivalued. It is present and nonNULL in every partition entry and cannot be directly modified. The attribute is updated as a side-effect of the operations that create, destroy, or change the replication of a partition.

The LDAP server does not currently support the Replica Pointer syntax. LDAP clients cannot access this attribute.

Replica Up To

Identifies the changes the object has sent out to other replicas of a partition.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR
DS_PER_REPLICA
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.175

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Resource

Contains a list of resources associated with the object.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.89

Used In

Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute lists resources that are managed by the server.

Revision

Specifies the revision number of the object.

NDS Operational: Yes

LDAP Name

revision

Syntax

Counter (page 936)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SCHEDULE_SYNC_NEVER

ASN.1 ID

2.16.840.1.113719.1.1.4.1.122

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Role Occupant

Specifies the name of an object that fulfills an organizational role.

NDS Operational: Yes

LDAP Name

roleOccupant

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.33

Used In

Organizational Role (page 212)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

roomNumber

Specifies a room number.

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.6

Used In

inetOrgPerson (page 291)

Organizational Person (page 207)

Person (page 222)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Run Setup Script

Specifies whether a setup script is to be run when the User objects are created with this Template object.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.58.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

S (State or Province Name)

Specifies the name of a state or province.

NDS Operational: Yes

LDAP Name

st
stateOrProvinceName

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (1..128)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.8

Used In

External Entity (page 125)
Locality (page 141)
Organization (page 202)
Organizational Person (page 207)
Organizational Role (page 212)
Organizational Unit (page 215)
Person (page 222)
residentialPerson (page 332)
Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When used as a component of an NDS name, a state or province name identifies a geographical subdivision in which the named object is either physically located in, or associated with, in some other important way.

An LDAP schema query requires extensions to return the NDS attribute constraints.

SA (Street Address)

Specifies a site for local distribution and physical delivery of mail in the form of a postal address (that is, the street name, place, avenue, and house number).

NDS Operational: Yes

LDAP Name

street

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SIZED_ATTR (1..128)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.9

Used In

External Entity (page 125)

Locality (page 141)

Organization (page 202)

Organizational Person (page 207)

Organizational Role (page 212)

Organizational Unit (page 215)

residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When used as a component of an NDS name, a street address identifies the address at which the named object is either located in, or associated with, in some other important way. The attribute value for street address is a string (for example, "Amulfstrae 60").

An LDAP schema query requires extensions to return the NDS attribute constraints.

SAP Name

Contains the name used by a print server when advertising itself using the NetWare Service Advertising Protocol (SAP).

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (1..47)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.91

Used In

Print Server (page 230)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

SAS:Security DN

In NetWare 5.x, identifies the distinguished name of the security container.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SERVER_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.39.4.1.1

Used In

Tree Root (page 266)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SAS:Service DN

Identifies the SAS:Service object that belongs to the NCP Server object.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SERVER_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.39.4.1.2

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute on the NCP Server works in conjunction with the Host Server attribute on the SAS:Service object. If you know the NCP Server object, then you can use the SAS:Service DN attribute to find the SAS:Service object. If you know the SAS:Service object, you can use the Host Server attribute to find the NCP Server object.

searchGuide

Specifies an attribute used in a search filter.

LDAP Name

searchGuide

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.14

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for backwards compatibility and has been replaced by the enhancedSearchGuide (page 423) attribute. For more information, see RFC 2256.

searchSizeLimit

Specifies the maximum number of objects that an LDAP search can return.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0..2147483647)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.194

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was redefined for NDS 8 to allow a 0 value to represent unlimited searches.

LDAP Services for NDS returns object data until the object number exceeds the value of this attribute. The LDAP server then stops sending object data and considers the request complete. The default value is 500.

searchTimeLimit

Specifies the maximum number of milliseconds that an LDAP search can take before returning.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0..2147483647)
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.195

Used In

LDAP Server (page 135)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is an LDAP Services for NDS configuration parameter. The default is 3600 seconds. It was redefined for NDS 8 to allow a 0 value to represent an unlimited time.

The LDAP server continues processing the request until the search time has expired. It then returns an error reporting that the search time was exceeded. The LDAP server considers the request complete.

Security Equals

Specifies group membership and security equivalences of an object.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SERVER_READ

DS_SYNC_IMMEDIATE

DS_WRITE_MANAGED

ASN.1 ID

2.16.840.1.113719.1.1.4.1.92

Used In

Server (page 254)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Security Flags

Specifies the NCP Packet Signature level of the object.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.165

Used In

Server (page 254)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NCP Packet Signature uses the following four security flags:

0 = No signing

1 = Signing allowed

2 = Signing preferred

3 = Signing mandatory

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

See Also

Specifies the names of NDS objects that can reflect other aspects of the same real world object.

NDS Operational: Yes

LDAP Name

seeAlso

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.34

Used In

applicationEntity (page 80)

applicationProcess (page 83)

Device (page 113)

External Entity (page 125)

Group (page 128)

List (page 138)

Locality (page 141)

NSCP:groupOfCertificates (page 180)

Organization (page 202)

Organizational Role (page 212)

Organizational Unit (page 215)

Person (page 222)

Profile (page 238)
Resource (page 245)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Serial Number

Specifies the serial number of a device.

NDS Operational: Yes

LDAP Name

serialNumber

Syntax

Printable String (page 971)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (1..64)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.5

Used In

AFP Server (page 74)
Device (page 113)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extensions to return the NDS attribute constraints.

Server

Specifies a list of servers.

NDS Operational: Yes

LDAP Name

server

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SERVER_READ

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.95

Used In

Computer (page 101)

Queue (page 241)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Server Holds

Contains the number of accounting charges pending while a server performs a chargeable action.

NDS Operational: Yes

Syntax

Hold (page 946)

Constraints

DS_NONREMOVABLE_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.96

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The Server Holds attribute is used when NetWare Accounting is active. Each time a server is about to perform an action that will be charged against a user's account, the server makes sure the account has a sufficient balance. To do this, the server places a hold against an object's balance, which is an estimate of what the final charge will be. If the hold is successful, meaning sufficient balance remains, the action is performed. When the action is completed, the hold is cancelled and a true charge for the actual amount is made against the object's balance.

Up to 16 different servers can have holds pending against a user's balance simultaneously. When a hold is pending, this attribute contains the name of the server requesting the hold and the total hold amount.

The LDAP server does not currently support the Hold syntax. LDAP clients cannot access this attribute.

Set Password After Create

Determines whether the network administrator is prompted to create a password for the User when the User object is created with this Template object.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.61.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Setup Script

Specifies a setup script that can be used to configure the User objects created with this Template object.

Syntax

Stream (page 977)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.57.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Status

Specifies the operational state of the specified object.

NDS Operational: Yes

LDAP Name

status

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.98

Used In

Computer (page 101)
Printer (page 234)
Server (page 254)
Volume (page 281)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Operational state values for Status might be Up (1) and Down (0). The values vary with the object.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

supportedAlgorithms

Specifies supported algorithms in binary format.

LDAP Name

supportedAlgorithms

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.52

Used In

userSecurityInformation (page 279)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

supportedApplicationContext

Specifies the identifiers for the OSI application contexts in the application layer.

LDAP Name

supportedApplicationContext

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.30

Used In

applicationEntity (page 80)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

Supported Connections

Identifies the number of concurrent connections a server allows.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.100

Used In

AFP Server (page 74)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Supported Gateway

Contains a list of gateways supported by the associated object.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVEABLE_ATTR

DS_SIZED_ATTR (1..4096)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.173

Used In

Messaging Server (page 150)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute provides messaging connectivity between the MHS messaging system and external messaging systems. This attribute supports existing third-party gateway products.

The gateway names consist of two sub-attributes: gateway name and gateway protocol type. The gateway names have the convention of eight or fewer characters, followed by "/", followed by another eight or fewer characters.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Supported Services

Contains a list of services that the associated object supports.

NDS Operational: Yes

LDAP Name

supportedServices

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SIZED_ATTR (1..64)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.101

Used In

Messaging Server (page 150)

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

Supported Typefaces

Identifies the typefaces supported by a printer. Multiple typefaces should be represented as multiple values.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_SIZED_ATTR (1..64)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.102

Used In

Printer (page 234)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An example of a value for this attribute is

Supported Typefaces=computer modern, American modern, CM, AM

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Surname

Specifies the name an individual inherits from a parent (or assumes by marriage) and by which the individual is commonly known.

NDS Operational: Yes

LDAP Name

sn

surname

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_PUBLIC_READ

DS_SIZED_ATTR (1..64)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.4

Used In

Person (page 222)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The value depends upon the culture. For example, in the American name “Robert Jones,” “Jones” is the surname. However, in the Chinese name “Sun Yat-sen,” “Sun” is the surname.

An LDAP schema query requires extensions to return NDS attribute constraints.

In NDS 8 and above, Surname is multi-valued. In previous releases of NDS, Surname is single-valued.

Synchronization Tolerance

Determines how old the timestamps in the partition's Purge Vector can be and still allow the partition to be considered synchronized with the other replicas of the partition.

NDS Operational: Yes

Syntax

Timestamp (page 983)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_PUBLIC_READ
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.184

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is new in NetWare 5.x.

The LDAP server does not currently support the Timestamp syntax. LDAP clients cannot access this attribute.

Synchronized Up To

Contains a list of time stamps that indicate the last time all servers holding a copy of the specified replica were synchronized.

NDS Operational: Yes

Syntax

Timestamp (page 983)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_PER_REPLICA
DS_SPARSE_REQUIRED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.33

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The LDAP server does not currently support the Timestamp syntax. LDAP clients cannot access this attribute.

T (Tree Name)

Specifies the name of a Tree Root object.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_OPERATIONAL

DS_SIZED_ATTR (1..32)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.181

Used In

Tree Root (page 266)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is new in NetWare 5.x. For the Tree Root class, this attribute is mandatory. A tree name can be 1 to 32 characters long.

Telephone Number

Specifies a telephone number associated with an object.

NDS Operational: Yes

LDAP Name

telephoneNumber

Syntax

Telephone Number (page 979)

Constraints

DS_NONREMOVABLE_ATTR
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.20

Used In

Organization (page 202)
Organizational Role (page 212)
Organizational Unit (page 215)
Person (page 222)
residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A value for this attribute is a string that complies with the internationally agreed format for showing international telephone numbers, Recommendation E.123 (for example, "+ 44 582 10101").

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

telexNumber

Specifies a telex number.

LDAP Name

telexNumber

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.21

Used In

Organizational Person (page 207)
residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

telexTerminalIdentifier

Specifies an identifier for a telex terminal.

LDAP Name

telexTerminalIdentifier

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.22

Used In

Organizational Person (page 207)
residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

Timezone

Used to set the time zone offset for a user (or other objects that log in) so that the login restrictions can maintain appropriate login time restrictions by adjusting to daylight saving and time zone differences.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.178

Used In

Server (page 254)
User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Title

Specifies the designated position or function of an object within an organization.

NDS Operational: Yes

LDAP Name

title

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SIZED_ATTR (1..64)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

Used In

External Entity (page 125)

Organizational Person (page 207)

ASN.1 ID

2.5.4.12

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An attribute value for this attribute is a string. For example:

"Manager, Distributed Applications."

An LDAP schema query requires extensions to return the NDS attribute constraints.

Transitive Vector

Used on partition root objects so NDS can manage the synchronization process and determine what needs to be sent to other replicas.

NDS Operational: Yes

Syntax

Octet String (page 962)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SCHEDULE_SYNC_NEVER
DS_SPARSE_REQUIRED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.180

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDS 6.xx and 7.xx, this attribute has the following flags: DS_SYNC_IMMEDIATE and DS_PER_REPLICA. In NDS 8 and above, these flags have been removed.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Octet String syntax.

Trustees Of New Object

Specifies which NDS objects have NDS rights to the User objects created with this Template object.

Syntax

Object ACL (page 957)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.54.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Type Creator Map

Identifies an object as a Macintosh file-system client.

NDS Operational: Yes

Syntax

Stream (page 977)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.107

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used internally by “NetWare for Macintosh” to identify clients that should be given type and creator descriptions when file information is requested.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

UID (User ID)

Specifies a unique user ID for use by UNIX clients.

NDS Operational: Yes

Syntax

Integer (page 948)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.108

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

uniqueID

Contains an RDN-type name that can be used to create a unique name in the NDS tree.

NDS Operational: Yes

LDAP Name

uid

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SIZED (1...64)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.1

Used In

inetOrgPerson (page 291)
Organizational Person (page 207)
Person (page 222)

Remarks

LDAP applications generate uniqueID names and ensure their uniqueness.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Unknown

Contains an attribute value for an attribute that is no longer defined by the schema.

NDS Operational: Yes

Syntax

Unknown (page 989)

Constraints

DS_NONREMOVABLE_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.109

Used In

(Special)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is provided for use by the server in restoring objects. If an attribute definition has been deleted and an object that is being restored has a value of the deleted attribute type, the now undefined attribute will be restored as Unknown.

Unknown Auxiliary Class

Stores the name of an unknown auxiliary class.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints)

DS_READ_ONLY_ATTR

DS_SIZED (1...32)

DS_STRING_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.504

Used In

Top (page 263)

Remarks

When NDS declares an object unknown that contains attributes from what appears to be an auxiliary class, NDS adds this attribute to the object and stores the names of the auxiliary classes in this attribute.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Unknown Base Class

Stores the class an object was before it was changed to an unknown class.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (1..32)

DS_STRING_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.110

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons.

Used By

Contains the new Distributed Reference Links that NetWare 5.x uses instead of back links.

NDS Operational: Yes

Syntax

Path (page 966)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SERVER_READ
DS_SPARSE_REQUIRED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.186

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Distributed Reference Links manage references throughout the NDS tree and are automatically maintained by NDS. These links identify a partition root, and that specified partition contains an object that references the object with the Used By attribute. The Path syntax contains three uint32 fields and a data field. The Used By attribute uses only the first two fields. The first field identifies whether the attribute points to a partition (0) or to a volume (1). The second field is the partition root ID.

For LDAP clients to access this attribute, the LDAP server must be configured to map the attribute to a name without spaces or colons. The LDAP server in NDS eDirectory supports the Path syntax.

User

Contains a list of users associated with the object.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SERVER_READ

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.111

Used In

Queue (page 241)

Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute contains a list specifying which users are authorized to use the queue or server.

userCertificate

Specifies a certificate for certificate management.

LDAP Name

userCertificate
userCertificate;binary

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.36

Used In

inetOrgPerson (page 291)
pkiUser (page 228)
strongAuthenticationUser (page 258)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is used by Certificate Authorities (Novell and third party).

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

userPassword

Allows an entry's password to be set or changed by LDAP clients.

NDS Operational: Yes

LDAP Name

userPassword

Syntax

Octet String (page 962)

ASN.1 ID

2.5.4.35

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is accessible only through LDAP and is a write-only attribute.

Passwords in NDS are stored as a RSA public and private key pairs. The Novell LDAP server uses the userPassword attribute to generate these key pairs for an LDAP client.

- ♦ NDS 8.17 or higher is required for users to change their passwords.
- ♦ NDS 7.xx and higher is required for an administrator to change user passwords.

To change a password for a user, administrators are required only to submit a new value for the userPassword attribute.

For a user to change his or her own password, the user must submit a delete request for the userPassword attribute with the current value and an add request for the userPassword attribute with the new value.

When creating a user, a value must be set for the userPassword attribute in order for the user to log in to the directory.

When using an LDAP compare function to verify a userPassword value, the function can return true and the client can still be locked out of the account. NDS uses the following attributes, not just the userPassword attribute, to control access to an account:

- ♦ Locked By Intruder (page 499)
- ♦ Login Allowed Time Map (page 501)
- ♦ Login Disabled (page 502)
- ♦ Login Expiration Time (page 503)
- ♦ Login Maximum Simultaneous (page 510)
- ♦ Password Expiration Interval (page 617)
- ♦ Password Required (page 622)

If the password is verified as valid, these other attributes should be checked to determine why the client cannot access the account.

Uses

Indicates when a resource is referencing another object outside the NDS tree.

NDS Operational: Yes

Syntax

Path (page 966)

Constraints

DS_NONREMOVABLE_ATTR
DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SERVER_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.187

Used In

Resource (page 245)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is new in NetWare 5.x and is similar to the Used By attribute, but is managed by external resource agents rather than NDS. The Volume agent uses this attribute.

Version

Contains a string that describes the version identifier of the software associated with the object.

NDS Operational: Yes

Syntax

Case Ignore String (page 929)

Constraints

DS_NONREMOVABLE_ATTR
DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..64)
DS_STRING_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.112

Used In

inetOrgPerson (page 291)
LDAP Server (page 135)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Volume

Contains the distinguished name of a volume on which a queue is located.

NDS Operational: Yes

Syntax

Distinguished Name (page 938)

Constraints

DS_NONREMOVABLE_ATTR

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.1.4.1.113

Used In

Queue (page 241)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Volume Space Restrictions

Specifies the location and the amount of the space restrictions that are assigned to the User objects created with this Template object.

Syntax

Path (page 966)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.60.4.1.1

Used In

Template (page 260)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

WANMAN:Cost

Contains the traffic cost that a range of network addresses uses.

Syntax

Octet String (page 962)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.38.4.1.3

Used In

NCP Server (page 155)

WANMAN:LAN Area (page 285)

Remarks

This attribute contains three fields:

- ♦ Cost.

Specifies the cost of traffic on the range of addresses. The network administrator enters this value, which can be expressed in time, money, or some other measurement.

- ♦ Start Address.

Specifies the beginning range of the network address. This field also keeps track of the type of address (for example, IPX or IP).

- ♦ Stop Address.

Specifies the ending range of the network address. This address can be the same as the Start Address, indicating that the cost applies only to a single network.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

WANMAN:Default Cost

Identifies the cost for networks when the network address is not found in the WANMAN:Cost attribute.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.38.4.1.4

Used In

NCP Server (page 155)

WANMAN:LAN Area (page 285)

Remarks

This attribute allows the network administrator to set a default cost value for all networks that do not require a unique cost value.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

WANMAN:LAN Area Membership

Identifies the LAN area group that this server belongs to.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.38.4.1.2

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

WANMAN:WAN Policy

Contains a WAN policy that is used to determine when traffic should be sent.

Syntax

Octet List (page 960)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.38.4.1.1

Used In

NCP Server (page 155)

WANMAN:LAN Area (page 285)

Remarks

When this attribute is assigned to an NCP Server object, the policy applies just to that server. When the attribute is assigned to the WANMAN:LAN Area object, the policy applies to a group of servers.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

x121Address

Specifies an address used in electronic data exchange.

LDAP Name

x121Address

Syntax

Numeric String (page 955)

Constraints

DS_SIZED_ATTR (1..15)

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.24

Used In

Organizational Person (page 207)

residentialPerson (page 332)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

For more information, see RFC 2256.

x500UniqueIdentifier

Specifies an identifier to use in distinguishing between objects when a DN has been reused.

LDAP Name

x500UniqueIdentifier

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.5.4.45

Used In

inetOrgPerson (page 291)
Organizational Person (page 207)
User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

An LDAP schema query requires extended terms (X_NDS) to return the NDS attribute constraints.

6

Novell Attribute Extensions

This chapter lists alphabetically all the attribute definitions that extend the schema in a default NetWare installation or from the downloadable schema files (<http://www.novell.com/products/nds/schema/index.html>). It also includes the obsolete attributes that have extended the schema in NDS releases previous to NDS eDirectory 8.5.

- ♦ For attribute definitions that are installed with NDS, see “Base Attribute Definitions” on page 361.
- ♦ For an explanation of the types of information included about each attribute, see “Reading NDS Attribute Type Definitions” on page 63.

audio

Contains an audio file in binary format.

LDAP Name

audio

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.55

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

carLicense

Contains the license plate number of the object's car.

LDAP Name

carLicense

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.1

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

Client Install Candidate

Indicates that the PSetup utility should be used to install the printer on the workstation.

Syntax

Boolean (page 922)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.10

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Color Supported

Indicates that the printer supports color.

Syntax

Boolean (page 922)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.24

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Database Dir Path

Contains some of the information required to locate the Printer Manager's Managed Object Database (MODB).

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0.4

Used In

NDPS Manager (page 299)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

See the Database Volume Name attribute for the other information required to locate the MODB.

Database Volume Name

Contains some of the information required to locate the Printer Manager's Managed Object Database (MODB).

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0.2

Used In

NDPS Manager (page 299)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

See the Database Dir Path attribute for the other information required to locate the MODB.

Datapool Location

Defines the volume object where spooling for the printer takes place.

Syntax

Typed Name (page 986)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.30

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Datapool Locations

Defines the volume objects where spooling for each printer takes place.

Syntax

Typed Name (page 986)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0.1

Used In

NDPS Manager (page 299)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Delivery Methods Installed

Defines the notification method NLMs that are loaded with the Notification Service.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.0

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

displayName

Contains the object's preferred name for displaying in a list of objects.

LDAP Name

displayName

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.241

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support. Since the CN attribute is multi-valued, CN cannot be used to set a name preference.

Employee ID

Specifies an ID that has been assigned to an employee.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

Reserved

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For NetWare 5.0, LDAP added this attribute for Netscape support. LDAP maps requests for the employeeNumber attribute to this attribute. For more information, see the documentation for the Netscape SuiteSpot schema.

In NDS 8, LDAP no longer uses this attribute. See NCP:employeeNumber.

Entrust:AttributeCertificate

Replaced by attributeCertificate (page 367).

Entrust:User

Specifies an Entrust user.

Syntax

Case Exact String (page 924)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.254

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

LDAP added this attribute for Entrust support. The LDAP server maps requests for the `entrustUser` attribute to this attribute. For more information, see the documentation for the Entrust schema.

GW API Gateway Directory Path

Defines the GroupWise API Gateway's directory path.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.12

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

GW API Gateway Directory Volume

Defines the GroupWise API Gateway's volume.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUE_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.13

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

IPP URI

Defines the Internet Printing Protocol's Uniform Resource Identifier list.

Syntax

Case Ignore List (page 926)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.34

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute contains the printer's name in the IPP environment.

IPP URI Security Scheme

Defines the Internet Printing Protocol's security schemes that correspond to each Uniform Resource Identifier in the IPP URL.

Syntax

Case Ignore List (page 926)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUE_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.35

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

jpegPhoto

Contains a JPEG photo of the object.

LDAP Name

jpegPhoto

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.60

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support. In NDS, the JPEG file cannot be larger than 64 K.

labeledUri

Contains the object's Uniform Resource Identifier (URI).

LDAP Name

labeledUri

Syntax

Case Ignore String (page 929)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.57

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support. Currently, Netscape expects the URI to be a URL.

LDAP Class Map

Is reserved.

Syntax

Stream (page 977)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

None

Used In

None

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

IdapPhoto

Contains a photo of the object in binary format.

LDAP Name

photo

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.7

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

In NDS, the photo file cannot be larger than 64 K.

LDAPUserCertificate

Replaced by userCertificate (page 715).

LDAP:ARL

Replaced by authorityRevocationList (page 377).

LDAP:caCertificate

Replaced by cACertificate (page 392).

LDAP:CRL

Replaced by `certificateRevocationList` (page 396).

LDAP:crossCertificatePair

Replaced by `crossCertificatePair` (page 399).

MASV:Authorized Range

Contains a list of clearances that the object is authorized to use during login and authentication.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.1.6

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The session clearance requested during log in and authentication must be within the list of clearances defined by this attribute.

MASV:Default Range

Defines the session clearance to assign if one is not specified during login and authentication.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.1.5

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This value must be within the list of authorized clearances as specified in the MASV:Authorized Range attribute.

MASV:Domain Policy

Defines security policy information which is global to an NDS tree.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.1.7

Used In

MASV:Security Policy (page 144)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The global policy information includes default classifications for resources and default clearances for logged-in objects. These default values are used only if classifications and clearances are not explicitly defined for an object.

MASV:Label

Defines the classification of all NDS objects contained on the partition.

Syntax

Octet String (page 962)

Constraints

DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.1.3

Used In

Partition (page 219)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Once a label is explicitly assigned, the partition label cannot be modified.

MASV:Proposed Label

Defines the classification to be assigned to a partition that is to be created by splitting an existing partition.

Syntax

Octet String (page 962)

Constraints

DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.31.4.1.4

Used In

Top (page 263)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A proposed label is not a label, but rather a value that will be assigned when the partition is split to create a new subordinate partition. If a proposed label is not assigned when a partition is created, the label of the new partition is set to the parent partition’s label. The MASV:Proposed Label attribute may be modified up until the new partition is created.

Maximum Speed

Represents the maximum printing speed of the printer's engine.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..10000)

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.44

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The Maximum Speed Units attribute defines the speed units.

Maximum Speed Units

Represents a value which is interpreted as a rate of printing.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED (1..1000000)

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.43

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The speed unit can represent a value such as pages per minute or characters per second.

MHS Send Directory Path

Defines the MHS Send directory path

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.1

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

MHS Send Directory Volume

Defines the MHS Send directory volume.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.2

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Accountant Role

Identifies who has accountant rights to the printer object.

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.34

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When rights to this attribute are granted to a User or container object, then the User or the users in the container become accountants on the NDPS printer that owns this attribute.

NDPS Control Flags

Controls the automatic printer installation features of NDPS.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0..4294967295)

ASN.1 ID

2.16.840.1.113719.1.4.2.3.7

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Database Saved Timestamp

Records the time of the last backup of the NDPS database.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUE_ATTR
DS_SIZED_ATTR (0..4294967295)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0.5

Used In

NDPS Manager (page 299)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A complete backup of the NDPS database includes the backup of the two NDPS stream attributes: the NDPS Database Saved Data Image attribute and the NDPS Database Saved Index Image attribute.

NDPS Database Saved Data Image

Contains the last backup of the NDPS database and is paired with the NDPS Database Saved Index Image attribute.

Syntax

Stream (page 977)

Constraints

DS_SINGLE_VALUE_ATTR
DS_SIZED_ATTR (0..4294967295)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0.6

Used In

NDPS Manager (page 299)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Database Saved Index Image

Contains the index of the last backup of the database and is paired with the NDPS Database Saved Data Image attribute.

Syntax

Stream (page 977)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (0..4294967295)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0.7

Used In

NDPS Manager (page 299)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Default Printer

Controls the behavior of the automatic printer installation features of NDPS.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.3.1

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Default Public Printer

Controls the behavior of the automatic printer installation features of NDPS.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.3.2

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Job Configuration

Contains named job configurations that can be used to affect (jobs submitted to) installed, controlled-access printers.

Syntax

Stream (page 977)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATT

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.23

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Manager Status

Contains a status in the form of an OID which represents the current status of the NDPS Manager object.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0.3

Used In

NDPS Manager (page 299)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Operator Role

Indicates who has operator rights to the printer in question.

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.0

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When rights to this attribute are granted to a User or a container object, then the User or the users in the container become operators on the NDSP printer that owns this attribute.

NDPS Printer Install List

Controls the behavior of the automatic printer installation features of NDPS.

Syntax

Typed Name (page 986)

Constraints

DS_PULIC_READ

ASN.1 ID

2.16.840.1.113719.1.4.2.3.3

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Printer Install Timestamp

Controls the behavior of the automatic printer installation features of NDPS.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.3.5

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Printer Queue List

Describes the QMS queues which are serviced by the NDPS Printer object which owns this attribute.

Syntax

Typed Name (page 986)

Constraints

DS_PUBLIC_READ

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.26

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Printer Siblings

Identifies other NDPS Printer objects which are used to allow access to a particular piece of printer hardware.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.31

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute allows multiple printer objects to share the same piece of printer hardware. This feature allows the configurations and limitations of one printer object to be isolated from those of another while allowing diverse groups of people to share the same printer.

NDPS Public Printer Install List

Controls the behavior of the automatic printer installation features of NDPS.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ

ASN.1 ID

2.16.840.1.113719.1.4.2.3.4

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS Replace All Client Printers

Controls the behavior of the automatic printer installation features of NDPS.

Syntax

Boolean (page 922)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.3.6

Used In

Organization (page 202)
Organizational Unit (page 215)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS SMTP Server

Identifies the mail server through which to send notification when the SMTP method is used.

Syntax

Case Exact String (page 924)

Constraints

DS_SIZED_ATTR (1..2000)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.17

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDPS User Role

Identifies who can use the printer.

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.1

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When rights to this attribute are granted to a User, Group, or container object, then the User or the users in the Group or container object can use the NDPS Printer object that owns this attribute.

NDSCat:Actual All Attributes

Specifies whether the catalog dredger finished its dredge operation.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.255

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

If TRUE, all attributes of the objects found in dredge are stored in the catalog.

NDSCat:Actual Attribute Count

Specifies the number of attributes found and stored in the catalog.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.256

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Actual Attributes

Specifies the names of the attributes stored in the catalog if the value of NDSCat:All Attributes is FALSE.

Syntax

Case Exact String (page 924)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.257

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Actual Base Object

Lists all the entries that were searched.

Syntax

Distinguished Name (page 938)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.258

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A base object is a entry that indicates where a search should start. It can be any container or leaf object in the NDS tree.

NDSCat:Actual Catalog Size

Specifies the size of the catalog in bytes.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.259

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Actual End Time

Specifies the time that the dredge completed.

Syntax

Time (page 981)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.260

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Actual Filter

Specifies the filter that the dredger used to created the last completed catalog.

Syntax

Case Exact String (page 924)

Constraints

DS_SINGLE_VALUED_ATTR
DS_STRING_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.261

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Actual Object Count

Specifies the number of objects added to the catalog database.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.262

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Actual Return Code

Specifies the return code from the last completed dredge.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.263

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

A zero (0) return code indicates a completed dredge with no errors.

NDSCat:Actual Scope

Specifies the scope used for the last dredge.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.264

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDS Catalog Services uses the following values in this attribute:

0 = search a single object

1 = search the immediate subordinates of an object, but not the object

2 = search the object and all of its subordinates

NDSCat:Actual Search Aliases

Specifies whether aliases were dereferenced during the last completed dredge.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.265

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When set to TRUE, NDS Catalog Services dereferences aliases and places them in the catalog with the name of the object they reference.

NDSCat:Actual Start Time

Specifies the time the last dredge started.

Syntax

Time (page 981)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.266

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Actual Value Count

Specifies the number of values added to the catalog database during the last completed dredge.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.267

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:All Attributes

Specifies whether the dredger should include all attributes or only those stored in NDSCat:Attributes.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.273

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The default value for this attribute is FALSE, indicating to the dredger to store only the attributes listed in NDSCat:Attributes (page 798).

NDSCat:AttrDefTbl

Specifies a list, in table format, of all the attributes and their assigned identification numbers which have been defined to be included in the catalog.

Syntax

Octet List (page 960)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.268

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Attributes

Contains a list of the attributes the dredger should store in the catalog database.

Syntax

Case Exact String (page 924)

Constraints

DS_STRING_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.274

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

If no attributes are listed and NDSCat:All Attributes is set to FALSE, the dredger still stores the following about all objects in the NDS database:

- ♦ NDS Canonical Name (the slash format of the distinguished name)
- ♦ Object class of the entry
- ♦ Distinguished name of the entry in dot format

The dredger stores the following about the attributes that have been selected from the NDS database:

- ♦ Attribute name
- ♦ Attribute syntax
- ♦ Attribute value

NDSCat:Auto Dredge

Specifies whether the dredger uses the NDSCat:Start Time and NDSCat:Dredge Interval attributes to determine when the next dredge occurs.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.275

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When set to TRUE, the dredger schedules its next dredge according to the NDSCat:Start Time and NDSCat:Dredge Interval attributes.

When set to FALSE, the dredger has no schedule for dredging and waits until the NDSCat:Auto Dredge attribute is set to TRUE.

NDSCat:Base Object

Contains a list of distinguished names of the containers or subtrees to search.

Syntax

Distinguished Name (page 938)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.276

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

If this attribute has no value, the dredger searches the entire NDS tree.

NDSCat:CatalogDB

Contains the catalog database written out on the last completed dredge.

Syntax

Stream (page 977)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.269

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Catalog List

Contains the list of catalog databases maintained by a specified server.

Syntax

Distinguished Name (page 938)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.288

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This is a multi-valued attribute that provides the distinguished names of the catalogs maintained by a specified server. The DSCat NLM accesses this list when creating or updating a catalog.

NDSCat:Dredge Interval

Specifies the number of seconds between dredges.

Syntax

Interval (page 951)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.277

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

If the dredge interval is smaller than the time required to perform a dredge, the dredger uses the value in the NDSCat:Start Time attribute to schedule the next dredge.

NDSCat:Filter

Contains the string that specifies the query parameters.

Syntax

Case Exact String (page 924)

Constraints

DS_SINGLE_VALUED_ATTR

DS_STRING_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.278

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:IndexDefTbl

Contains a table of all indexes for the given catalog with their assigned index numbers.

Syntax

Octet List (page 960)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.270

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The first field is the index name, and the second field is the index number stored in hi-lo order.

NDSCat:Indexes

Contains a list of indexes to be used in the creation of the catalog database.

Syntax

Case Ignore String (page 929)

Constraints

DS_STRING_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.279

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

By default, three indexes are always created:

- ♦ DSOBJECTNAME - the fully distinguished name of the object in dot format, from the leaf to the root container
- ♦ DSCANONICALNAME - the fully distinguished name of the object in slash format, from the leaf to the root container
- ♦ DSBASECLASS - the object class

NDSCat:Label

Specifies when the first dredge starts.

Syntax

Case Ignore List (page 926)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.271

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute allows users to configure the catalog to dredge during off-hour cycles.

NDSCat:Log

Contains a running log of information returned by the dredger for one or more dredges.

Syntax

Stream (page 977)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.272

Used In

NDSCat:Catalog (page 305)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Master Catalog

Specifies the name of the object containing the source catalog database.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.287

Used In

NDSCat:Slave Catalog (page 311)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The slave catalog is an exact duplicate of the master catalog. After the dredger writes the master catalog, it writes any specified slave catalogs. Slave catalogs can be located any where in the NDS tree, including local partitions for performance enhancement.

NDSCat:Max Log Size

Specifies the maximum size a log file can attain before being truncated.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.280

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The dredger inspects the size of the log file it maintains locally and the size of the log file maintained on the catalog object. If the sum of these two sizes is greater than the NDSCat:Max Log Size attribute, the dredger truncates the catalog's log file before writing out the log file of the last dredge.

NDSCat:Max Retries

Specifies the number of retries the dredger attempts to obtain information on any container.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.281

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Max Threads

Specifies the maximum number of threads in the dredger thread pool.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SIZED_ATTR (1..20)

ASN.1 ID

2.16.840.1.113719.1.1.4.1.290

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute limits how much server bandwidth the dredger can use.

NDSCat:Retry Interval

Specifies the amount of time between retries.

Syntax

Interval (page 951)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.282

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Scope

Specifies whether the search is to include the object, the object's immediate subordinates, or the entire subtree.

Syntax

Integer (page 948)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.283

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The default value for this attribute is to search the subtree.

NDS Catalog Services uses the following values in this attribute:

0 = search a single object

1 = search the immediate subordinates of an object, but not the object

2 = search the object and all of its subordinates

NDSCat:Search Aliases

Specifies whether the dredger dereferences aliases.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.284

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

If set to TRUE, the dredger dereferences aliases.

NDSCat:Slave Catalog List

Contains a list of slave catalogs the dredger is to write to when the dredge is completed for the master catalog.

Syntax

Distinguished Name (page 938)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.1.4.1.285

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NDSCat:Start Time

Specifies the time the dredger begins the first dredge.

Syntax

Time (page 981)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.286

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute allows the user to schedule the dredger to create the catalog during off-hour cycles. The value is modified at the end of each dredge to indicate the next dredge time.

NDSCat:Synch Interval

Specifies how often the dredger refreshes its internal list of catalogs.

Syntax

Interval (page 951)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.1.4.1.289

Used In

NDSCat:Master Catalog (page 308)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

When this interval expires, the dredger reads its own internal list of catalog objects and compares it with the list in the NDSCat:Catalog List attribute. If any discrepancies exist, the dredger updates its own list. The dredger also refreshes its list of catalog object attributes, including information concerning both master and slave catalogs.

NLS:Common Certificate

Contains the certificate's usage, its assignment, and activation key.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ

ASN.1 ID

2.16.840.1.113719.1.51.4.1.5

Used In

NLS:License Certificate (page 317)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Current Installed

Contains the actual number of licenses installed.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.9

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Current Peak Installed

Contains the peak number of license certificates installed during the current hour.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.7

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Current Peak Used

Contains the peak number of license certificates used during the current hour.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.6

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Current Used

Contains the actual number of licenses being used.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.8

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Hourly Data Size

Specifies how big the NLS:Peak Used Data and NLS:Peak Installed Data attributes can be.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.3

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The default value is 109020 which allows 15 months of data to be stored in the attributes.

NLS:License Database

Not currently used.

Syntax

Boolean (page 922)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.6

Used In

NLS:License Server (page 320)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:License ID

Contains an identifier for the license certificate.

Syntax

Case Exact String (page 924)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.3

Used In

NLS:License Certificate (page 317)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This identifier can be unique, but it is not required to be unique.

This attribute is a naming attribute for the NLS:License Certificate object.

NLS:License Service Provider

Contains the distinguished name of the NLS:License Server object that contains the NLS licensing configuration information.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.3.2

Used In

NCP Server (page 155)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:LSP Revision

Contains the revision number of the NLSLSP NLM.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.3.1

Used In

NLS:License Server (page 320)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Owner

Not currently used.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.4

Used In

NLS:License Certificate (page 317)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Peak Installed Data

Contains the peak number of installed license certificates, for each hour, for the last 15 months.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.5

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Peak Used Data

Contains the peak number of used certificates, for each hour, for the last 15 months.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.4

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Product

Contains the name of the product which is using the license certificates.

Syntax

Case Exact String (page 924)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.1

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is a naming attribute for the NLS:Product Container object.

NLS:Publisher

Contains the name of the company which publishes the software that is using the license certificates.

Syntax

Case Exact String (page 924)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.0

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is a naming attribute for the NLS:Product Container object.

NLS:Revision

Contains the revision of the data.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.3.0

Used In

NLS:License Certificate (page 317)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute supplies information that allows for backwards compatibility between revisions of data format. NetWare 5.0 has a revision number of 34.

NLS:Search Type

Specifies the range of the search.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.0

Used In

NLS:License Server (page 320)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Two search types have been defined:

0	Indicates a search from the user’s context to the root of the NDS tree. This is the default value.
1	Indicates a search from the user’s context to the root of the partition.

NLS:Summary Update Time

Contains when the NLS:Product Container object was last updated.

Syntax

Time (page 981)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.2

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The NLS:Product Container object is updated when licenses are added, deleted, or assigned.

NLS:Summary Version

Not currently used.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.2.1

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Transaction Database

Indicates whether transaction information is stored in a database.

Syntax

Boolean (page 922)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.7

Used In

NLS:License Server (page 320)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The transaction database stores events such as who received a license, who released a license, and who added a license.

NLS:Transaction Log Name

Contains the name of the database which stores NLS transactions.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.8

Used In

NLS:License Server (page 320)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Transaction Log Size

Specifies the maximum size of the transaction log file.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.9

Used In

NLS:License Server (page 320)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NLS:Version

Contains the version of the product which is using the license certificates.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.51.4.1.2

Used In

NLS:Product Container (page 323)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute is a naming attribute for the NLS:Product Container object.

Notification Consumers

Identifies each User object to notify.

Syntax

Typed Name (page 986)

Constraints

None

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.21

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The User object is tied to a unique identifier that indicates the user’s notification profile. The notification profile is used for notifications which are persistent.

Notification Profile

Contains a set of notification profiles which are tied through IDs to the Notification Consumers attribute.

Syntax

Stream (page 977)

Constraints

DS_SINGLE-VALUE_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.0.0.2.0

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Notification Service Enabled

Defines the state of the notification service as the Broker NLM is loaded.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.3

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Notification Srv Net Addr

Contains the advertising address that can be used to contact the notification service.

Syntax

Net Address (page 953)

Constraints

DS_PUBLIC_READ

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.14

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

The Broker NLM starts the notification service. For NDPS v 2.0 or higher, it can advertise the IP address and port, the IPX address and socket, or both.

Notification Srv Net Address

Contains the advertising IPX address that can be used to contact the notification service.

Syntax

Net Address (page 953)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.9

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NRD:Registry Data

Holds the NetWare Registry Database.

Syntax

Stream (page 977)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.15.4.1

Used In

User (page 272)

Remarks

This attribute is used by NetWare Administrator and ZENworks to store Windows-type registry information.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NRD:Registry Index

Contains the index to the NetWare Registry Database.

Syntax

Stream (page 977)

Constraints

DS_SINGLE-VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.15.4.2

Used In

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

NSCP:mailAccessDomain

Replaced by NSCP:AmailAccessDomain (page 566).

NSCP:mailAlternateAddress

Replaced by NSCP:AmailAlternateAddress (page 567).

NSCP:mailAutoReplyMode

Replaced by NSCP:AmailAutoReplyMode (page 568).

NSCP:mailAutoReplyText

Replaced by NSCP:AmailAutoReplyText (page 569).

NSCP:mailDeliveryOption

Replaced by NSCP:AmailDeliveryOption (page 570).

NSCP:mailForwardingAddress

Replaced by NSCP:AmailForwardingAddress (page 571).

NSCP:mailHost

Replaced by NSCP:AmailHost (page 572).

NSCP:mailMessageStore

Replaced by NSCP:AmailMessageStore (page 573).

NSCP:mailProgramDeliveryInfo

Replaced by NSCP:AmailProgramDeliveryInfo (page 574).

NSCP:mailQuota

Replaced by NSCP:AmailQuota (page 575).

NSCP:ngComponent

Specifies a Netscape component.

Syntax

Distinguished Name (page 938)

Constraints

None

ASN.1 ID

2.16.840.1.113730.3.1.196

Used In

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NetWare 5, LDAP added this attribute for Netscape support. NDS 8 replaces this attribute with the NSCP:ngcomponentCIS attribute. For more information, see the documentation for the Netscape schema.

NSCP:nsLicenseEndTime

Replaced by NSCP:AnsLicenseEndTime (page 576).

NSCP:nsLicensedFor

Replaced by NSCP:AnsLicensedFor (page 577).

NSCP:nsLicenseStartTime

Replaced by NSCP:AnsLicenseStartTime (page 578).

Page Description Languages

Identifies the different page description languages supported by the NDPS printer.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.42

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute indicates whether the printer supports such page description languages as PCL or PostScript.

preferredLanguage

Contains the object's preference for written or spoken language.

LDAP Name

preferredLanguage

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.39

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

Primary Notification Service

Identifies the NDPS Broker object that contains the notification service the printer can use.

Syntax

Typed Name (page 986)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.38

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Primary Resource Service

Identifies the NDPS Broker object that contains the resource management service that the printer can use.

Syntax

Typed Name (page 986)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.39

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Printer Agent Name

Contains the Unicode name of the Printer Agent that was converted into this object.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.51

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Printer Manufacturer

Contains the name of manufacturer of the printer.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.21

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute contains the name of a company that manufactured the printer, a company such as Epson, Hewlett Packard, Lexmark, or Xerox.

Printer Mechanism Types

Represents the printer's mechanism for printing, such as laser or ink jet.

Syntax

Integer (page 948)

Constraints

DS_P[UBLIC_READ
DS_SIZED_ATTR (1..2147483647

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.69

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Printer Model

Contains the model name of the printer.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.0.3.0.3

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Printer Status

Contains the status in the form of an OID that represents the current status of the printer.

Syntax

Octet String (page 962)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.2.0.3

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Printer to PA ID Mappings

Contains the mappings between NDPS Printer objects and their internal printer agent IDs that the NDPS Printer Manager objects use.

Syntax

Typed Name (page 986)

Constraints

ASN.1 ID

2.16.840.1.113719.1.4.2.1.0.0

Used In

NDPS Manager (page 299)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

PSM Name

Identifies the NDPS Printer Manager object that manages this printer.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.14

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Registry Advertising Name

Contains the SAP name for the service registry service which this NDPS Broker object is running.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..48)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.4

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Registry Service Enabled

Defines the state of the service registry service as the Broker NLM is loaded.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.5

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Registry Srv Net Addr

Contains the advertising address that can be used to contact the service registry service.

Syntax

Net Address (page 953)

Constraints

DS_PUBLIC_READ

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.15

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

For NDPS v2.0 or higher, this attribute can contain the IP address and port, the IPX address and socket number, or both.

Registry Srv Net Address

Contains the IPX address and socket number that is advertised so that the service registry service can be contacted.

Syntax

Net Address (page 953)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.10

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Resolution

Defines the dots-per-inch value which the printer is capable of.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..1000000)

ASN.1 ID

2.16.840.1.113719.1.0.1.1.52

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Resource Mgmt Srv Net Addr

Contains the advertising address used to contact the resource management service started by the NDPS Broker NLM.

Syntax

Net Address (page 953)

Constraints

DS_PUBLIC_READ

ASN.1 ID

2.16.840.1.113719.1.0.1.1.53

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

In NDPS v2.0 or higher, the address can be an IP address and port, an IPX address and socket number, or both.

Resource Mgmt Svc Net Address

Contains the advertising IPX address and socket that can be used to contact the resource management service started by the NDPS Broker NLM.

Syntax

Net Address (page 953)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.0.1.1.54

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Resource Mgmt Service Enabled

Defines the state of the resource management service when the Broker NLM is loaded.

Syntax

Boolean (page 922)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.0.1.1.55

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Resource Mgr Database Path

Defines the path to the root directory for all the file resources which the resource management service manages.

Syntax

Case Ignore String (page 929)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..128)

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.7

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

Resource Mgr Database Volume

Defines the volume for the root directory for all file resources which the resource management service manages.

Syntax

Distinguished Name (page 938)

Constraints

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.4.2.1.1.8

Used In

NDPS Broker (page 296)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

secretary

Contains the name of the object's secretary or administrative assistant.

LDAP Name

secretary

Syntax

Distinguished Name (page 938)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

0.9.2342.19200300.100.1.21

Used In

inetOrgPerson (page 291)

User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

Sides Supported

Defines how many sides of the paper the printer can print on.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SIZED_ATTR (1..100)

ASN.1 ID

2.16.840.1.113719.1.4.0.3.2.59

Used In

NDPS Printer (page 302)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Attribute

Contains the names of SLP attributes assigned to the SLP Service object.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.11

Used In

SLP Service (page 342)

Remarks

For a list of defined SLP attributes, see RFC 2165. A typical attribute has the following format:

```
(svcname_ws:fileserver_name)
```

Multiple SLP attributes can be stored in a single string because the SLP Directory Agent understands the SLP attribute syntax.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Cache Limit

Currently not used.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ

DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.49.4.1.5

Used In

SLP Directory Agent (page 339)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP DA Back Link

Contains the distinguished name of the NCP Server object which the SLP Directory Agent is running on.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.6

Used In

SLP Directory Agent (page 339)

Remarks

See also the SLP Directory Agent DN (page 889) attribute, which links the NCP Server object to the SLP Directory Agent object.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Directory Agent DN

Contains the distinguished name of the SLP Directory Agent object.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.16

Used In

NCP Server (page 155)

Remarks

When the SLP Directory Agent is started, it uses this attribute to find the SLP Directory Agent object that stores its configuration information. For the attribute that contains the link from the SLP Directory Agent to the NCP Server, see the SLP DA Back Link (page 888) attribute.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Language

Contains the language used by an SLP Service object.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.12

Used In

SLP Service (page 342)

Remarks

For a list of language strings defined for SLP, see RFC 2165.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Lifetime

Specifies how long an SLP Service object should remain in the service list of the SLP Directory Agent.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.13

Used In

SLP Service (page 342)

Remarks

SLP supports lifetimes of 1 second to 17 hours.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Scope Name

Contains the name of the SLP scope.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.49.4.1.2

Used In

SLP Scope Unit (page 336)

Remarks

SLP uses scope as a method for scaling a network. It is similar to domains in UNIX and zones in AppleTalk.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Scope Unit DN

Contains the distinguished name of the SLP Scope Unit object.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.9

Used In

SLP Directory Agent (page 339)

Remarks

This is a multivalued attribute since SLP Directory Agents can be assigned to multiple SLP scopes.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Start Purge Hour

Contains the time to purge SLP Service objects which have not updated their lifetimes.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.8

Used In

SLP Directory Agent (page 339)

Remarks

When a SLP Service object lets its lifetime expire, the SLP Directory Agent removes it from the list service objects used to answer service queries, but the service object is not removed from the master list. When the purge hour arrives, the SLP Directory Agent removes the service objects with expired lifetimes from the master list.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Status

Contains the status of the SLP Directory Agent object.

Syntax

Integer (page 948)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.7

Used In

SLP Directory Agent (page 339)

Remarks

The SLP Directory Agent object uses the following values for status:

- ♦ 0 down
- ♦ 1 up

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP SU Back Link

Contains the distinguished name of the SLP Directory Agent object.

Syntax

Distinguished Name (page 938)

Constraints

DS_PUBLIC_READ
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.4

Used In

SLP Scope Unit (page 336)

Remarks

Each SLP Scope Unit object is associated with an SLP Directory Agent object, and this attribute contains that assignment.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP SU Type

Contains all of the registered service types.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.3

Used In

SLP Scope Unit (page 336)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP Type

Contains the type of service provided by an SLP Service object.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.15

Used In

SLP Service (page 342)

Remarks

SLP does not define strings for various services. The service defines its type and then registers it with the SLP Directory Agent.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SLP URL

Contains the URL address of an SLP Service object.

Syntax

Case Ignore String (page 929)

Constraints

DS_PUBLIC_READ
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.10

Used In

SLP Service (page 342)

Remarks

A typical URL address has the following format:

```
Service:Bindery.Novell:///fileserver_name
```

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SMS Protocol Address

Contains the network address of the SMS SMDR object.

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.50.4.1.2

Used In

SMS SMDR Class (page 345)

Remarks

This attribute contains the list of protocols that the SMDR can use and the network address for each supported protocol.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SMS Registered Service

Contains the list of services for which the SMS SMDR object has registered.

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.50.4.1.1

Used In

SMS SMDR Class (page 345)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SU

Contains the name of the SLP Scope Unit object.

Syntax

Case Ignore String (page 929)

Constraints

DS_SIZED_ATTR (1..27)

DS_STRING_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113719.1.49.4.1.1

Used In

SLP Scope Unit (page 336)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SvcInfo

Contains miscellaneous information about the program that registers the service.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

None

Used In

Device (page 113)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SvcType

Contains the service type name.

Syntax

Case Ignore String (page 929)

Constraints

DS_INGLE_VALUED_ATTR

DS_SYNC_IMMEDIATE

ASN.1 ID

None

Used In

Device (page 113)

Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

SvcTypeID

Contains the globally unique identifier that is associated with the service type.

Syntax

Octet String (page 962)

Constraints

DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN.1 ID

None

Used In

Device (page 113)
NetSvc (page 314)
Server (page 254)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

userSMIMECertificate

Contains the object's certificate for Netscape Communicator for S/MIME.

LDAP Name

userSMIMECertificate

Syntax

Octet String (page 962)

Constraints

DS_SYNC_IMMEDIATE

ASN.1 ID

2.16.840.1.113730.3.1.40

Used In

inetOrgPerson (page 291)
User (page 272)

Remarks

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

This attribute was added for Netscape support.

7

LDAP Operational Attributes

This chapter lists alphabetically all the LDAP operational attribute definitions that are in the base schema for NDS.

- ♦ For base schema attribute definitions, see “Base Attribute Definitions” on page 361.
- ♦ For the attribute definitions which become part of the standard schema during a default installation, see “Novell Attribute Extensions” on page 729.
- ♦ For an explanation of the types of information included about each attribute, see “Reading NDS Attribute Type Definitions” on page 63.

The information contained in the LDAP operational attributes is read-only. Clients cannot modify the attribute values. All of these operational attributes except `creatorsName` and `modifiersName` are available only through LDAP (see the DSI flags for equivalent information through NDAP). `creatorsName` and `modifiersName` are available through LDAP and NDAP.

They are not returned in search results unless explicitly requested by name. They can be requested for all entries in an NDS directory.

createTimeStamp

Contains when the entry was created.

Added: NDS eDirectory 8.5

LDAP Name

createTimeStamp

Syntax

Timestamp (page 983)

Constraints

DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR

ASN1 ID

2.5.18.1

Remarks

This attribute is accessible only through LDAP. Through NDAP, this information is available with the NWDSGetObjectInfo function which accesses the DCK_DSI_FLAGS.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

creatorsName

Contains the name of the user that created this entry.

Added: NDS eDirectory 8.5

LDAP Name

creatorsName

Syntax

Case Ignore String (page 929)

Constraints

DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN1 ID

2.5.18.3

Used In

Top (page 263)

Remarks

This attribute is accessible through LDAP and NDAP.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

entryFlags

Contains information about the entry's state, for example, whether the entry is an alias, partition, or a container.

Added: NDS eDirectory 8.5

LDAP Name

entryFlags

Syntax

Integer (page 948)

Constraints

DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR

ASN1 ID

2.16.840.1.113719.1.27.4.48

Remarks

This attribute is accessible only through LDAP. Through NDAP, this information is available with the NWDSGetObjectNameAndInfo function which accesses the DSI_ENTRY_FLAGS through the DCK_DSI_FLAGS key.

For possible values, see DSI_ENTRY_FLAGS Values.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

federationBoundary

Contains where the federation boundary begins.

Added: NDS eDirectory 8.5

LDAP Name

federationBoundary

Syntax

Distinguished Name (page 938)

Constraints

DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR

ASN.1 ID

2.16.840.1.113719.1.27.4.51

Remarks

This attribute is accessible only through LDAP.

The tree name for all DNS-rooted trees is T=DNS. This attribute returns the full context for a DNS-rooted tree, and the full context contains the external references, such as dc=novell, dc=com. The namingcontexts attribute on the root DSE will also contain the federation boundary.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

localEntryID

Contains the record number for the entry in the server's local database.

Added: NDS eDirectory 8.5

LDAP Name

localEntryID

Syntax

Integer (page 948)

Constraints

DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR

ASN1 ID

2.16.840.1.113719.1.27.4.50

Remarks

This attribute is accessible only through LDAP, and through LDAP, it is only accessible if you know the name. This attribute is not returned in a call to read the schema.

In an LDAP search, the localEntryID can be used in place of the dn to request information about a specific entry.

Through NDAP, this information is available with the NWDSGetObjectNameAndInfo function which accesses the DCK_DSI_FLAGS key.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

modifiersName

Contains the distinguished name of the last user that modified this entry.

Added: NDS eDirectory 8.5

LDAP Name

modifiersName

Syntax

Distinguished Name (page 938)

Constraints

DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR
DS_SYNC_IMMEDIATE

ASN1 ID

2.5.18.4

Used In

Top (page 263)

Remarks

This attribute is accessible through LDAP and NDAP.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

modifyTimeStamp

Contains when the entry was last modified.

Added: NDS eDirectory 8.5

LDAP Name

modifyTimeStamp

Syntax

Timestamp (page 983)

Constraints

DS_OPERATIONAL
DS_READ_ONLY_ATTR
DS_SINGLE_VALUED_ATTR

ASN1 ID

2.5.18.2

Remarks

This attribute is accessible only through LDAP. Through NDAP, this information is available with the NWDSGetObjectNameAndInfo function which accesses the DCK_DSI_FLAGS.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

structuralObjectClass

Contains the base class of the entry.

Added: NDS eDirectory 8.5

LDAP Name

structuralObjectClass

Syntax

Class Name (page 934)

Constraints

DS_OPERATIONAL

ASN1 ID

2.5.21.9

Remarks

This attribute is accessible only through LDAP. Through NDAP, this information is available with the NWDSGetObjectNameAndInfo function which accesses the DCK_DSI_FLAGS.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

subordinateCount

Contains the number of entries immediately subordinate to this entry.

Added: NDS eDirectory 8.5

LDAP Name

subordinateCount

Syntax

Integer (page 948)

Constraints

DS_OPERATIONAL

ASN1 ID

2.16.840.1.113719.1.27.4.49

Remarks

This attribute does not contain the count of entries for an entire subtree. It contains the count for only the nodes below this entry. For leaf entries, it returns zero.

This attribute is accessible only through LDAP. Through NDAP, this information is available with the NWDSGetObjectNameAndInfo function which accesses the DCK_DSI_FLAGS.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

subschemaSubentry

Contains the LDAP name for the schema.

Added: NDS eDirectory 8.5

LDAP Name

subschemaSubentry

Syntax

Distinguished Name (page 938)

Constraints

DS_OPERATIONAL

ASN1 ID

2.5.18.10

Remarks

This attribute is accessible only through LDAP. It returns the same value as the subschemaSubentry on the root DSE.

For help in understanding the attribute definition template, see “Reading NDS Attribute Type Definitions” on page 63.

8

Attribute Syntax Definitions

This chapter lists alphabetically all the attribute syntax definitions that are in the operational schema.

For the NWDS functions and those built on top of them, the `nwdsdefs.h` file contains an enumerated data type for each syntax. The last member of the structure is `SYNTAX_COUNT`, which is not syntax type, but a count that returns the number of syntaxes defined in the enumeration.

For LDAP, the syntax types use the following data types that are specific to LDAP. UTF8 (Universal Transformation Format) is a method for converting 16-bit Unicode characters into 8-bit characters.

Data Type	Description
<code>dstring</code>	A UTF8 encoded string.
<code>distinguishedname</code>	LDAP string format of the distinguished name (CN=Kim, OU=Sales, O=Widget, Inc.). See RFC 2253 for more information.
<code>LDAPDN</code>	LDAP format of the distinguished name (CN=Kim, OU=Sales, O=Widget, Inc.), BER encoded
<code>LDAPString</code>	A type and length preceded string, BER encoded
<code>octetstring</code>	Opaque data in an application-defined format
<code>uint16string</code>	A numeric string with an upper bound of 65535
<code>uint32string</code>	A numeric string with an upper bound of 429467295

For an explanation of the types of information included about each syntax, see “Reading Syntax Definitions” on page 66.

Back Link

Is used for the Back Link attribute, which NDS uses for its internal management.

Syntax ID

```
#define SYN_BACK_LINK 23
```

LDAP Name

Tagged Name

ASN1.ID

2.16.840.1.113719.1.1.5.1.23

API Data Structure

```
typedef struct
{
    uint32    remoteID;
    pnstr8    objectName;
} Back_Link_T;
```

Transfer Format

uint32	Length
uint32	remoteID
unicode	Distinguished Name

LDAP Format

String

```
taggedName = uint32string "#" distinguishedname
```

Binary

```
taggedName ::= SEQUENCE {  
    name      LDAPDN,  
    number    uint32  
}
```

Matching Rules

Equality

Approximate—Not currently supported through LDAP

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

The remoteID field identifies the back linked object on the server, and the objectName field identifies the server holding an external reference.

NDS verifies the DN field and must therefore contain a DN, but the number field can be used to hold any application specific value.

Used In

Back Link (page 382)

Boolean

Is used for attributes whose values represent true (1) or false (0).

Syntax ID

```
#define SYN_BOOLEAN 7
```

LDAP Name

Boolean

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.7

API Data Structure

```
typedef uint8 Boolean_T;
```

Transfer Format

```
uint32      Length = 1
BYTE        Content
```

LDAP Format

```
boolean = "TRUE" / "FALSE"
```

Matching Rules

Equality

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Two boolean attributes match for equality if they are both TRUE or both FALSE.

TRUE is represented as one (1), while FALSE is represented as zero (0).

Any attribute defined using this syntax is single-valued.

Used In

Allow Unlimited Credit (page 365)
Detect Intruder (page 413)
LDAP Allow Clear Text Password (page 464)
LDAP Enable SSL (page 469)
LDAP Enable TCP (page 470)
LDAP Enable UDP (page 471)
Locked By Intruder (page 499)
Lockout After Detection (page 500)
Login Disabled (page 502)
NDSCat:Actual All Attributes (page 783)
NDSCat:Actual Search Aliases (page 793)
NDSCat:All Attributes (page 796)
NDSCat:Auto Dredge (page 799)
NDSCat:Search Aliases (page 815)
NLS:License Database (page 825)
NLS:Transaction Database (page 838)
Password Allow Change (page 616)
Password Required (page 622)
Password Unique Required (page 623)
Run Setup Script (page 662)
Set Password After Create (page 681)

Case Exact String

Is used in attributes whose values are strings for which the case (upper or lower) is significant when performing comparisons.

Syntax ID

```
#define SYN_CE_STRING 2
```

LDAP Name

IA5 String

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.26 (LDAP)
2.16.840.1.113719.1.1.5.1.2 (NDAP)

API Data Structure

```
typedef pnstr8 CE_String_T;
```

Transfer Format

uint32	Length
unicode	String

LDAP Format

string

Matching Rules

Equality
Substrings

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Attributes using this syntax can set size limits.

Two case exact strings match for equality when they are of same length and their corresponding characters are identical. For example, “Dundee” and “DUNDEE” do not match.

In comparing case exact strings, the following spaces are not significant:

- ♦ Leading spaces (precede the first printing character)
- ♦ Trailing spaces (follow the last printing character)
- ♦ Multiple consecutive internal spaces (equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

Used In

Entrust:User (page 742)
NDSCat:Actual Attributes (page 785)
NDSCat:Actual Filter (page 789)
NDSCat:Attributes (page 798)
NDSCat:Filter (page 804)
NLS:License ID (page 826)
NLS:Product (page 832)
NLS:Publisher (page 833)
NSCP:mailMessageStore (page 856)
NSCP:mailProgramDeliveryInfo (page 857)
NSCP:mailMessageStore (page 856)
NSCP:mailProgramDeliveryInfo (page 857)
NSCP:mgrpRFC822mailmember (page 583)

Case Ignore List

Is used for attributes whose values are ordered sequences of strings for which the case (upper or lower) is not significant when performing comparisons.

Syntax ID

```
#define SYN_CI_LIST 6
```

LDAP Name

Case Ignore List

ASN1.ID

2.16.840.1.113719.1.1.5.1.6

API Data Structure

```
typedef struct _ci_list
{
    struct _ci_list N_FAR *next;
    pnstr8 s;
} CI_List_T;
```

Transfer Format

```
uint32    Length of attribute value
uint32    Number of strings

/* For each string */
Align4
unicode   String
```

LDAP Format

String

```
caseIgnorelist = dstring *( "$" dstring)
```

Binary

`caseIgnorelist ::= SEQUENCE OF LDAPString`

Matching Rules

Equality

Approximate

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Two case ignore lists match for equality if, and only if, the number of strings in each is the same, and all corresponding strings match. For two corresponding strings in the list to match, they must be the same length and their corresponding characters must be identical (according to the rules for case ignore strings).

When comparing the strings in case ignore lists, the following spaces are regarded as not significant:

- ♦ Leading spaces (precede the first printing character)
- ♦ Trailing spaces (follow the last printing character)
- ♦ Multiple consecutive internal spaces (equivalent to a single space character).

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

The **NWDSGetAttrVal** function places successive CI_LIST elements in consecutive memory locations. The *value* parameter of the **NWDSGetAttrVal** function should point to enough memory to contain both the NULL-terminated strings along with a CI_List_T structure per list element. (The required length of value can be determined by calling the **NWDSComputeValSize** function.)

Used In

Language (page 460)

Case Ignore String

Is used in attributes whose values are strings for which the case (upper or lower) is not significant when performing comparisons.

Syntax ID

```
#define SYN_CI_STRING 3
```

LDAP Name

Directory String

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.15

API Data Structure

```
typedef pnstr8 CI_String_T;
```

Transfer Format

```
uint32      Length  
unicode     String
```

LDAP Format

UTF-8 encoded string

Matching Rules

Equality
Substrings

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Attributes using this syntax can set size limits.

Two case ignore strings match for equality when they are of the same length and their corresponding characters are identical in all respects except that of case. For example, as case ignore strings, “Dundee” and “DUNDEE” would be equal.

When comparing case ignore strings, the following spaces are not significant:

- ♦ Leading spaces (precede the first printing character)
- ♦ Trailing spaces (follow the last printing character)
- ♦ Multiple consecutive internal spaces (equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

Used In

businessCategory (page 389)
C (Country Name) (page 390)
Cartridge (page 395)
CN (Common Name) (page 400)
dc (page 406)
Description (page 410)
dmdName (page 415)
Employee ID (page 740)
Full Name (page 429)
generationQualifier (page 432)
Generational Qualifier (page 431)
Given Name (page 434)
Host Resource Name (page 448)
houseIdentifier (page 446)
Initials (page 451)

Internet EMail Address (page 453)
knowledgeInformation (page 457)
L (Locality Name) (page 458)
LDAP Backup Log Filename (page 467)
LDAP Log Filename (page 474)
LDAP Referral (page 478)
LDAP:keyMaterialName (page 493)
Mailbox ID (page 516)
Messaging Server Type (page 533)
NDSCat:Indexes (page 806)
NDSPKI:Given Name (page 537)
NDSPKI:Parent CA (page 543)
NDSPKI:Subject Name (page 552)
NLS:Transaction Log Name (page 839)
NLS:Version (page 841)
NNS Domain (page 562)
NSCP:administratorContactInfo (page 564)
NSCP:adminURL (page 565)
NSCP:mailAccessDomain (page 849)
NSCP:mailAlternateAddress (page 850)
NSCP:mailAutoReplyMode (page 851)
NSCP:mailAutoReplyText (page 852)
NSCP:mailDeliveryOption (page 853)
NSCP:mailForwardingAddress (page 854)
NSCP:mailHost (page 855)
NSCP:mailQuota (page 858)
NSCP:nsLicenseEndTime (page 860)
NSCP:nsLicensedFor (page 861)
NSCP:nsLicenseStartTime (page 862)
NSCP:employeeNumber (page 579)
NSCP:installationTimeStamp (page 580)
NSCP:mailAccessDomain (page 849)
NSCP:mailAlternateAddress (page 850)
NSCP:mailAutoReplyMode (page 851)
NSCP:mailAutoReplyText (page 852)
NSCP:mailDeliveryOption (page 853)

NSCP:mailForwardingAddress (page 854)
NSCP:mailHost (page 855)
NSCP:mailQuota (page 858)
NSCP:memberCertificateDesc (page 582)
NSCP:ngComponent (page 859)
NSCP:nsaclrole (page 585)
NSCP:nscreator (page 586)
NSCP:nsflags (page 587)
NSCP:nsLicenseEndTime (page 860)
NSCP:nsLicensedFor (page 861)
NSCP:nsLicenseStartTime (page 862)
NSCP:nsnewsACL (page 588)
NSCP:nsprettyname (page 589)
NSCP:serverHostName (page 590)
NSCP:serverProductName (page 591)
NSCP:serverRoot (page 592)
NSCP:serverVersionNumber (page 593)
NSCP:subtreeACI (page 594)
O (Organization Name) (page 595)
OU (Organizational Unit Name) (page 606)
Physical Delivery Office Name (page 627)
Postal Code (page 631)
Postal Office Box (page 633)
Queue Directory (page 652)
registeredAddress (page 655)
roomNumber (page 661)
S (State or Province Name) (page 663)
SA (Street Address) (page 665)
SAP Name (page 667)
SLP Attribute (page 886)
SLP Language (page 890)
SLP Scope Name (page 892)
SLP SU Type (page 897)
SLP Type (page 898)
SLP URL (page 899)
SU (page 902)

supportedApplicationContext (page 686)
Supported Gateway (page 688)
Supported Services (page 689)
Supported Typefaces (page 690)
Surname (page 691)
SvcType (page 904)
Title (page 701)
uniqueID (page 707)
Unknown Auxiliary Class (page 710)
Unknown Base Class (page 711)
Version (page 719)

Class Name

Is used for attributes whose values represent object class names.

Syntax ID

```
#define SYN_CLASS_NAME 20
```

LDAP Name

OID

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.38

API Data Structure

```
typedef pnstr8 Class_Name_T
```

Transfer Format

```
uint32      Length
unicode     String
```

LDAP Format

string

Matching Rules

Equality

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

The matching rule for values of Class Name are the same as those for Case Ignore String. That is, two Class Names match for equality when their lengths and corresponding characters are identical in all respects except that of case and spaces as described below.

In comparing attributes using the Class Name syntax, the following white space (spaces, tabs, etc.) is not significant:

- ♦ Leading spaces (those preceding the first printable character)
- ♦ Trailing spaces (those following the last printable character)
- ♦ Multiple consecutive internal spaces (these are taken as equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

In NDS, an object class name can be 32 characters long, and must be unique within its super class structure. Although NDS allows spaces and non-alphanumeric characters in the name, you should conform to LDAP naming conventions.

In LDAP, the name can be an alpha string with one hyphen (-) or a numeric string with a period (.) separating numbers. Spaces are not allowed. LDAP specifications recommend using an alpha string such as locality rather than a numeric string such as 2.5.6.3.

Used In

AuxClass Object Class Backup (page 379)

Object Class (page 601)

Counter

Is used for attributes whose values are signed integers.

Syntax ID

```
#define SYN_COUNTER 22
```

LDAP Name

Counter

ASN1.ID

2.16.840.1.113719.1.1.5.1.22

API Data Structure

```
typedef uint32 Counter_T;
```

Transfer Format

```
uint32    Length = 4
uint32    Content
```

LDAP Format

String

```
numeric string
```

Binary

```
counter ::=uint32
```

Matching Rules

Equality

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Any attribute defined using this syntax is single-valued. This syntax differs from Integer in that any value added to an attribute of this syntax is arithmetically added to the total, and any value deleted is arithmetically subtracted from the total.

Used In

Account Balance (page 363)

Login Grace Remaining (page 505)

Login Intruder Attempts (page 507)

Revision (page 659)

Distinguished Name

Is used for attributes whose values are the names of objects in the NDS tree.

Syntax ID

```
#define SYN_DIST_NAME 1
```

LDAP Name

DN

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.12

API Data Structure

```
typedef pnstr8 DN_T;
```

Transfer Format

```
uint32      Length
unicode     String
```

LDAP Format

```
string
```

Matching Rules

Equality

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

In NDAP, a Distinguished Name (DN) is transmitted on the wire as a Unicode string. The largest supported DN is 256 Unicode characters, not including the terminating NULL character. Distinguished Names are not case sensitive, even if one of the naming attributes is case sensitive.

NDAP supports a number of formats for the distinguished name:

- ♦ Typeful or typeless
- ♦ Back slashes or periods as delimiters

For example, NDAP supports the following:

```
\XYZ_tree\ACME\HR\JRoss
\XYZ_tree\O=ACME\OU=HR\CN=JRoss
JRoss.HR.ACME
CN=JRoss.OU=HR.O=Acme
```

LDAP supports only one format for a distinguished name: typeful with commas as the delimiter, for example:

```
CN=JRoss, OU=HR, O=Acme
```

Used In

Aliased Object Name (page 362)
Audit:File Link (page 372)
Audit:Link List (page 373)
Default Queue (page 407)
Device (page 414)
dn (page 416)
Equivalent To Me (page 424)
Group Membership (page 436)
Higher Privileges (page 440)
Host Device (page 447)
Host Server (page 449)
LDAP Anonymous Identity (page 465)
LDAP Group (page 472)

- LDAP Host Server (page 473)
- LDAP Server (page 483)
- LDAP Server List (page 486)
- LDAP Suffix (page 488)
- LDAP:bindCatalog (page 491)
- LDAP:searchCatalog (page 495)
- Mailbox Location (page 518)
- Member (page 525)
- Members Of Template (page 526)
- Message Routing Group (page 528)
- Message Server (page 529)
- Messaging Server (page 532)
- NDSCat:Actual Base Object (page 786)
- NDSPKI:Key Material DN (page 539)
- NDSPKI:Parent CA DN (page 544)
- NDSPKI:Tree CA DN (page 553)
- NSCP:ngComponent (page 859)
- NDSCat:Base Object (page 800)
- NDSCat:Catalog List (page 802)
- NDSCat:Master Catalog (page 809)
- NDSCat:Slave Catalog List (page 816)
- NLS:License Service Provider (page 827)
- NLS:Owner (page 829)
- Operator (page 603)
- Owner (page 608)
- Postmaster (page 635)
- Profile (page 645)
- Profile Membership (page 646)
- Reference (page 654)
- Resource (page 658)
- Role Occupant (page 660)
- SAS:Security DN (page 668)
- SAS:Service DN (page 669)
- Security Equals (page 673)
- See Also (page 675)
- Server (page 678)

SLP DA Back Link (page 888)
SLP Directory Agent DN (page 889)
SLP Scope Unit DN (page 893)
SLP SU Back Link (page 896)
User (page 714)
Volume (page 720)
WANMAN:LAN Area Membership (page 725)

Email Address

Is used for attributes whose values represent e-mail addresses.

Syntax ID

```
#define SYN_EMAIL_ADDRESS 14
```

LDAP Name

Tagged String

ASN1.ID

2.16.840.1.113719.1.1.5.1.14

API Data Structure

```
typedef struct
{
    uint32      type;
    pustr8      address;
} EMail_Address_T;
```

Transfer Format

```
uint32      Length = N
BYTE  [N]    Content
```

LDAP Format

String

```
taggedString = uint32string "#" IA5String
```

Binary

```
taggedString ::= SEQUENCE {
    number    uint32,
```

```
    string    IA5String
}
```

Matching Rules

Equality

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Two EMail Addresses match for equality when their lengths and corresponding characters are identical in all respects except that case is ignored and extra spaces as described below.

In comparing attributes using the EMail Address syntax, the following white space (spaces, tabs, etc.) is not significant:

- ♦ Leading spaces (those preceding the first printable character)
- ♦ Trailing spaces (those following the last printable character)
- ♦ Multiple consecutive internal spaces (these are taken as equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

NDS makes no assumption about the internal structure of the *address* field or the values in the *type* field.

Used In

EMail Address (page 419)

Facsimile Telephone Number

Specifies a string that complies with the internationally agreed format for showing international telephone numbers, E.123, and an optional bit string formatted according to Recommendation T.30.

Syntax ID

```
#define SYN_FAX_NUMBER 11
```

LDAP Name

Facsimile Telephone Number

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.22

API Data Structure

```
typedef struct
{
    pnstr8      telephoneNumber;
    Bit_String_T parameters;
} Fax_Number_T;
```

Transfer Format

```
uint32      Length
unicode     Telephone number
Align4
uint32      Bit Count = B
uint32      Bit String Length = M
BYTE (M)    Bit String
```

LDAP Format

```
fax-number    = printablestring [ "$" faxparameters ]
faxparameters = faxparm / ( faxparm "$" faxparameters )
faxparm = "twoDimensional" / "fineResolution" /
```

```
"unlimitedLength" /  
"b4Length" / "a3Width" / "b4Width" / "uncompressed"
```

The first printable string is the telephone number, and the faxparm tokens represent fax parameters.

Matching Rules

Equality

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Facsimile Telephone Number values are matched based on the telephone number field. The rules for matching fax telephone numbers are identical to those for the Case Exact syntax except that all space and hyphen (-) characters are skipped during the comparison.

This syntax combines a telephone number with an optional bit string. On the wire, the bit string is represented using an integral number of bytes, with padding bits (if needed) at the least-significant end of the final byte. The value M is the number of bytes needed to hold B bits, where the value B indicates the number of bits in the bit string. That is, the value of M is B/8 rounded up to the nearest integer.

Used In

Facsimile Telephone Number (page 427)

Hold

Is used for attributes whose values represent an object name/level pair.

Syntax ID

```
#define SYN_HOLD 26
```

API Data Structure

```
typedef struct
{
    pnstr8      objectName;
    nuint32     amount;
} Hold_T;
```

Transfer Format

```
uint32      Length
uint32      Amount
unicode     Subject
```

Matching Rules

Equality

Approximate—Not currently supported through LDAP

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

This syntax is an accounting quantity, which is an amount tentatively held against a subject's credit limit, pending completion of a transaction. In the wire format, the Subject field is the Distinguished Name of the object.

NDS treats the Hold amount similarly to the Counter syntax, with new values added to or subtracted from the base total. If the evaluated Hold amount goes to 0 (zero), the Hold record is deleted.

This syntax is not supported in LDAP.

Used In

Server Holds (page 679)

Integer

Is used for attributes whose values are signed integers.

Syntax ID

```
#define SYN_INTEGER 8
```

LDAP Name

INTEGER

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.27

API Data Structure

```
typedef nint32 Integer_T;
```

Transfer Format

```
uint32    Length = 4
uint32    Content
```

LDAP Format

numeric string

Matching Rules

Equality
Ordering

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Attributes using this syntax can set size limits.

The attributes for two integers match for equality if they are the same. The comparison for ordering uses signed integer rules.

Used In

Audit:Contents (page 370)
Audit:Type (page 376)
Bindery Object Restriction (page 383)
Convergence (page 403)
DS Revision (page 418)
GID (Group ID) (page 433)
Home Directory Rights (page 443)
LDAP Log Level (page 475)
LDAP Log Size Limit (page 477)
LDAP Screen Level (page 479)
LDAP Search Size Limit (page 481)
LDAP Search Time Limit (page 482)
LDAP Server Bind Limit (page 484)
LDAP Server Idle Timeout (page 485)
LDAP SSL Port (page 487)
LDAP TCP Port (page 489)
LDAP UDP Port (page 490)
LDAP:bindCatalogUsage (page 492)
LDAP:otherReferralUsage (page 494)
LDAP:searchCatalogUsage (page 496)
LDAP:searchReferralUsage (page 498)
Login Grace Limit (page 504)
Login Intruder Limit (page 508)
Login Maximum Simultaneous (page 510)
Memory (page 527)
Minimum Account Balance (page 534)

NDSCat:Actual Attribute Count (page 784)
NDSCat:Actual Catalog Size (page 787)
NDSCat:Actual Object Count (page 790)
NDSCat:Actual Return Code (page 791)
NDSCat:Actual Scope (page 792)
NDSCat:Actual Value Count (page 795)
NDSCat:Max Log Size (page 810)
NDSCat:Max Retries (page 811)
NDSCat:Max Threads (page 812)
NDSCat:Scope (page 814)
NLS:Current Installed (page 820)
NLS:Current Peak Installed (page 821)
NLS:Current Peak Used (page 822)
NLS:Current Used (page 823)
NLS:Hourly Data Size (page 824)
NLS:LSP Revision (page 828)
NLS:Revision (page 834)
NLS:Search Type (page 835)
NLS:Summary Version (page 837)
NLS:Transaction Log Size (page 840)
Password Minimum Length (page 621)
searchSizeLimit (page 671)
searchTimeLimit (page 672)
Security Flags (page 674)
SLP Cache Limit (page 887)
SLP Lifetime (page 891)
SLP Start Purge Hour (page 894)
SLP Status (page 895)
Status (page 683)
Supported Connections (page 687)
UID (User ID) (page 706)
WANMAN:Default Cost (page 724)

Interval

Is used for attributes whose values represent intervals of time in seconds.

Syntax ID

```
#define SYN_INTERVAL 27
```

LDAP Name

INTEGER

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.27

API Data Structure

```
typedef nint32 Integer_T;
```

Transfer Format

```
uint32    Length = 4
uint32    Content
```

LDAP Format

numeric string

Matching Rules

Equality
Ordering

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Attributes using this syntax can set size limits.

The attributes for two intervals match for equality if they are the same. The comparison for ordering uses signed integer rules.

The Interval value is the number of seconds in a time interval.

Used In

Certificate Validity Interval (page 398)
High Convergence Sync Interval (page 439)
Intruder Attempt Reset Interval (page 454)
Intruder Lockout Reset Interval (page 456)
Low Convergence Sync Interval (page 515)
NDSCat:Dredge Interval (page 803)
NDSCat:Retry Interval (page 813)
NDSCat:Synch Interval (page 818)
Password Expiration Interval (page 617)

Net Address

Is used for an internal number assigned by the network that specifies where a device (such as a server or workstation) can be located in the network cabling system. The address type indicates the type of communications protocol used (IPX, IP, etc.).

Syntax ID

```
#define SYN_NET_ADDRESS 12
```

LDAP Name

Tagged Data

ASN.1 ID

2.16.840.1.113719.1.1.5.1.12

API Data Structure

```
typedef struct
{
    uint32      addressType;
    uint32      addressLength;
    puint8      address;
} Net_Address_T;
```

Transfer Format

uint32	Length
uint32	Transport Type
uint32	Address Length = M
BYTE [M]	Address

LDAP Format

String

```
taggedData = uint32string "#" octetstring
```

Binary

```
taggedData ::= SEQUENCE {  
    type    uint32,  
    data    OCTET STRING  
}
```

Matching Rules

Equality

For two values of Net Address to match, the type, address length, and value of the address must match. The address length is the number of bytes. The address itself is stored as a binary string. This string is the literal value of the address. To display it as a hexadecimal value, you must convert each 4-bit nibble to the correct character (0,1,2,3,...F).

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

The address or transport type indicates the type of communications protocol used. For a list of flags defined for type, see *Network Address Types (NDS Core Services)*.

For IPX, the Address Length is 12. It includes the network number, host number, and socket, in that order. The address is in binary (not displayable) format.

Used In

Login Intruder Address (page 506)

Network Address (page 556)

Network Address Restriction (page 558)

Numeric String

Is used for attributes whose values are numeric strings as defined in CCITT X.208.

Syntax ID

```
#define SYN_NU_STRING 5
```

LDAP Name

Numeric String

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.36

API Data Structure

```
typedef pnstr8 NU_String_T;
```

Transfer Format

uint32	Length
unicode	String

LDAP Format

string

Matching Rules

Equality
Substrings

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

For two numeric strings to match for equality, the strings must be the same length and their corresponding characters must be identical. The following characters are in the numeric string character set:

- ♦ 0..9 digits
- ♦ Space character

When comparing numeric strings, the following spaces are not significant:

- ♦ Leading spaces (precede the first printing character)
- ♦ Trailing spaces (follow the last printing character)
- ♦ Multiple consecutive internal spaces (equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

Used In

Bindery Type (page 388)

internationaliSDNNumber (page 452)

x121Address (page 727)

Object ACL

Is used for attributes whose values represent ACL entries.

Syntax ID

```
#define SYN_OBJECT_ACL 17
```

LDAP Name

NDS ACL

ASN.1 ID

2.16.840.1.113719.1.1.5.1.17

API Data Structure

```
typedef struct
{
    pnstr8      protectedAttrName;
    pnstr8      subjectName;
    nuint32     privileges;
} Object_ACL_T;
```

Transfer Format

uint32	Length
unicode	Name of Protected Attribute
Align4	
unicode	Subject Name
Align4	
uint32	Privileges

LDAP Format

String

```
ndsAcl = privileges "#" scope "#" subjectname "#"
        protectedattrname

privileges = uint32string

scope = "entry" / "subtree"

subjectname = distinguishedname / "[Self]" / "[Creator]" /
            "[Public]" / "[Inheritance Mask]" / "[Root]"

protectedattrname = caseignorestring / "[Entry Rights]" /
            "[All Attribute Rights]"
```

Binary

```
ndsAcl ::= SEQUENCE {
    privileges          uint32,
    subjectName         LDAPDN,
    protectedAttrName   LDAPString
}
```

Matching Rules

Approximate
Equality

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

An Object ACL value can protect either an object or an attribute. The protected object is always the one that contains the ACL attribute. If an ACL entry is to apply to the object as a whole, the protected attribute name should be left empty (NULL). If a specific attribute is to be protected, it should be named in the ACL entry.

You can match an ACL value against either a subject (trustee) or a privilege set, or both. If the subject name is not to be considered in the comparison, specify it as NULL. If the privilege set is not to be considered in the

comparison, specify an “approximate match” with a privilege set value of zero.

The Object ACL syntax supports both matching for equality and approximate matching. The difference between matching for equality and approximate matching concerns the privileges field of the comparison value. When matching for equality, the privilege set must match exactly for the comparison to succeed. When approximate matching has been selected, any bits in the privilege field in the filter that are set must also be set in the target. Any other bits in the target are ignored.

Values with the same *protectedAttrName* and *subjectName* fields are considered to be duplicate, and so are not permitted.

For information on bit mask for the *privileges* field and on the special values available for *protectedAttrName* and *subjectName* fields, see the Object_ACL_T structure (*NDS Core Services*).

Used In

ACL (page 364)

Inherited ACL (page 450)

New Object’s DS Rights (page 559)

New Object’s Self Rights (page 561)

Trustees Of New Object (page 704)

Octet List

Is used to describe an ordered sequence of octet strings.

Syntax ID

```
#define SYN_OCTET_LIST 13
```

LDAP Name

Octet List

ASN.1 ID

2.16.840.1.113719.1.1.5.1.13

API Data Structure

```
typedef struct _octet_list
{
    struct _octet_list N_FAR *next;
    uint32                length;
    puint8                data;
} Octet_List_T;
```

Transfer Format

```
uint32    Length
uint32    Number of strings

/*For each string */
Align4
uint32    Component Length = K
BYTE [K]  Component Data
```

LDAP Format

This syntax can only be transmitted in binary form.

Binary

`octetList ::= SEQUENCE OF OCTET STRING`

Matching Rules

Approximate—one string in the list matches

Equality—number of strings is the same and each string matches

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

A presented octet list matches a stored list if the presented list is a subset of the stored list. Octet strings are so designated because they are not interpreted by NDS. They are simply a series of bits with no Unicode implications.

Approximate matching rules apply because when comparing two octet lists, only one octet string in the list needs to match.

The *length* is the number of bits divided by 8 and rounded to the nearest integer. Thus each octet represents eight bits of data. The number of data bits will always be evenly divisible by 8.

Used In

NDSCat:AttrDefTbl (page 797)

NDSCat:IndexDefTbl (page 805)

WANMAN:WAN Policy (page 726)

Octet String

Used for attributes whose values are byte strings and are not interpreted by NDS.

Syntax ID

```
#define SYN_OCTET_STRING 9
```

LDAP Name

Octet String

ASN1.ID

1.3.6.1.4.1.1466.115.121.1.40

API Data Structure

```
typedef struct  
{  
    nuint32    length;  
    pnuint8    data;  
} Octet_String_T;
```

Transfer Format

```
uint32    Length = N  
BYTE [N]  Content
```

LDAP Format

This syntax can only be transmitted in binary form.

Matching Rules

Equality
Ordering

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Attributes using this syntax can set size limits. Octet strings are not Unicode strings.

For two octet strings to match, they must be the same length and the corresponding bit sequence (octets) must be identical. When comparing two strings, the first pair of octets that do not match are used to determine the order of the strings.

Used In

attributeCertificate (page 367)
Audit:A Encryption Key (page 368)
Audit:B Encryption Key (page 369)
Audit:Current Encryption Key (page 371)
Audit:Policy (page 375)
authorityRevocationList (page 377)
Authority Revocation (page 378)
Bindery Property (page 384)
cACertificate (page 392)
CA Private Key (page 393)
CA Public Key (page 394)
certificateRevocationList (page 396)
Certificate Revocation (page 397)
crossCertificatePair (page 399)
Cross Certificate Pair (page 405)
deltaRevocationList (page 408)
enhancedSearchGuide (page 423)
Entrust:AttributeCertificate (page 741)
External Name (page 425)
External Synchronizer (page 426)
GUID (page 437)
LDAP ACL v11 (page 463)
LDAP Attribute Map v11 (page 466)
LDAP Class Map v11 (page 468)

LDAPUserCertificate (page 751)
LDAP:ARL (page 752)
LDAP:caCertificate (page 753)
LDAP:CRL (page 754)
LDAP:crossCertificatePair (page 755)
Login Allowed Time Map (page 501)
masvAuthorizedRange (page 520)
masvDefaultRange (page 521)
masvDomainPolicy (page 522)
masvLabel (page 523)
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NDSPKI:Certificate Chain (page 536)
NDSPKI:Key File (page 538)
NDSPKI:Private Key (page 545)
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NDSPKI:Public Key Certificate (page 547)
NLS:Common Certificate (page 819)
NLS:Peak Installed Data (page 830)
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Obituary (page 597)
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Other GUID (page 604)
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preferredDeliveryMethod (page 636)
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Printer Configuration (page 641)
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Replica Up To (page 657)
searchGuide (page 670)
SMS Protocol Address (page 900)
SMS Registered Service (page 901)
supportedAlgorithms (page 685)
SvcInfo (page 903)

SvcTypeID (page 905)
telexNumber (page 698)
telexTerminalIdentifier (page 699)
Timezone (page 700)
Transitive Vector (page 703)
userCertificate (page 715)
WANMAN:Cost (page 722)
x500UniqueIdentifier (page 728)

Path

Is used for attributes that represent a file system path.

Syntax ID

```
#define SYN_PATH 15
```

LDAP Name

Tagged Name And String

ASN.1 ID

2.16.840.1.113719.1.1.5.1.15

API Data Structure

```
typedef struct
{
    uint32    nameSpaceType;
    pnstr8    volumeName;
    pnstr8    path;
} Path_T;
```

Transfer Format

```
uint32    Length
uint32    Name Space
unicode    Volume
Align4
unicode    Path
```

LDAP Format

String

```
taggedNameAndString = distinguishedname "#" uint32string "#"
                        dstring
```

Binary

```
taggedNameAndString ::= SEQUENCE {  
    objectName    LDAPDN,  
    number        uint32,  
    string        LDAPString  
}
```

Matching Rules

Equality

Substrings

Approximate—Not currently supported through LDAP

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Attributes using this syntax can set size limits.

The string represented by the *path* field is compared for equality using the same rules that Case Exact String (page 924) uses. That is, two Paths match for equality when their lengths and corresponding characters, including case, are identical.

In comparing two Paths, the following white space (spaces, tabs, etc.) is not significant:

- ♦ Leading spaces (those preceding the first printable character)
- ♦ Trailing spaces (those following the last printable character)
- ♦ Multiple consecutive internal spaces (these are taken as equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

The *volumeName* field must refer to a distinguished name of an object that exists in the NDS tree. NDS verifies that this field refers to an existing object, but does not ensure that the object is a volume object. Thus, this syntax can be used to refer to any type of object in the NDS tree.

When this syntax is used to specify a path to a file system object, the *volumeName* field must refer to a volume object that already exists in the NDS tree. In searches and comparisons, the *path* field can specify a path, or you can set the path to “*” to match any path on the volume.

For information about the fields in the structure, see the Path_T structure (*NDS Core Services*).

Used In

Audit:Path (page 374)
Home Directory (page 441)
Messaging Database Location (page 530)
New Object’s FS Rights (page 560)
Path (page 625)
Used By (page 712)
Uses (page 718)
Volume Space Restrictions (page 721)

Postal Address

Is used for attributes whose values are postal addresses.

Syntax ID

```
#define SYN_PO_ADDRESS 18
```

LDAP Name

Postal Address

ASN.1 ID

1.3.6.1.4.1.1466.115.121.1.41

API Data Structure

```
typedef pnstr8 Postal_Address_T[6];
```

Transfer Format

```
uint32    Length of attribute value
uint32    Number of strings

/* For each string */
Align4
unicode   String
```

LDAP Format

```
dstring *5 ($ dstring)
```

Matching Rules

Equality

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

An attribute value for Postal Address will be typically composed of selected attributes from the MHS Unformatted Postal O/R Address version 1 according to Recommendation F.401. The value is limited to 6 lines of 30 characters each, including a Postal Country Name. Normally the information contained in such an address could include a name, street address, city, state or province, postal code and possibly a postal office box number, depending on the specific requirements of the named object.

NDS requires 6 strings; blank strings must include at least a space. LDAP supports 1 to 6 strings, and the Novell LDAP server handles the differences between NDS and LDAP.

The matching rules for values of this type are the same as those for the Case Ignore List (page 926) attribute. That is, two postal addresses match for equality if, and only if, the number of strings in each is the same and all corresponding strings match. For two corresponding strings in the list to match, they must be the same length and their corresponding characters must be identical (see the matching rules for the Case Ignore List (page 926) attribute).

When comparing the strings in a Postal Address, the following white space (spaces, tabs, etc.) is regarded as not significant:

- ♦ Leading spaces (those preceding the first printable character)
- ♦ Trailing spaces (those following the last printable character)
- ♦ Multiple consecutive internal spaces (these are taken as equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

Used In

Postal Address (page 629)

Printable String

Is used in attributes whose values are printable strings as defined in CCITT X.208. The case (upper or lower) is significant when comparing printable strings.

Syntax ID

```
#define SYN_PR_STRING 4
```

LDAP Name

Printable String

ASN.1 ID

1.3.6.1.4.1.1466.115.121.1.44

API Data Structure

```
typedef pnstr8 PR_String_T;
```

Transfer Format

uint32	Length
unicode	String

Matching Rules

Equality
Substrings

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Attributes using this syntax can set size limits.

The following characters are in the printable string character set:

A..Z	upper case alphabetic characters
a..z	lower case alphabetic characters
0..9	digits
	space character
'	apostrophe
"	quotation
(	left parenthesis
)	right parenthesis
+	plus sign
,	comma
-	hyphen
.	full stop (period)
/	solidus (slash)
:	colon
=	equal sign
?	question mark

Two printable strings match for equality when they are the same length and their corresponding characters are identical. Case (upper or lower) is significant when comparing printable strings. For example, as printable strings, “Dundee” and “DUNDEE” do not match.

When comparing printable strings, the following spaces are not significant:

- ◆ Leading spaces (precede the first printing character)
- ◆ Trailing spaces (follow the last printing character)
- ◆ Multiple consecutive internal spaces (equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

Used In

destinationIndicator (page 412)

dnQualifier (page 417)

Page Description Language (page 610)

Serial Number (page 677)

Replica Pointer

Is used for attributes whose values represent partition replicas.

Syntax ID

```
#define SYN_REPLICA_POINTER 16
```

LDAP Name

NDS Replica Pointer

ASN.1 ID

2.16.840.1.113719.1.1.5.1.16

API Data Structure

```
typedef struct
{
    pnstr8          serverName;
    nint32          replicaType;
    nint32          replicaNumber;
    nuint32         count;
    Net_Address_T   replicaAddressHint[1];
} Replica_Pointer_T;
```

Transfer Format

```
uint32          Length
unicode         Server Name
Align4
uint16          Replica Type
uint16          Replica State
uint32          Replica Number
uint32          Number of Addresses = N
Address Record[N]  Addresses

/* For each address record */
Align 4
uint32          Transport Type
uint32          Address Length = M
```

LDAP Format

This syntax can only be transmitted in binary form.

Binary

```
ndsReplicaPointer ::= SEQUENCE {
    serverName          LDAPDN,
    replicaType         uint16,
    replicaState        uint16,
    replicaNumber       uint32,
    replicaAddressHint  SEQUENCE OF NetAddress
}

NetAddress ::= SEQUENCE {
    transportType      uint32,
    addressValue       OCTET STRING
}
```

Matching Rules

Equality

Approximate—Not currently supported through LDAP

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Each value of this syntax is composed of five parts:

- ◆ The complete name of the server that stores the replica.
- ◆ A value describing the capabilities of this copy of the partition: master, secondary, read-only, subordinate reference.
- ◆ A value indicating the current state of the replica (new, dying, locked, changing state, splitting, joining, moving).
- ◆ A number representing the replica. All replicas of a partition have different numbers which are assigned when the replicas are created.

- ♦ A referral which contains a count of the addresses and then one or more network addresses that hint at the node at which the server probably resides. Since servers are accessible over different protocols, the server may have an address for each supported protocol.

The matching rules for values of Replica Pointer are based on the replica server name field only, and are the same as those for Distinguished Name (page 938).

The length of the *replicaAddressHint* structure is variable and cannot be calculated in a block, because address types are not all the same length. For more information, see the *Net_Address_T* structure (*NDS Core Services*).

Used In

Replica (page 656)

Stream

Is used for login scripts and other stream attributes.

Syntax ID

```
#define SYN_STREAM 21
```

LDAP Name

Binary

ASN.1 ID

1.3.6.1.4.1.1466.115.121.1.5

API Data Structure

```
typedef Octet_String_T    Stream_T;
```

Transfer Format

```
uint32    Length = N  
BYTE[N]   Content
```

LDAP Format

BER-encoded

Matching Rules

None

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Streams are files of information. The data stored in a stream file has no syntax enforcement of any kind. It is purely arbitrary data, defined by the application that created and uses it.

Any attribute defined with this syntax is single-valued. When you try to read an attribute of this type or search on it, the value behaves like an empty string. That is, when you return from read or search, it returns an empty octet string. When you try to add a value, the request is ignored.

You must read or write attributes of this syntax by calling the **NWDSOpenStream** function, followed by calling standard file Read and Write functions.

Used In

- LDAP Class Map (page 749)
- Login Script (page 511)
- NDSCat:CatalogDB (page 801)
- NDSCat:Log (page 808)
- NRD:Registry Data (page 847)
- NRD:Registry Index (page 848)
- Print Job Configuration (page 638)
- Printer Control (page 642)
- Setup Script (page 682)

Telephone Number

Is used for attributes whose values are telephone numbers.

Syntax ID

```
#define SYN_TEL_NUMBER 10
```

LDAP Name

Telephone Number

ASN.1 ID

1.3.6.1.4.1.1466.115.121.1.50

API Data Structure

```
typedef pnstr8 TN_String_T;
```

Transfer Format

```
uint32    Length
unicode   String
```

LDAP Format

```
printable string
```

Matching Rules

Equality
Substrings

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Attributes using this syntax can set size limits. The length of telephone number strings must be between 1 and 32.

The rules for matching telephone numbers are identical to those for the Case Exact String attribute syntax except that all space and hyphen (-) characters are skipped during the comparison.

Two telephone numbers match for equality when their lengths and corresponding characters, including case, are identical.

In comparing case exact strings, the following spaces are not significant:

- ♦ Leading spaces (precede the first printing character)
- ♦ Trailing spaces (follow the last printing character)
- ♦ Multiple consecutive internal spaces (equivalent to a single space character)

In matching attributes that conform to this syntax, NDS omits those spaces that are not significant (as defined above). NDS stores insignificant spaces with the attribute value.

Used In

Telephone Number (page 696)

Time

Is used for attributes whose values represent time.

Syntax ID

```
#define SYN_TIME 24
```

LDAP Name

Generalized Time

ASN.1 ID

1.3.6.1.4.1.1466.115.121.1.24

API Data Structure

```
typedef nint32 Integer_T;
```

LDAP Format

printable string

Transfer Format

```
uint32    Length = 4
uint32    Content
```

Matching Rules

Equality
Ordering

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

In NDS, the time is stored as an integer. Zero equals 12:00 midnight, January 1, 1970, UTC, and the value equals the number of whole seconds since that time.

The LDAP server converts the NDS time to the LDAP format specified by X.208 which includes the year, month, day, hour, minute, optionally seconds, and time zone (GMT is recommended for the time zone and uses Z as its symbol). An attribute with an LDAP time syntax would have a value similar to the following:

```
199412161032Z or 19941216103200Z
```

Two Time values are compared by comparing the Integer_T values.

Used In

Last Login Time (page 461)
Login Expiration Time (page 503)
Login Intruder Reset Time (page 509)
Login Time (page 513)
Low Convergence Reset Time (page 514)
NDSCat:Actual End Time (page 788)
NDSCat:Actual Start Time (page 794)
NDSCat:Start Time (page 817)
NLS:Summary Update Time (page 836)
Password Expiration Time (page 618)

Timestamp

Is used for attributes whose values mark the time when a particular event occurred or will occur.

Syntax ID

```
#define SYN_TIMESTAMP 19
```

LDAP Name

NDS Timestamp

ASN.1 ID

2.16.840.1.113719.1.1.5.1.19

API Data Structure

```
typedef struct
{
    nuint32    wholeSeconds;
    nuint16    replicaNum;
    nuint16    eventID;
} TimeStamp_T;
```

Transfer Format

```
uint32          Length = 8
TIMESTAMP       Timestamp Value

/* The Timestamp Value is represented as follows */

uint32    Seconds
uint16    Replica Number
uint16    Event
```

LDAP Format

String

```
ndsTimestamp = wholeseconds "#" replicanum "#" event

wholeseconds = uint32string ; 0 = 12:00 midnight Jan 01 1970,
UTC

replicanum = uint16string

event = uint16string
```

Binary

```
ndsTimestamp ::= SEQUENCE {
    wholeSeconds    uint32,
    replicaNum      uint16,
    eventID         uint16
}
```

Matching Rules

Equality
Ordering

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

A timestamp value has three components:

- ♦ The wholeSeconds field consists of the whole number of seconds, where zero equals 12:00 midnight, January 1, 1970, UTC.
- ♦ The replicaNum field identifies the server that created the timestamp. A replica number is assigned whenever a replica is created on a server.
- ♦ The eventID field is an integer that orders events occurring within the same whole-second interval. The event number restarts at one for each new second.

The initial NULL value of a timestamp has seconds=1 and event=0. Values can be skipped, but must not be reused. An “unknown” event is coded as 0xFFFF.

Time stamps can be compared for equality and for ordering. Two Timestamp values are matched for equality by comparing the whole Seconds fields and then the Event fields. If the whole Seconds fields are unequal, order is determined by that field alone. If the Seconds fields are equal and the Event fields are unequal, order is determined by the Event fields. If both fields are equal, the time stamps are equal. For ordering comparisons, the Timestamp value is treated as a 64-bit unsigned integer, with Seconds as the most significant.

Used In

Replica Up To (page 657)
Partition Creation Time (page 614)
Purge Vector (page 650)
Received Up To (page 653)
Synchronization Tolerance (page 693)
Synchronized Up To (page 694)

Typed Name

Is used for attributes whose values represent a level and an interval associated with an object.

Syntax ID

```
#define SYN_TYPED_NAME 25
```

LDAP Name

Typed Name

ASN.1 ID

2.16.840.1.113719.1.1.5.1.25

API Data Structure

```
typedef struct
{
    pnstr8      objectName;
    nuint32     level;
    nuint32     interval;
} Typed_Name_T;
```

Transfer Format

uint32	Length
uint32	Level
uint32	Interval
unicode	Distinguished Name

LDAP Format

String

```
typedname = objectname "#" level "#" interval  
  
objectname = distinguishedname  
  
level = uint32string  
  
interval = uint32string
```

Binary

```
typedName ::= SEQUENCE {  
    LDAPDN      objectName,  
    uint32      level,  
    uint32      interval  
}
```

Matching Rules

Equality

Approximate—Not currently supported through LDAP

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

This syntax names an NDS object and attaches two numeric values to it:

- ♦ The level of the attribute indicates the priority.
- ♦ The interval indicates the frequency of reference.

The objectName or Distinguished Name identifies the NDS object referred to by the Typed Name and must refer to a distinguished name of an object that exists in the NDS tree. NDS verifies that this field refers to an existing object.

The values of level and interval are user-assigned and relative. NDS does not check their values and thus applications can use them to order, enforce priorities, or implement iterative intervals.

In comparisons and searches using equality, all three parts of the field must be equal for attributes to be equal.

In comparisons and searches using NDS approximate matching, the level and interval fields are ignored and only the objectName fields must be equal.

Used In

Notify (page 563)

Partition Control (page 612)

Print Server (page 639)

Printer (page 640)

Queue (page 651)

Unknown

Is used for attributes whose attribute definition was deleted from the schema.

Syntax ID

```
#define SYN_UNKNOWN 0
```

LDAP Name

Unknown

ASN.1 ID

2.16.840.1.113719.1.1.5.1.0

API Data Structure

```
typedef struct
{
    pnstr8      attrName;
    nuint32     syntaxID;
    nuint32     valueLen;
    nptr        value;
} Unknown_Attr_T;
```

Transfer Format

```
uint32      Length = N
BYTE [N]    Content
```

LDAP Format

Matching Rules

Equality

Ordering

Remarks

For help in understanding the syntax definition template, see “Reading Syntax Definitions” on page 66.

Used In

Auxiliary Class Flag (page 380)
Password Management (page 619)
Unknown (page 709)

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2.16.840.1.113719.1.1.6.1.32 266

2.16.840.1.113719.1.1.6.1.33	135, 159	2.16.840.1.113719.1.4.0.3.2.44	761
2.16.840.1.113719.1.1.6.1.34	132	2.16.840.1.113719.1.4.0.3.2.51	867
2.16.840.1.113719.1.1.6.1.4	101	2.16.840.1.113719.1.4.0.3.2.59	885
2.16.840.1.113719.1.1.6.1.42	305	2.16.840.1.113719.1.4.0.3.2.69	869
2.16.840.1.113719.1.1.6.1.43	308	2.16.840.1.113719.1.4.2.1.0	299
2.16.840.1.113719.1.1.6.1.44	311	2.16.840.1.113719.1.4.2.1.0.0	872
2.16.840.1.113719.1.1.6.1.7	116	2.16.840.1.113719.1.4.2.1.0.1	737
2.16.840.1.113719.1.1.6.1.8	128	2.16.840.1.113719.1.4.2.1.0.2	735
2.16.840.1.113719.1.12.4.1.0	368	2.16.840.1.113719.1.4.2.1.0.3	773
2.16.840.1.113719.1.12.4.2.0	369	2.16.840.1.113719.1.4.2.1.0.4	734
2.16.840.1.113719.1.12.4.3.0	370	2.16.840.1.113719.1.4.2.1.0.5	767
2.16.840.1.113719.1.12.4.4.0	376	2.16.840.1.113719.1.4.2.1.0.6	768
2.16.840.1.113719.1.12.4.5.0	371	2.16.840.1.113719.1.4.2.1.0.7	769
2.16.840.1.113719.1.12.4.6.0	372	2.16.840.1.113719.1.4.2.1.1	296
2.16.840.1.113719.1.12.4.7.0	373	2.16.840.1.113719.1.4.2.1.1.0	738
2.16.840.1.113719.1.12.4.8.0	374	2.16.840.1.113719.1.4.2.1.1.1	763
2.16.840.1.113719.1.12.4.9.0	375	2.16.840.1.113719.1.4.2.1.1.10	877
2.16.840.1.113719.1.12.6.1.0	86	2.16.840.1.113719.1.4.2.1.1.12	743
2.16.840.1.113719.1.15.4.1	847	2.16.840.1.113719.1.4.2.1.1.13	744
2.16.840.1.113719.1.15.4.2	848	2.16.840.1.113719.1.4.2.1.1.14	845
2.16.840.1.113719.1.27.4.48	910	2.16.840.1.113719.1.4.2.1.1.15	876
2.16.840.1.113719.1.27.4.49	916	2.16.840.1.113719.1.4.2.1.1.17	781
2.16.840.1.113719.1.27.4.50	912	2.16.840.1.113719.1.4.2.1.1.2	764
2.16.840.1.113719.1.27.4.51	911	2.16.840.1.113719.1.4.2.1.1.3	844
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2.16.840.1.113719.1.31.4.1.4	524, 760	2.16.840.1.113719.1.4.2.1.1.5	875
2.16.840.1.113719.1.31.4.1.5	521, 757	2.16.840.1.113719.1.4.2.1.1.7	882
2.16.840.1.113719.1.31.4.1.6	520, 756	2.16.840.1.113719.1.4.2.1.1.8	883
2.16.840.1.113719.1.31.4.1.7	522, 758	2.16.840.1.113719.1.4.2.1.1.9	846
2.16.840.1.113719.1.31.6.1.1	144	2.16.840.1.113719.1.4.2.2.0	302
2.16.840.1.113719.1.38.4.1.1	726	2.16.840.1.113719.1.4.2.2.0.0	774
2.16.840.1.113719.1.38.4.1.2	725	2.16.840.1.113719.1.4.2.2.0.1	782
2.16.840.1.113719.1.38.4.1.3	722	2.16.840.1.113719.1.4.2.2.0.10	732
2.16.840.1.113719.1.38.4.1.4	724	2.16.840.1.113719.1.4.2.2.0.21	842
2.16.840.1.113719.1.38.6.1.4	285	2.16.840.1.113719.1.4.2.2.0.23	772
2.16.840.1.113719.1.39.4.1.1	668	2.16.840.1.113719.1.4.2.2.0.26	777
2.16.840.1.113719.1.39.4.1.2	669	2.16.840.1.113719.1.4.2.2.0.3	871
2.16.840.1.113719.1.39.6.1.1	248	2.16.840.1.113719.1.4.2.2.0.30	736
2.16.840.1.113719.1.39.6.1.2	251	2.16.840.1.113719.1.4.2.2.0.31	778
2.16.840.1.113719.1.4.0.0.2.0	843	2.16.840.1.113719.1.4.2.2.0.34	745, 765
2.16.840.1.113719.1.4.0.3.0.3	870	2.16.840.1.113719.1.4.2.2.0.35	746
2.16.840.1.113719.1.4.0.3.2.14	873	2.16.840.1.113719.1.4.2.3.1	770
2.16.840.1.113719.1.4.0.3.2.21	868	2.16.840.1.113719.1.4.2.3.2	771
2.16.840.1.113719.1.4.0.3.2.24	733	2.16.840.1.113719.1.4.2.3.3	775
2.16.840.1.113719.1.4.0.3.2.38	865	2.16.840.1.113719.1.4.2.3.4	779
2.16.840.1.113719.1.4.0.3.2.39	866	2.16.840.1.113719.1.4.2.3.5	776
2.16.840.1.113719.1.4.0.3.2.42	863	2.16.840.1.113719.1.4.2.3.6	780
2.16.840.1.113719.1.4.0.3.2.43	762	2.16.840.1.113719.1.4.2.3.7	766

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2.16.840.1.113730.3.1.193 589
2.16.840.1.113730.3.1.194 587

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2.16.840.1.113730.3.1.196	584, 859	2.5.4.2	457
2.16.840.1.113730.3.1.199	582	2.5.4.20	696
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2.16.840.1.113730.3.1.3	579	2.5.4.25	452
2.16.840.1.113730.3.1.30	583	2.5.4.26	655
2.16.840.1.113730.3.1.36	577	2.5.4.27	412
2.16.840.1.113730.3.1.37	578	2.5.4.28	636
2.16.840.1.113730.3.1.38	576	2.5.4.29	637
2.16.840.1.113730.3.1.39	864	2.5.4.3	400
2.16.840.1.113730.3.1.4	422	2.5.4.30	686
2.16.840.1.113730.3.1.40	906	2.5.4.31	525
2.16.840.1.113730.3.1.47	581	2.5.4.32	608
2.16.840.1.113730.3.1.69	594	2.5.4.33	660
2.16.840.1.113730.3.1.70	592	2.5.4.34	675
2.16.840.1.113730.3.1.71	591	2.5.4.35	716
2.16.840.1.113730.3.1.72	593	2.5.4.36	715
2.16.840.1.113730.3.1.73	580	2.5.4.37	392
2.16.840.1.113730.3.1.74	564	2.5.4.38	377
2.16.840.1.113730.3.1.75	565	2.5.4.39	396
2.16.840.1.113730.3.1.76	590	2.5.4.4	691
2.16.840.1.113730.3.2.10	193	2.5.4.40	399
2.16.840.1.113730.3.2.2	272, 291	2.5.4.42	434
2.16.840.1.113730.3.2.24	189	2.5.4.43	451
2.16.840.1.113730.3.2.26	197	2.5.4.44	432
2.16.840.1.113730.3.2.3	187	2.5.4.45	728
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2.5.18.3	909	2.5.4.50	525
2.5.18.4	913	2.5.4.51	446
2.5.21.9	915	2.5.4.52	685
2.5.4.0	601	2.5.4.53	408
2.5.4.1	362	2.5.4.54	415
2.5.4.10	595	2.5.4.58	367
2.5.4.11	606	2.5.4.6	390
2.5.4.12	701	2.5.4.7	458
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Revision History

The following table lists all changes made to the NDS Schema Reference:

February 2001	Changed operational (attributes and classes) to base and base (attributes and classes) to standard to avoid confusion with LDAP operational attributes. Put the LDAP operational attributes in a separate chapter.
September 2000	Updated the schema registration link and added NDS version information to the Surname and Transitive Vector attributes.
July 2000	Added LDAP operational attributes and the LDAP userPassword attribute.
May 2000	Added LDAP names and OIDs to the NDS syntax definitions.

